# Companion notes: degree correction and additional likelihood results

A working collection accompanying the SBM article

spectral-lrt research project

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**Abstract**

These notes collect material separated from the ordinary-SBM article. The first part gives the bounded-parameter degree-corrected model, the distinction between profile and known-target-degree oracles, and the regularized fitted Bernoulli profile classifier. Its current risk bound requires only $np \to \infty$ and $np^2 \le C_1$, with no logarithmic lower bound. The DC-specific nuisance estimation, profile reduction, and tilted cancellation are retained here. The small-floor geometry and its probability bounds are proved here; sharp-transfer, spectral-row-tail, and local deleted-graph tools are shared with the adjacency-based SBM article. Their applicability to this normalized DC encoder is checked explicitly. The later parts preserve the separate earlier unregularized decoders, the wider-probability ordinary free-profile result, an obstruction for the unrestricted known-target-degree oracle, and auxiliary arguments. This is a working companion, not a second article with a unified scope. Historical hypotheses are attached to their original decoders and do not restrict the current regularized theorem.

**Reading map.** The first article is *Adjacency spectral likelihood classification in stochastic block models*. It is independent of this companion. The source package contains the following overflow from the combined revision.

| Original component | Present location |
|---|---|
| Ordinary adjacency theorem and shared sharp/local tools | The first article; small-floor DC geometry proved in Section 3 here. |
| DC model, current regularized algorithm and DC fitting | Sections 1–5; bounded degree parameters and no logarithmic lower bound. |
| Earlier normalized geometry, ordinary free-profile decoder and probability closure | Sections 6–8 and 10; their separate historical algorithms and full proofs. |
| Earlier DC decoder and classical oracle bridge | Section 9, with its probability closure in Section 10. |
| Known-target obstruction and zero-degree fallback | Section 11 and Section 12.1. |
| Optional exponential-moment transfer and retained-spectrum identity | Sections 3.9 and 12.2. |
| Implementation scope, sparse history and audit records | Sections 12.3–12.4; source directories `audits/` and `archive/`. |

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# 1 Degree-corrected models and oracle benchmarks

The target is a constant oracle-risk bound

$$\mathbb{E}\ell(\widehat{z}, z) \le CR_*(n), \tag{1}$$

with $C$ independent of $n$. The companion retains ordinary-SBM notation where needed by the historical free-profile analysis. The independent ordinary-SBM article contains the current ordinary adjacency block-LRT theorem and its complete proof.

## 1.1 Bernoulli models and fixed structural constants

There are $n$ vertices and $K \ge 2$ groups, with fixed $K$. Let $z \in [K]^n$, let $Z$ be its indicator matrix, and let $n_a$ be the size of group $a$. Throughout $n_a \ge \pi_- n$ for a fixed $\pi_- > 0$. The edges above the diagonal are independent, the adjacency matrix $A$ is symmetric, and $A_{ii} = 0$. In DC-SBM,

$$A_{ij} \sim \mathrm{Bern}(\theta_i\theta_j pC_{z_i z_j}), \quad i < j, \qquad d = np, \qquad \Omega = \Theta Z(pC)Z^\top\Theta. \tag{2}$$

Here $\Theta = \mathrm{diag}(\theta_i)$, $C = C^\top$, and the fixed structural bounds are

$$0 < \theta_- \le \theta_i \le \theta_+ < \infty, \quad \sum_{i:z_i=a} \theta_i = n_a, \quad 0 < c_- \le C_{ab} \le c_+ < \infty, \quad \sigma_{\min}(C) \ge \kappa > 0. \tag{3}$$

The model class is nonempty only when $1 \in [\theta_-, \theta_+]$. All constants in asymptotic bounds may depend on these structural bounds, but not on $n$. One may take a fixed $C$, or a sequence in this same compact class. Ordinary SBM is the special case $\theta_i = 1$, with block probability matrix $P = pC$.

The completed matrix $\Omega$ includes a diagonal for algebraic convenience; the actual mean is $\mathbb{E}A = \Omega - \mathrm{diag}(\Omega_{ii})$. That deterministic difference has norm $O(p)$ and is included in all error bounds. Put $D = \mathrm{diag}(A\mathbf{1})$ and $\lambda_i = (\mathbb{E}A\mathbf{1})_i$. Thus $\lambda_i \asymp d$ and $\Omega\mathbf{1}$ equals $d\theta_i S_{z_i}$ in coordinate $i$, where

$$\pi_a = n_a/n, \qquad S_a = \sum_b \pi_b C_{ab}. \tag{4}$$

The loss for the estimated partition is

$$\ell(\widehat{z}, z) = \frac{1}{n}\min_{\sigma\in\mathfrak{S}_K}\sum_i \mathbf{1}\{\sigma(\widehat{z}_i) \ne z_i\}.$$

## 1.2 Three explicitly different local oracles

All oracle classifiers below use the raw incident row $A_{i,-i}$ and the true neighbor labels. They are local likelihood classifiers: an exact global block-size constraint is not used to determine the one missing label. For simplicity candidate weights are uniform. Any fixed weights bounded above and below by positive constants give the same conclusions.

For ordinary SBM, the oracle chooses the maximizer of

$$\ell_{i,*}(a) = \sum_{j\ne i}\{A_{ij}\log P_{az_j} + (1 - A_{ij})\log(1 - P_{az_j})\}. \tag{5}$$

Its average error probability is $R_{\mathrm{SBM},*}$. For DC-SBM, the *known-target-degree oracle* replaces $P_{az_j}$ in (5) by $\theta_i\theta_j pC_{az_j}$ and is supplied with every true $\theta_j$, including $\theta_i$. Its risk is $R_{\theta,*}$.

The *profile oracle* is supplied with true neighbor parameters and $pC$, but profiles the target degree separately for every candidate:

$$\ell_{i,\mathrm{prof},*}(a) = \sup_{t\ge 0}\sum_{j\ne i}\{A_{ij}\log(t\theta_j pC_{az_j}) + (1 - A_{ij})\log(1 - t\theta_j pC_{az_j})\}. \tag{6}$$

We restrict candidate edge probabilities to at most $q_{\max} = 1/2$; the constraint is inactive on the events used in the sparse theorem. Write $R_{\mathrm{prof},*}$ for its average error probability. This is a full-edge generalized likelihood-ratio reference. It is not described as a Bayes test between two fixed known-$\theta_i$ laws. In all three definitions, the risk averages the actual directional error at each fixed true vertex. Ties are resolved by a fixed ordering; strict-tail arguments below cover any such ordering.

## 2 The current regularized DC-SBM result

Fix sufficiently small $a_0, \alpha > 0$ and sufficiently large $\beta > 0$, depending only on the fixed structural parameter bounds. Choose them so that the population degrees lie strictly inside the propensity-clamping interval, and the population floor is inactive. Put

$$\mathbf{d} = A\mathbf{1}, \quad \bar{D} = n^{-1}\mathbf{1}^\top \mathbf{d}, \quad \tau = a_0 \bar{D}, \quad G_i = \max(D_i, \tau).$$

The all-zero graph has a fixed output. Otherwise $G$ is positive, even when some vertices are isolated. Retain the $K$ eigenpairs with largest absolute eigenvalues of $S_G = G^{-1/2} A G^{-1/2}$ and the raw degrees. There is no requirement that a Perron degree vector be one of these eigenvectors after flooring. Compute

$$R_0 = G^{1/2} U \Lambda U^\top G^{1/2} = A H_G, \qquad H_G = G^{-1/2} U U^\top G^{1/2},$$

and put, with $r = \mathrm{diag}(R_1)$ understood as a column vector,

$$R_1 = R_0 + (\mathbf{d} - R_0\mathbf{1})\mathbf{1}^\top / n, \qquad R^h = R_1 + \frac{r\mathbf{1}^\top}{n-1} - \frac{n}{n-1}\,\mathrm{diag}(r). \tag{7}$$

Thus $R^h\mathbf{1} = \mathbf{d}$ and $\mathrm{diag}(R^h) = 0$ exactly. Only $(\mathbf{d}, U, \Lambda)$ are used after encoding; $G$ is recovered from $\mathbf{d}$.

Initialize by normalizing the rows of $U$ and using the deterministic radius-1/8, deletion-radius-1/4 count procedure in lemma 9.2. A zero row is assigned any fixed unit direction for this initialization, without stopping other vertices.

Define

$$T_i = \min\{\max(D_i, \alpha\bar{D}), \beta\bar{D}\}, \qquad \widehat{\theta}_i = T_i / \overline{T}_{\widehat{z}_i}, \qquad \overline{T}_a = \widehat{n}_a^{-1} \sum_{j:\widehat{z}_j = a} T_j.$$

Fit the symmetric block matrix by the exact signed Bernoulli working criterion (110), with $R^h$ and these degree estimates, under the same fixed candidate-probability cap. Classify each row by exact Bernoulli target profiling, with the degree parameter profiled separately for every candidate community.

At least one fit-and-LRT update is performed. The procedure may use any finite number of these hard refits, including finite data-dependent stopping. The invariant basin below covers every such endpoint simultaneously. An undefined empty-group or parameter-fit operation may have a fixed fallback; those operations are well-defined on the proved aggregate event. No individual zero degree triggers a graph-wide fallback.

**Theorem 2.1** (Regularized DC-SBM at every diverging expected-degree scale)**.** *Under the fixed structural assumptions (2)–(3), suppose*

$$d = np \longrightarrow \infty, \qquad np^2 \le C_1 < \infty.$$

*The regularized algorithm of section 2, using the DC-SBM profile LRT, satisfies*

$$\mathbb{E}\ell(\widehat{z}, z) \le C R_{\mathrm{prof},*}(n)$$

*for all sufficiently large $n$. The constant may depend on the fixed structural bounds, clamping constants and $C_1$, but not on $n$. Under the finite-$n$ equal-exposure condition (54), the right side may be replaced by $CR_{\theta,*}(n)$. In particular, the classical balanced planted partition with fixed separated $q/p$ is included. There is no lower bound proportional to $\log n$ in this statement.*

The reference $R_{\mathrm{prof},*}$ uses the full incident edge row. The second comparison is available only under the stated equal-exposure condition. The impossibility result in theorem 11.1 shows why a known-target-degree oracle is not a valid unrestricted benchmark. Only retained degrees and one eigensystem enter the algorithm; deleted graphs below are proof devices.

# 3 Shared tools and their degree-corrected specialization

The ordinary-SBM article now uses a high-degree-capped adjacency encoder. Its sharp Bernoulli transfer, same-threshold spectral-row tail, and deterministic local scalar comparison also apply here. We recall those tools with their degree-corrected premises. The small-floor geometry and its exponentially accurate event are proved in full below, since they differ from the large-threshold adjacency geometry in that article. The DC classifier in these notes retains its normalized encoder; no large-threshold DC extension is asserted.

## 3.1 Sharp likelihood transfer

**Lemma 3.1** (Bounded tilted error preserves the sharp prefactor)**.** *Let $X_j$ be independent Bernoulli variables with probabilities $p_j$, and let*

$$T = b_0 + \sum_j w_j X_j, \qquad K(s) = \log \mathbb{E} e^{-sT}.$$

*Suppose $|w_j| \le W$, the minimizer $s_*$ satisfies $\tau \le s_* \le 1-\tau$ for a fixed $\tau > 0$, and under the probability measure $Q$ proportional to $e^{-s_* T}$,*

$$\mathbb{E}_Q T = 0, \quad vd \le \mathrm{Var}_Q(T) \le V_0 d, \quad \sum_j r_j \le V_0 d, \tag{8}$$

*where $r_j = Q(X_j = 1)$. Put $I = -K(s_*)$. For every fixed $L < \infty$ and all sufficiently large $d$, uniformly over $|x| \le L$,*

$$c_L d^{-1/2} e^{-I} \le \mathbb{P}(T < x) \le \mathbb{P}(T \le x) \le C_L d^{-1/2} e^{-I}. \tag{9}$$

*Let $\mathcal{W}$ be independent of $(X_j)_j$ and let*

$$\widehat{T} = T + b + \sum_j e_j X_j$$

*have coefficients measurable with respect to $\mathcal{W}$. On a measurable training event suppose*

$$\max_j |e_j| \le M, \qquad \left| b + \sum_j r_j e_j \right| \le B, \qquad \sum_j r_j e_j^2 \le V. \tag{10}$$

*For fixed $M, V$ and every real $x$,*

$$\mathbb{P}(\widehat{T} \le x \mid \mathcal{W}) \le C \exp\{s_* x + B + e^M V/2\}\, d^{-1/2} e^{-I}. \tag{11}$$

*The constant is uniform over the training event. In particular $B$ can be a nonnegative training-measurable random bound, provided its indicated exponential moment is integrated when removing the conditioning.*

*Proof reference.* Lemma A.1 of the SBM article proves this by Berry–Esseen under the exact Bernoulli tilt, followed by a second tilt for the independent-training perturbation. That argument makes no block-model assumption. It therefore applies directly to all scores used below. See also Zhou and Li [5] for sharp Chernoff comparisons in community detection.

### 3.2 Application to the actual Bernoulli SBM oracle

For a true label $a$ and a competitor $b$, the oracle contrast is

$$T_{i,ab} = \ell_{i,*}(a) - \ell_{i,*}(b) = b_{i,ab} + \sum_{j \neq i} w^{ab}_{z_j} A_{ij},$$

where

$$w^{ab}_c = \log \frac{P_{ac}(1 - P_{bc})}{P_{bc}(1 - P_{ac})}, \qquad b_{i,ab} = \sum_{j \neq i} \log \frac{1 - P_{az_j}}{1 - P_{bz_j}}. \tag{12}$$

Its exact Chernoff information is

$$I_{i,ab} = - \min_{0 \leq s \leq 1} \sum_{j \neq i} \log \left[ P^{1-s}_{az_j} P^s_{bz_j} + (1 - P_{az_j})^{1-s} (1 - P_{bz_j})^s \right]. \tag{13}$$

On the compact positive full-rank classes used here, probabilities are comparable to $p$ and bounded away from one, the weights are bounded, $s_*$ stays away from the endpoints, and the tilted variance is comparable to $d$. To verify the last assertions, the derivatives of the transform at its endpoints are respectively minus and plus the two directional Bernoulli KL divergences, both of order $d$. Its second derivative is bounded by $Cd$. The row separation implied by $\sigma_{\min}(P) \geq \kappa p$ gives a lower bound $cd$ on that derivative throughout $[0, 1]$: tilted probabilities are comparable to $p$ and the sum of squared log-odds differences is at least $cn$. Thus lemma 3.1 applies uniformly to every pair.

The same conclusion holds for the profile contrasts defined below, with their own directional information $J_{i,a \to b}$. It is not necessary for those two directional exponents to coincide.

### 3.3 Fixed-$K$ comparison and label alignment

Let $q_i$ be the error probability of the specified oracle at vertex $i$. For any coherent family of its candidate scores, strict pairwise loss to a competitor implies an oracle error; an oracle error implies non-strict loss to some competitor. Hence

$$\max_{b \neq a} \mathbb{P}(T_{i,ab} < 0) \leq q_i \leq \sum_{b \neq a} \mathbb{P}(T_{i,ab} \leq 0).$$

By (9),

$$q_i \asymp \frac{1}{\sqrt{d}} \sum_{b \neq z_i} e^{-I_{i,ab}} \tag{14}$$

for the SBM oracle; replace $I$ by $J$ for the profile oracle. The constants may include a factor depending on fixed $K$. Fitted partitions are aligned with the true groups by a single common permutation on the good event. Uniform score comparisons under that alignment imply

$$\mathbb{E}[\ell(\widehat{z}, z) \mathbf{1}\{\mathcal{G}\}] \leq \frac{1}{n} \sum_i \mathbb{P}(\text{aligned fitted error at } i, \mathcal{G}).$$

The ensuing pairwise bounds apply to fixed true coefficient vectors; a data-dependent matching does not become an independence assumption. Finally,

$$\mathbb{E}\ell(\widehat{z}, z) \leq C R_* + \mathbb{P}(\mathcal{G}^c). \tag{15}$$

Section 10 proves, for the historical decoders, $\mathbb{P}(\mathcal{G}^c) = O(R_*)$ under the stated model conditions. It is this comparison, rather than merely $\mathbb{P}(\mathcal{G}^c) = o(1)$, that completes each theorem.

### 3.4 Empirical-floor geometry for bounded degree parameters

Throughout this section $D_0 = \operatorname{diag}(\Omega\mathbf{1})$, $G_i = \max(D_i, a_0\bar{D})$, and $\|v\|_G^2 = v^\top G v$. The floor multiplier is fixed and sufficiently small.

**Proposition 3.2** (Sparse degree-floor geometry)**.** *Under the fixed structural conditions (2)–(3), suppose $d = np \to \infty$ and $np^2 \le C_1$. For every fixed $r > 0$, with probability at least $1 - C_r n^{-r}$, $cd \le G_i$, $cnd \le \sum_i D_i$, and*

$$\left\|G^{-1/2}(A-\Omega)G^{-1/2}\right\|_{\mathrm{op}} \le C_r d^{-1/2}, \qquad \sum_i G_i \le C_r nd, \tag{16}$$

$$\sum_i \frac{(D_i - (D_0)_{ii})^2}{G_i} + \sum_i \frac{(G_i - (D_0)_{ii})^2}{G_i} \le C_r n. \tag{17}$$

*Let $U$ contain the $K$ eigenvectors of $S_G$ with largest absolute eigenvalues, let $\Pi = UU^\top$, and let $H = G^{-1/2}\Pi G^{1/2}$. For all sufficiently large $n$, on the same event the retained eigenvalues have a fixed separation from zero, the remaining eigenvalues have magnitude at most $C_r/\sqrt{d}$, and*

$$\left\|(I-\Pi)G^{1/2}Z\right\|_F \le C_r\sqrt{n}. \tag{18}$$

*For every block-constant $f$ with $\|f\|_\infty \le C_f$,*

$$\|(I-H)f\|_G \le C_r C_f\sqrt{n}, \qquad \|(I-H)f\|_2^2 \le C_r C_f^2 n/d, \quad \|v\|_G^2 = v^\top G v. \tag{19}$$

*After alignment, the orthonormal population basis*

$$V = G^{-1/2}\Theta Z(Z^\top\Theta G^{-1}\Theta Z)^{-1/2}O$$

*satisfies $\|U-V\|_F \le C_r/\sqrt{d}$ and $\sqrt{G_i}\,\|V_{i\cdot}\|_2 \le C_r\sqrt{d/n}$. The same assertions hold simultaneously for recomputed graphs after any fixed number of deletions, after increasing the probability parameter. No lower bound proportional to $\log n$ is imposed.*

*Proof.* The Bernoulli transform of the total number of edges gives $cnd \le \sum_i D_i \le Cnd$ except with probability $e^{-cnd}$. For a sufficiently small fixed $a_0$, on this event

$$cd \le \tau = a_0\bar{D} \le Cd, \qquad \tau < \min_i (D_0)_{ii}.$$

Let $d_* = n\max_{i\neq j}\mathbb{E}A_{ij} \asymp d$, and put $W_{h,ii} = \sqrt{\min(1, 2d_*/D_i)}$, with value one at zero degree. The set $\mathcal{S} = \{i : D_i > 2d_*\}$ has size at most $10n/d_*$ except with probability $e^{-cn}$. Indeed, if $m = \lfloor 10n/d_*\rfloor + 1 \le n$ vertices exceed the threshold, some set $T$ of size $m$ satisfies $\sum_{i\in T} D_i > 2md_*$. This is an independent-edge sum with coefficients in $\{1,2\}$ and mean at most $md_*$. At $t = (\log 2)/2$, its exponential transform bounds this probability by $\exp\{-(\log 2 - 1/2)md_*\}$, since $(e^{tc}-1)/c \le (e^{2t}-1)/2 = 1/2$ for $c = 1, 2$. The union over $T$ costs at most $\exp\{m\log(en/m)\} = \exp\{o(md_*)\}$. If $m > n$, the assertion is automatic.

Every row of $W_hAW_h$ has squared Euclidean norm at most $W_{h,ii}^2 D_i \le 2d_*$. The regularized-adjacency theorem of Le, Levina and Vershynin [4, Theorem 2.1 and Remark 2.2], in its row-norm form, therefore gives

$$\|W_hAW_h - \mathbb{E}A\|_{\mathrm{op}} \le C_r\sqrt{d}$$

with failure probability at most $C_r n^{-r}$. It permits these observed-graph-dependent weights. The difference $W_h\Omega W_h - \Omega$ is supported on rows or columns in $\mathcal{S}$, and every entry has magnitude at most $Cp$; hence

$$\|W_h\Omega W_h - \Omega\|_{\mathrm{op}} \le Cp\sqrt{n|\mathcal{S}|} \le C\sqrt{d}.$$

The completed-mean diagonal costs $O(p)$. Thus $\|W_h(A-\Omega)W_h\|_{\mathrm{op}} \le C_r\sqrt{d}$. The exact factorization

$$G^{-1/2} = (2d_*)^{-1/2} F W_h, \qquad F_{ii} = \sqrt{\frac{\max(D_i, 2d_*)}{G_i}}$$

has uniformly bounded $F$, because $G_i \ge cd$ and $G_i \ge D_i$. Conjugating proves the noise bound in (16). Also $G_i \le D_i + \tau$ gives $\sum_i G_i \le Cnd$.

Applying the noise bound to $G^{1/2}\mathbf{1}$ gives

$$\sum_i \frac{(D_i - (D_0)_{ii})^2}{G_i} = \left\|G^{-1/2}(A-\Omega)\mathbf{1}\right\|_2^2 \le C_r n.$$

Since $\tau < (D_0)_{ii}$, the maximum defining $G_i$ only reduces the coordinatewise distance to $(D_0)_{ii}$. This proves the second energy bound in (17).

Let $J = Z^\top \Theta G^{-1} \Theta Z$. In each block, Cauchy–Schwarz, the bounded degree parameters, and comparable block masses give

$$\frac{cn}{d} \le \frac{(\sum_{i:z_i=a}\theta_i)^2}{\sum_{i:z_i=a} G_i} \le J_{aa} = \sum_{i:z_i=a} \frac{\theta_i^2}{G_i} \le \frac{Cn}{d}.$$

The nonzero spectrum of $G^{-1/2}\Omega G^{-1/2}$ is that of $pJ^{1/2}CJ^{1/2}$, which has a fixed separation from zero. Subspace perturbation and (16) give the claimed eigenvalue and Frobenius bounds. The formula for $V$ and the lower bound on $J$ give its weighted row bound.

The identity $(D_0)_{ii} = d\theta_i S_{z_i}$ shows that $G^{-1/2}D_0 Z$ belongs to $\mathrm{col}(G^{-1/2}\Theta Z)$. Moreover

$$\left\|G^{1/2}Z - G^{-1/2}D_0 Z\right\|_F^2 = \sum_i \frac{(G_i - (D_0)_{ii})^2}{G_i} \le C_r n, \qquad \left\|G^{1/2}Z\right\|_F^2 \le C_r nd.$$

Combining this approximation with the $O(d^{-1/2})$ projector error proves (18). The matrix $H$ is the $G$-orthogonal projector corresponding to $\Pi$. Writing $f = Zb$ then gives the first bound in (19); the second follows from $G_i \ge cd$.

Deleting a fixed number of vertices changes each block mass and each sum of degree parameters by $O(1)$. The preceding argument uses their comparability, not exact mean-one normalization on a minor, and applies with its recomputed degrees and threshold. Increasing the fixed probability parameter before a finite-depth union bound proves the simultaneous assertion. □

### 3.5 Spectral-row tails and local deleted-graph comparison

Let $\mathcal{E}_r$ denote the aggregate event just described, including recomputed versions after at most three deletions. Each deleted graph also has its own training-measurable event. The same-threshold row-tail proof of Lemma C.1 in the adjacency SBM article applies to any fixed multiplier in $\tau/\bar{D}$, provided the aggregate premises hold. Indeed, with $W = \tau^{1/2}G^{-1/2}$ and $\widetilde{A} = WAW$, the preceding proposition gives $\|W(A-\Omega)W\|_{\mathrm{op}} \le C\sqrt{d}$ and an informative gap of order $d$. Also

$$W_i \le 1, \quad W_i^2 D_i \le \tau, \quad \left\|\widetilde{A}\right\|_{\mathrm{op}} \le \tau, \quad \|\Omega_{i\cdot}\|_2 \le Cd/\sqrt{n}.$$

These bounds and the exact threshold change $\tau^{(-i)} - \tau = a_0(\bar{D} - 2D_i)/(n-1)$ are all the weight and population inputs to that proof. It uses neither $\tau \ge \max_i (D_0)_{ii}$ nor equality of the degree parameters. Thus the small multiplier $a_0$ used here gives

**Lemma 3.3** (Truncated energy for the degree-floor eigenvectors)**.** *There are fixed constants $R_0, C, c > 0$ such that, for every fixed $R \ge R_0$ and all sufficiently large $n$,*

$$\mathbb{E}\left[\mathbf{1}\{\mathcal{E}_r\}\sum_{j=1}^n \|U_{j\cdot}\|_2^2\, \mathbf{1}\{\sqrt{n}\,\|U_{j\cdot}\|_2 > R\}\right] \le CR^2 e^{-cd\log R}. \tag{20}$$

*The constants $R_0, C, c$ depend on the fixed structural bounds, $a_0$, and $r$, but not on $R$ or $n$. The sufficiently-large-$n$ condition may depend on the chosen fixed $R$. The same result holds for each recomputed deleted graph with its training-measurable geometry event.*

The proof keeps the empirical-threshold perturbation of order $1/n$, uses unrotated training coefficients in its conditional Bennett bound, and absorbs the recurrence at the same truncation threshold. It therefore also applies on the stronger aggregate event below. For an all-zero training graph its proof-only convention is a deterministic orthonormal fallback and identity weights; no conditioning on a full-graph event is introduced.

**Corollary 3.4** (Exceptional coefficients are hit with exponentially small mass)**.** *On a training graph with the preceding floor, put $H = G^{-1/2}UU^\top G^{1/2}$. Let $f$ be any training-measurable vector satisfying $\|f\|_\infty \le F_0$, and let $e = (H - I)f$. For every prescribed fixed $H_0 > 0$, there is a fixed $M < \infty$ such that, for any independent tested Bernoulli row with probabilities $p_j \le Cd/n$,*

$$\mathbb{E}\left[\mathbf{1}\{\mathcal{E}_r\} \sum_{j:|e_j|>M} p_j\right] \le C_{H_0} e^{-H_0 d}. \tag{21}$$

*Here $\mathcal{E}_r$ is the training-measurable geometry event. No exchangeability among degree parameters is required.*

*Proof.* On that event, $G_j \ge a_- d$ and $\sum_j G_j \le C_G nd$. Consequently

$$|e_j| \le F_0 + \frac{\|U_{j\cdot}\|_2}{\sqrt{G_j}} \left\|U^\top G^{1/2} f\right\|_2 \le F_0 + F_0\sqrt{C_G/a_-}\sqrt{n}\,\|U_{j\cdot}\|_2 .$$

For $M_R = F_0 + F_0\sqrt{C_G/a_-}R$, the event $|e_j| > M_R$ implies $\sqrt{n}\,\|U_{j\cdot}\|_2 > R$. Write $H_R(U) = \sum_j \|U_{j\cdot}\|_2^2 \mathbf{1}\{\sqrt{n}\,\|U_{j\cdot}\|_2 > R\}$. Then lemma 3.3 gives

$$\mathbb{E}\left[\mathbf{1}\{\mathcal{E}_r\}\sum_j \mathbf{1}\{|e_j| > M_R\}\right] \le \frac{n}{R^2}\mathbb{E}[H_R(U)\mathbf{1}\{\mathcal{E}_r\}] \le Cne^{-cd\log R}. \tag{22}$$

Multiplication by $Cd/n$ bounds the expected hit mass by $Cde^{-cd\log R}$. Choose a fixed $R$ with $c\log R > H_0 + 1$ and absorb the factor $d$. Independence is required only to interpret the conditional probability that the tested row hits the training exceptional set; no independence between the training geometry event and its eigenvectors is asserted. □

**Corollary 3.5** (Average exceptional target-row mass)**.** *For every fixed degree cutoff $L$ and every fixed $H_0 > 0$, there is a fixed $L_0$ such that*

$$\frac{1}{n}\sum_i \mathbb{P}\left\{\sqrt{G_i}\,\|U_{i\cdot}\|_2 > L_0\sqrt{d/n},\quad D_i \le Ld,\quad \mathcal{E}_r\right\} \le C_{H_0,L}e^{-H_0 d}. \tag{23}$$

*Proof.* On the displayed event, $G_i \le C_L d$. Choose $L_0 = R\sqrt{C_L}$. The target-row exception then implies $\sqrt{n}\,\|U_{i\cdot}\|_2 > R$. Apply (20) and divide its corresponding expected count by $n$. Choosing fixed $R$ sufficiently large gives (23). □

### 3.6 The local scalar comparison

**Lemma 3.6** (A local floored scalar comparison)**.** *Write $A = \left(\begin{smallmatrix} 0 & x^\top \\ x & B \end{smallmatrix}\right)$, $k = \sum_j x_j \le Ld$, and let $\Delta$ be the diagonal degree floor recomputed from $B$. Suppose*

$$cd \le \Delta_j, \qquad \sum_j \Delta_j \le Cnd, \qquad S_B = \Delta^{-1/2}B\Delta^{-1/2} = U_B\Lambda_B U_B^\top + N,$$

*where* $NU_B = 0$, $\|N\|_{\mathrm{op}} \le C/\sqrt{d}$, *and the* $K$ *retained eigenvalues have a fixed gap from zero. Let the full floor be* $G = \mathrm{diag}(g, \Delta + E)$, *with*

$$cd \le g \le Cd, \qquad E = E_{\mathrm{loc}} + E_{\mathrm{glob}}, \qquad 0 \le E_{\mathrm{loc}} \le \mathrm{diag}(x), \quad \|E_{\mathrm{glob}}\|_{\mathrm{op}} \le Cd/n.$$

*Here* $d \to \infty$ *and* $d \le Cn$. *Suppose only on hit coordinates that*

$$\left\|\Delta_j^{-1/2}(U_B)_{j\cdot}\right\|_2 \le R/\sqrt{nd} \quad \textit{whenever } x_j = 1. \tag{24}$$

*Let* $U, \Lambda$ *be the full informative eigensystem of* $G^{-1/2}AG^{-1/2}$, *and put*

$$H_G = G^{-1/2}UU^\top G^{1/2}, \qquad H_B = \Delta^{-1/2}U_B U_B^\top \Delta^{1/2}.$$

*For every* $\|w\|_\infty \le W$, *uniformly over the stated inputs,*

$$(AH_G w)_i = x^\top H_B w_{-i} + O_{L,R,W}(1). \tag{25}$$

*No bound is assumed on unhit rows of* $U_B$ *or on* $\max_j \Delta_j$.

*Proof reference.* Lemma D.1 of the SBM article is deterministic under precisely these premises. Its generalized eigenvector equations contain the floored top degree $g$, not the raw incident degree $k$. Consequently its proof applies unchanged here.

**Corollary 3.7** (Exact-degree and hollow comparison)**.** *Under the same local premises, the reconstruction* (7) *satisfies, for* $\bar{w}_{-i} = (n-1)^{-1}\sum_{j\neq i} w_j$,

$$(R^h w)_i = x^\top\{H_B(w_{-i} - \bar{w}_{-i}\mathbf{1}) + \bar{w}_{-i}\mathbf{1}\} + O_{L,R,W}(1). \tag{26}$$

The same deterministic proof, including its local diagonal bound, is given in Corollary D.2 of the SBM article. Uniformity in $w$ is algebraic; fitted coefficients are first replaced by true block contrasts in proposition 4.8 before conditioning on the deleted graph.

### 3.7 Aggregate probabilities over all degree scales

**Lemma 3.8** (Exponentially accurate aggregate geometry above a logarithmic scale)**.** *Fix any* $A_0 > 0$ *and* $H > 0$. *Under the fixed positive compact structural conditions, if* $d \ge A_0 \log n$ *and* $np^2 \le C_1$, *there is an aggregate event* $\mathcal{E}_H$ *with*

$$\mathbb{P}(\mathcal{E}_H^c) \le C_H e^{-Hd}$$

*on which the empirical-floor geometry, weighted degree-energy bounds, two-sided total-degree bounds, and true block-total bounds used in eqs.* (32) *to* (34) *hold for the full graph and every graph obtained by deleting at most three vertices. The constants may depend on* $A_0, H, C_1$ *and the structural bounds. No condition is imposed on the smallest raw degree or the largest spectral row.*

*Proof.* Bounded degree parameters give independent upper-triangular entries with probabilities at most $Cd/n$. The centered bounded-entry norm inequality of Bandeira–van Handel [2, Corollary 3.12 and Remark 3.13] gives

$$\mathbb{P}\{\|A - \Omega\|_{\mathrm{op}} > L_H\sqrt{d}\} \le e^{-(H+2)d}$$

after choosing a sufficiently large fixed $L_H$. Indeed, the row variance sum is $O(d)$, the entry bound is one, and the prefactor $n$ in the norm-tail inequality is absorbed using $\log n \le d/A_0$. The completed-mean diagonal has norm $O(p)$ and is absorbed in the same threshold. No minimum raw degree is imposed.

The Bernoulli transform of the total number of edges gives, with failure probability at most $Ce^{-cnd}$,

$$cnd \le \sum_i D_i \le Cnd.$$

For each of the fixed $K^2$ true block totals, Bernstein's inequality for the independent upper-triangular edges, whose coefficients have absolute value at most two, gives

$$\mathbb{P}\left\{\left|[Z^\top(A-\Omega)Z]_{ab}\right| > Ln\right\} \le 2\exp\left\{-\frac{cL^2n}{d+L}\right\}.$$

The deterministic completed diagonal costs $O(d)$ and is included by increasing $L$. Since $d \to \infty$ and $d^2/n \le C_1$, the exponent is at least $c'L^2d/C_1$ for all sufficiently large $n$. Increasing the fixed $L = L_H$ therefore makes the failure of all block totals at most $C_He^{-(H+2)d}$. The total-degree failure is also bounded by $Ce^{-(H+2)d}$ for large $n$.

These three full-graph events imply the required events on every fixed-depth minor without a degree-wise probability union. To see this, let $J = [n] \setminus T$ with $|T| \le 3$, set $A_J = A_{JJ}$ and $\Omega_J = \Omega_{JJ}$, and recompute all degrees and the empirical floor on $J$. The centered matrix $A_J - \Omega_J$ is a principal submatrix of $A - \Omega$, so its norm is at most $L_H\sqrt{d}$. Removing $T$ changes the total raw degree by at most $2|T|n$; as $d \to \infty$, the two-sided bound by fixed multiples of $nd$ therefore persists uniformly over $T$. Removing $T$ changes a true block total of $A - \Omega$ by $O(|T|n)$ deterministically, since the entries have bounded absolute value. Its $O_H(n)$ bound persists as well. The surviving group masses and sums of degree parameters differ from their original values by $O(1)$ and remain comparable to $n$.

The recomputed average degree is consequently comparable to $d$. For the fixed sufficiently small floor multiplier $a_0$, its threshold $\tau_J = a_0\overline{D}_J$ lies between positive fixed multiples of $d$ and below every coordinate of $D_{0,J} = \Omega_J\mathbf{1}$. Thus $G_{J,j} = \max(D_{J,j}, \tau_J) \ge cd$ and $\sum_j G_{J,j} \le Cnd$. The raw norm bound immediately gives

$$\left\|G_J^{-1/2}(A_J - \Omega_J)G_J^{-1/2}\right\|_{\mathrm{op}} \le C_Hd^{-1/2},$$
$$\sum_{j\in J}\frac{(D_{J,j} - D_{0,J,j})^2}{G_{J,j}} = \left\|G_J^{-1/2}(A_J - \Omega_J)\mathbf{1}\right\|_2^2 \le C_Hn.$$

Since the population floor is inactive, $|G_{J,j} - D_{0,J,j}| \le |D_{J,j} - D_{0,J,j}|$; this proves the other weighted energy bound. The fixed signal gap and the residual $\left\|(I - U_JU_J^\top)G_J^{1/2}Z_J\right\|_F \le C_H\sqrt{n}$ now follow by the deterministic Gram-matrix and projection arguments in proposition 3.2. Those arguments use only the floor, total degree, normalized noise, and degree energy, all just proved. Intersecting the stated aggregate consequences defines $\mathcal{E}_H$. The three full-graph events imply it simultaneously for every indicated minor, proving its claimed failure bound.

For a deleted graph used as independent training data, form the analogous event using that graph and its further minors. The same argument gives failure probability $C_He^{-Hd}$, and this event is measurable from the training graph alone. The full-graph common event is used only for deterministic implications, never as a condition that would preserve independence of an incident row. □

### 3.8 The degree-corrected tilted cancellation

Work on a deleted graph so that all coordinates below index actual neighbors of a separate tested vertex. For the bounded block-constant linear candidate contrast used in the DC profile reduction, the exact tilted Bernoulli probabilities have the decomposition

$$r = D_0h + \epsilon, \qquad h \text{ block constant}, \quad \|h\|_\infty \le C/n, \quad \|\epsilon\|_\infty \le Cp^2. \tag{27}$$

For example the leading term of the exact tilt is $\theta_i\theta_j pC_{a,z_j}e^{-sw_{z_j}}$, while $(D_0)_{jj} = \theta_j p\sum_k \theta_k C_{z_j,z_k}$. Dividing the former by the latter gives the required block-constant $h_j$. The denominator in the exact Bernoulli tilt contributes $O(p^2)$ uniformly. This uses the original Bernoulli law throughout. The target coordinate is absent; artificially appending $r_i = 0$ would instead create an $O(p)$ error at that one coordinate.

For bounded block-constant $f$, put $e = (H - I)f$. The $G$-orthogonality of $H$ gives the exact cancellation

$$r^\top e = -\langle (I-H)h, (I-H)f\rangle_G + ((D_0 - G)h)^\top e + \epsilon^\top e. \tag{28}$$

The first term is $O(1)$ by (18), since the block coefficients of $h$ are $O(1/n)$. For the second, Cauchy–Schwarz and (17) give

$$|((D_0 - G)h)^\top e| \le \|h\|_\infty \left(\sum_j \frac{((D_0)_{jj} - G_j)^2}{G_j}\right)^{1/2} \|e\|_G \le C.$$

The last term is at most $Cp^2\sqrt{n}\,\sqrt{n/d} \le Cnp^2$. Also $r_j \le Cp$, so

$$|r^\top e| \le C(1 + np^2), \qquad \sum_j r_j e_j^2 \le Cp\,\|e\|_2^2 \le C. \tag{29}$$

Thus the first-order bias cancellation and bounded quadratic error do not require logarithmic degree. The replacement for a uniform coefficient bound is supplied by lemma 3.3 and corollary 3.4.

### 3.9 Two sharp transfers that do not assume every coefficient is bounded

The next two lemmas give concrete alternatives to a global row-incoherence event. They concern a row-independent, training-measurable score, just as in lemma 3.1.

**Lemma 3.9** (Exponential-moment coefficient control)**.** *Assume the oracle hypotheses of lemma 3.1. Let $\widehat{T} = T + b + \sum_j e_j X_j$, with $(b, e)$ measurable from independent training data. On a measurable training event suppose*

$$\left|b + \sum_j r_j e_j\right| \le B, \qquad \sum_j r_j e_j^2 e^{|e_j|} \le V,$$

*where $B, V$ are fixed. Then, for all real $x$ and all sufficiently large $d$,*

$$\mathbb{P}(\widehat{T} \le x \mid \textit{training}) \le Ce^{s_* x + B + V/2} d^{-1/2} e^{-I}.$$

*The constant is independent of the largest coefficient.*

*Proof.* Taylor's inequality gives, under the original oracle tilt $Q$,

$$\log \mathbb{E}e^{-s_*\widehat{T}} \le -I - s_*\left(b + \sum_j r_j e_j\right) + \frac{1}{2}\sum_j r_j e_j^2 e^{|e_j|} \le -I + B + V/2.$$

Tilt again by $e^{-s_*\widehat{T}}$. The new probabilities satisfy

$$\widehat{r}_j = \frac{r_j e^{-s_* e_j}}{1 - r_j + r_j e^{-s_* e_j}} \le r_j e^{s_*|e_j|}.$$

For $|e_j| \le 1$, this bounds the new centered third absolute moments by $C_W \sum_j r_j = O(d)$. For $|e_j| > 1$, use $s_* \le 1 - \tau$ and

$$e^{s_*|u|}(W + |u|)^3 \le C_{W,\tau} u^2 e^{|u|} \qquad (|u| > 1).$$

The remaining third moments are therefore $O(V)$. The upper variance bound follows from the same estimates. There is no restriction that every $e_j$ be bounded.

For the lower variance, restrict to $|e_j| \le 1$. The original variance discarded by this restriction is at most $W^2V$, and the weighted $\ell_2$ norm of the remaining perturbation is at most $\sqrt{V}$. Thus the square root of the original-weight variance of the remaining $(w_j + e_j)$ is at least $\sqrt{vd - W^2V} - \sqrt{V}$. On this restricted set, the new and old Bernoulli variances differ by only fixed factors. The new total variance is therefore at least $cd$ for large $d$. Berry–Esseen gives a $C/\sqrt{d}$ bound for every unit interval under the new tilt. Summing those intervals, using the first displayed exponential-moment bound, proves the assertion as in lemma 3.1. □

**Lemma 3.10** (Paying exceptional coefficients only when the row hits them)**.** *Assume the oracle hypotheses of lemma 3.1, and write $p_j$ for the un-tilted row probabilities. Suppose $p_j \le C_p r_j$, as holds here for bounded oracle likelihood coefficients. On an independent training event $\mathcal{E}$, suppose*

$$\left| b + \sum_j r_j e_j \right| \le B, \qquad \sum_j r_j e_j^2 \le V.$$

*Here $B, V$ are fixed constants. For any fixed $M > 0$ define*

$$h_M = \sum_{j:|e_j|>M} p_j, \qquad \eta_M = \sum_j r_j e_j^2 \mathbf{1}\{|e_j| > M\}.$$

*Then, for every fixed bounded $x$,*

$$\mathbb{P}(\widehat{T} \le x) \le C_M d^{-1/2} e^{-I} + \mathbb{E}[\mathbf{1}\{\mathcal{E}\} h_M] + \mathbb{P}(\mathcal{E}^c). \tag{30}$$

*Since $h_M \le C_p \eta_M / M^2$, an expected truncated quadratic error of order $d^{-1/2}e^{-I}$ suffices. A bound $Ce^{-Hd}$ is a convenient sufficient condition whenever $I \le C_I d$ and $H > C_I$. The expected hit-mass condition on $h_M$ is weaker and is itself enough.*

*Proof.* For analysis only, replace $e_j$ by $e'_j = e_j \mathbf{1}\{|e_j| \le M\}$. The truncated coefficients satisfy

$$\left| b + \sum_j r_j e'_j \right| \le B + \eta_M/M \le B + V/M, \qquad \sum_j r_j (e'_j)^2 \le V.$$

Hence lemma 3.1 bounds their score error by $C_M d^{-1/2} e^{-I}$ on $\mathcal{E}$. The original and truncated random scores are identical whenever $X_j = 0$ for every $j$ with $|e_j| > M$. Conditional on training, the probability that this fails is at most

$$\sum_{j:|e_j|>M} p_j = h_M \le \frac{C_p}{M^2} \eta_M.$$

Integrate this bound on $\mathcal{E}$ and pay at most $\mathbb{P}(\mathcal{E}^c)$ outside it. No coefficient truncation is an additional input or computation required of the implemented decoder. □

For the degree-floor projection there is a concrete reduction to a weighted row count. Define

$$\zeta_j = \sqrt{\frac{nd}{G_j}} \, \|U_{j\cdot}\|_2 .$$

If $e = (H - I)f$, $\|f\|_\infty \le C_f$ and $\sum_j G_j \le C_G nd$, then deterministically

$$|e_j| \le \frac{\|U_{j\cdot}\|_2}{\sqrt{G_j}} \left\| U^\top G^{1/2} f \right\|_2 + C_f \le C_f \sqrt{C_G} \, \zeta_j + C_f.$$

Choosing $M = C_f\sqrt{C_G}L + C_f$ thus gives $\{|e_j| > M\} \subseteq \{\zeta_j > L\}$. Since $p_j \le Cd/n$, the sufficient expected-count estimate

$$\mathbb{E}\left[\mathbf{1}\{\mathcal{E}\}\sum_j \mathbf{1}\{\zeta_j > L\}\right] \le Cne^{-Hd}, \qquad H > C_I, \tag{31}$$

would imply $\mathbb{E}[\mathbf{1}\{\mathcal{E}\}h_M] \le Cde^{-Hd} = o(d^{-1/2}e^{-I})$ whenever $I \le C_I d$. Equation (31) is not a consequence of aggregate geometry alone. It is proved by the same-threshold truncated-energy argument in lemma 3.3 and the floor lower bound.

# 4 DC fitting from aggregate geometry and local row bounds

This section supplies the deterministic fitting module for the degree-floor decoder. It uses the exact Bernoulli working criterion and the exact-degree reconstruction $R^h$ in (7). In particular, its rank-one degree correction and simple row hollowing are different from the earlier correction $\mathrm{diag}(s)(H - I)$. One profile refinement suffices. The optional finite repeated hard-refit version is covered explicitly by lemma 4.9.

Put $d = np$, $B = pC$, $\Omega = \Theta Z B Z^\top \Theta$ and $D_{0,i} = (\Omega\mathbf{1})_i$. As before, $\mathbf{d} = A\mathbf{1}$ is the raw degree vector, $D_i = \mathbf{d}_i$, and

$$G_i = \max(D_i, a_0\overline{D}), \qquad S = G^{-1/2}AG^{-1/2}, \qquad R_0 = G^{1/2}U\Lambda U^\top G^{1/2}.$$

The constants $a_0 > 0$ and all subsequent clipping constants are fixed using only the structural parameter bounds. They do not depend on $n$, the unknown labels, or individual unobserved parameters. Work on the two-sided total-degree event on which $\overline{D} \asymp d$ and the population floor is inactive. The aggregate geometric and block-total premises used below are

$$\left\|S - G^{-1/2}\Omega G^{-1/2}\right\|_{\mathrm{op}} \le C/\sqrt{d}, \qquad \sum_i G_i \le Cnd, \tag{32}$$

$$\sum_i \frac{(G_i - D_{0,i})^2}{G_i} + \sum_i \frac{(D_i - D_{0,i})^2}{G_i} \le Cn, \qquad \left\|(I - UU^\top)G^{1/2}Z\right\|_F \le C\sqrt{n}, \tag{33}$$

$$\max_{a,b}\left|[Z^\top(A - \Omega)Z]_{ab}\right| \le Cn. \tag{34}$$

Here $G_i \ge cd$, $D_{0,i} \asymp d$, and the retained signal has a fixed gap from zero. The degree-floor proposition already supplies (32)–(33). The fixed-$K$ Bernoulli block-total bound (34) has failure probability $C\exp\{-cL^2n/(d+L)\}$ when its fixed threshold constant is $L$. Under $np^2 = d^2/n \le C_1$, increasing $L$ pays this failure with any prescribed fixed exponential rate in $d$. No degree-wise union bound is needed for this block event.

Write $\Pi = UU^\top$. An aligned orthonormal basis of the random population space has the exact form

$$V = G^{-1/2}\Theta Z(Z^\top\Theta G^{-1}\Theta Z)^{-1/2}O,$$

where $O$ is orthogonal. The diagonal Gram matrix has entries comparable to $n/d$. The signal gap and (32) imply

$$\|U - V\|_F \le C/\sqrt{d}, \qquad \sqrt{G_i}\,\|V_{i\cdot}\|_2 \le C\sqrt{d/n}. \tag{35}$$

An orthogonal rotation used in this display does not change any row norm or projection. Also $\|S\|_{\mathrm{op}} \le 1$: the inequality $|x^\top Ax| \le \sum_i D_i x_i^2 \le \sum_i G_i x_i^2$ proves this directly. Hence $\|\Lambda\|_{\mathrm{op}} \le 1$.

**Lemma 4.1** (Uniform weighted mass of wrong-label sets)**.** *Fix $M < \infty$. For every possibly data-dependent set $E$ with $|E| \le Mn/d$, on the aggregate event,*

$$\sum_{i\in E} G_i \le C_M n, \qquad \sum_{i\in E}\sqrt{G_i}\,\|U_{i\cdot}\|_2 \le C_M\sqrt{n/d}. \tag{36}$$

*Moreover $\sum_i \sqrt{G_i}\,\|U_{i\cdot}\|_2 \le C\sqrt{nd}$.*

*Proof.* The elementary inequality $G_i \le 2D_{0,i} + (G_i - D_{0,i})^2/G_i$ gives $\sum_{i\in E} G_i \le C(d|E| + n) \le C_M n$. Put $a_i = \sqrt{G_i}\,\|U_{i\cdot}\|_2$. By (35) and Cauchy–Schwarz,

$$\sum_{i\in E} a_i \le C|E|\sqrt{d/n} + \left(\sum_{i\in E} G_i\right)^{1/2} \|U - V\|_F \le C_M\sqrt{n/d}.$$

The last assertion follows from $\sum_i a_i \le (\sum_i G_i)^{1/2}\,\|U\|_F$ and fixed $K$. All estimates hold simultaneously for all such sets; they impose no independence requirement on the selected labels. □

**Lemma 4.2** (Radius-count initialization under aggregate geometry)**.** *The deterministic radius-count initializer specified earlier has at most $Cn/d$ errors on the aggregate event, after one common permutation. This assertion requires neither a uniform degree upper bound nor uniform spectral-row incoherence.*

*Proof.* The normalized rows of $V$ are the $K$ orthogonal population centers. Choose fixed $b_0 > 2\max_i D_{0,i}/d$, uniformly over the structural class. For every $G_i > b_0 d$, $(G_i - D_{0,i})^2/G_i \ge G_i/4 \ge b_0 d/4$. Thus at most $Cn/d$ such rows occur by (33). They contribute at most $4Cn/d$ to the squared directional error. For the remaining rows, $\|V_{i\cdot}\|_2 \ge c/\sqrt{n}$. The row-normalization inequality and (35) bound their total squared directional error by $Cn\,\|U - V\|_F^2 \le Cn/d$. The existing radius-count lemma now applies with total squared error $Cn/d$ and fixed positive community proportions. Its proof is deterministic and unchanged. □

**Lemma 4.3** (Degree-corrected block totals and absolute mass)**.** *For $R^h$ as in (7),*

$$\sum_{i,j} |R^h_{ij}| \le Cnd. \tag{37}$$

*Uniformly over every fitted partition $\widehat{Z}$ having at most $Mn/d$ errors under the common label matching,*

$$[\widehat{Z}^\top R^h \widehat{Z}]_{ab} = n_a n_b B_{ab} + O_M(n). \tag{38}$$

*The omitted self-loops are included in the remainder.*

*Proof.* Let $S_{\text{tail}} = (I-\Pi)S(I-\Pi)$ and $F_1 = (I-\Pi)G^{1/2}\mathbf{1}$. Spectral invariance and the signal gap give $\|S_{\text{tail}}\|_{\text{op}} \le C/\sqrt{d}$ and $\|F_1\|_2 \le C\sqrt{n}$. Writing $v = \mathbf{d} - R_0\mathbf{1}$, one has exactly $v = G^{1/2}S_{\text{tail}}F_1$. Hence

$$\sum_i |v_i| \le \left(\sum_i G_i\right)^{1/2} \|S_{\text{tail}}F_1\|_2 \le Cn.$$

Since $\Pi_{ii} \le 1$ and $\sum_i \Pi_{ii} = K$,

$$\sum_i |(R_0)_{ii}| \le \sum_i G_i\Pi_{ii} \le 2(\max_i D_{0,i})K + \sum_i \frac{(G_i - D_{0,i})^2}{G_i} \le C(d+n) \le Cn.$$

Consequently $\sum_i |(R_1)_{ii}| \le Cn$ and the simple hollowing has total absolute cost at most $2\sum_i |(R_1)_{ii}| \le Cn$. With $a_i$ as in the preceding lemma, $|(R_0)_{ij}| \le a_i a_j$, so $\sum_{i,j} |(R_0)_{ij}| \le (\sum_i a_i)^2 \le Cnd$. The rank-one correction has total absolute mass $\sum_i |v_i| \le Cn$. This proves (37).

Put $\widehat{F} = (I-\Pi)G^{1/2}\widehat{Z}$. By (36),

$$\left\|\widehat{F}\right\|_F \le \left\|(I-\Pi)G^{1/2}Z\right\|_F + \left\|G^{1/2}(\widehat{Z} - Z)\right\|_F \le C_M\sqrt{n}.$$

The exact quadratic identity

$$\widehat{Z}^\top(R_0 - A)\widehat{Z} = -\widehat{F}^\top S_{\text{tail}}\widehat{F}$$

therefore has norm at most $C_M n/\sqrt{d}$. Moving the mislabeled vertices changes a raw block total by at most a fixed multiple of $\sum_{i\in E} D_i \le \sum_{i\in E} G_i \le C_M n$. Equation (34) and the exact within-block mean normalization of $\theta$ prove (38) with $R_0$ in place of $R^h$. The rank-one correction changes a fitted block total by $[\widehat{Z}^\top v]_a \widehat{n}_b/n$, where $\widehat{Z}^\top v = \widehat{F}^\top S_{\text{tail}}F_1 = O_M(n/\sqrt{d})$. The already bounded hollowing cost completes the proof. □

### 4.1 A bounded, normalized degree-parameter estimator

Choose fixed positive $\alpha$ sufficiently small and $\beta$ sufficiently large, using the structural bounds, so that on the two-sided total-degree event

$$\alpha\overline{D} \le \tfrac{1}{2}\min_i D_{0,i}, \qquad \beta\overline{D} \ge 2\max_i D_{0,i}.$$

Define

$$T_i = \min\{\max(D_i, \alpha\overline{D}), \beta\overline{D}\}, \qquad \widehat{\theta}_i = \frac{T_i}{\widehat{n}_{\widehat{z}_i}^{-1}\sum_{j:\widehat{z}_j=\widehat{z}_i} T_j}. \tag{39}$$

This changes the nuisance estimator only. The retained degrees and the degree identity $R^h\mathbf{1} = \mathbf{d}$ still use the original raw degrees. The fitted parameters lie in $[\alpha/\beta, \beta/\alpha]$, and their mean in every fitted block is exactly one.

**Lemma 4.4** (Clamped nuisance estimates)**.** *Uniformly over the $Mn/d$ count basin,*

$$\frac{\left\|\widehat{\theta} - \theta\right\|_2}{\sqrt{n}} \le C_M d^{-1/2}, \qquad \widehat{n}_a^{-1}\sum_{\widehat{z}_i=a} T_i = dS_a + O_M(1), \tag{40}$$

*where $D_{0,i} = dS_{z_i}\theta_i$.*

*Proof.* The chosen clamp interval contains $D_{0,i}$ with a fixed margin. A direct split into the unclamped, lower-clamped, and upper-clamped cases gives

$$|T_i - D_i| \le C\frac{(D_i - D_{0,i})^2}{G_i}, \qquad |T_i - D_{0,i}|^2 \le Cd\frac{(D_i - D_{0,i})^2}{G_i}.$$

For the first inequality, an active lower clamp implies a degree deviation of order $d$, and an active upper clamp implies a deviation of order $D_i$. For the second, the unclamped range has $G_i \le Cd$; projection onto an interval containing $D_{0,i}$ cannot increase the deviation. Above the upper clamp, $D_i((T_i - D_{0,i})/(D_i - D_{0,i}))^2 \le T_i \le Cd$. Summing yields

$$\sum_i |T_i - D_i| \le Cn, \qquad \sum_i |T_i - D_{0,i}|^2 \le Cnd. \tag{41}$$

Since $\mathbf{1} = Z\mathbf{1}_K$, the raw block-total event implies $\sum_{z_i=a}(D_i - D_{0,i}) = O(n)$. Clamping changes that sum by $O(n)$, and moving $Mn/d$ labels changes a clamped degree total by at most $Cd(Mn/d) = O_M(n)$. The fitted size differs from $n_a$ by $O_M(n/d)$. Together with $\sum_{z_i=a} D_{0,i} = n_a dS_a$, these facts prove the group-mean assertion in (40). For correctly labeled vertices, division of $T_i - D_{0,i}$ by this fitted mean, followed by (41), gives squared error $O_M(n/d)$. The incorrectly labeled vertices number $O_M(n/d)$ and have bounded theta error. This proves the first assertion. □

### 4.2 The exact Bernoulli block likelihood

Fit a positive symmetric $B$ by maximizing the ordered Bernoulli working criterion with $R^h$ and (39), divided by two, under the same fixed probability cap $\widehat{\theta}_i\widehat{\theta}_j B_{\widehat{z}_i\widehat{z}_j} \le q_{\max} < 1$. The next argument uses aggregate absolute mass instead of the stronger and unnecessary assertion that every reconstructed entry is less than one.

**Lemma 4.5** (Unique Bernoulli block fit and its rate)**.** *The fitted block roots are unique and interior for all large $n$. Uniformly over the $Mn/d$ basin,*

$$\max_{a,b}\left|\frac{\widehat{B}_{ab}}{B_{ab}} - 1\right| \le C_M(d^{-1} + p) \le C_M d^{-1}. \tag{42}$$

*Proof.* For the ordered pairs $\mathcal{J}_{ab}$ associated with an unordered block pair, put

$$E_{ab} = \sum_{\mathcal{J}_{ab}} R^h_{ij}, \qquad t_{ij} = \widehat{\theta}_i \widehat{\theta}_j, \qquad T_{ab} = \sum_{\mathcal{J}_{ab}} t_{ij}, \qquad \nu_{ab} = 2 - \mathbf{1}\{a = b\}.$$

The preceding block-total lemma and exact fitted normalization imply

$$E_{ab} = \nu_{ab} n_a n_b B_{ab} + O_M(n) \asymp n^2 p > 0, \qquad T_{ab} = \nu_{ab} \widehat{n}_a \widehat{n}_b + O(n).$$

The exact score equation is $E_{ab} = F_{ab}(b)$, where

$$F_{ab}(b) = b \sum_{\mathcal{J}_{ab}} \frac{(1 - R^h_{ij}) t_{ij}}{1 - b t_{ij}}.$$

For every admissible $b$, bounded positive theta, the probability cap, and (37) give

$$F'_{ab}(b) = \sum_{\mathcal{J}_{ab}} \frac{(1 - R^h_{ij}) t_{ij}}{(1 - b t_{ij})^2} \geq cn^2 - C \sum_{\mathcal{J}_{ab}} |R^h_{ij}| \geq cn^2 - Cnd > 0.$$

Here $p = d/n \to 0$. At zero, $F_{ab}(0) = 0$; its value at the positive cap endpoint is of order $n^2$, while $E_{ab} \asymp n^2 p$. There is a unique interior root. The likelihood derivative is $(E_{ab} - F_{ab}(b))/b$, so the root is its unique global maximum.

For $b \asymp p$,

$$\sum_{\mathcal{J}_{ab}} \frac{(1 - R^h_{ij}) t_{ij}}{1 - b t_{ij}} = T_{ab} + O(n^2 p).$$

The error follows by expanding the denominator and using $\sum |R^h_{ij}| \leq Cnd = Cn^2 p$. Bracketing the root between fixed multiples of $p$, then using this expansion, gives $\widehat{B}_{ab} = (E_{ab}/T_{ab})\{1 + O(p)\}$. The block-total and group-size errors prove the first inequality in (42). The second uses $dp = np^2 \leq C_1$. □

### 4.3 A local row condition suffices for every fitted stage

For a tested vertex $i$, impose only the local conditions

$$D_i \leq bd, \qquad a_i := \sqrt{G_i}\, \|U_{i\cdot}\|_2 \leq L\sqrt{d/n}, \tag{43}$$

where $b, L$ are fixed. These are not imposed on every neighbor simultaneously.

**Lemma 4.6** (Local absolute mass and wrong-label stability)**.** *On the aggregate event and* (43)*,*

$$\sum_j |R^h_{ij}| \leq C_{b,L} d, \qquad \sup_{|E| \leq Mn/d} \sum_{j \in E} |R^h_{ij}| \leq C_{M,b,L}. \tag{44}$$

*Proof.* Since $|(R_0)_{ij}| \leq a_i a_j$,

$$\sum_j |(R_0)_{ij}| \leq a_i \sum_j a_j \leq Cd, \qquad \sum_{j \in E} |(R_0)_{ij}| \leq a_i \sum_{j \in E} a_j \leq C_M$$

by lemma 4.1. Also $|v_i| \leq D_i + \sum_j |(R_0)_{ij}| \leq Cd$. The rank-one correction therefore adds at most $Cd$ to the first bound and $|v_i||E|/n \leq C_M$ to the second. Finally $|(R_1)_{ii}| \leq a_i^2 + |v_i|/n \leq Cp$. The simple row hollowing changes either row bound by at most $2|(R_1)_{ii}| \leq Cp$. This proves the claim. □

**Lemma 4.7** (Bernoulli profile reduction for a signed row)**.** *Let $r$ be a hollow signed row satisfying $\sum_j r_j = D_i \geq 0$ and $\sum_j |r_j| \leq Ld$. Let the positive neighbor intensities $b_j$ lie between fixed multiples of $p$. Maximizing the exact Bernoulli row criterion*

$$\sum_j \{r_j \log(tb_j) + (1 - r_j) \log(1 - tb_j)\}$$

*over $t \geq 0$ under the fixed cap $tb_j \leq q_{\max} < 1$ gives the usual normalized block-profile contrast, with remainder $O(np^2)$. No entrywise bound on $r_j$ is required.*

*Proof.* After combining the coefficient of $\log t$, the derivative is

$$D_i/t - \sum_j \frac{(1 - r_j)b_j}{1 - tb_j}.$$

Throughout the admissible interval,

$$\sum_j \frac{(1 - r_j)b_j}{1 - tb_j} \geq cnp - Cp \sum_j |r_j| \geq cd - CLdp \geq c'd.$$

Thus every maximizing $t$ lies in a fixed bounded interval, because $D_i \leq Ld$. This reasoning does not require individual nonedge weights $1 - r_j$ to be nonnegative. On that fixed interval, the exact Bernoulli expansion is uniform:

$$\ell_r(t) = D_i \log t + \sum_j r_j \log b_j - t \sum_j b_j + O(np^2).$$

Restoring the edge terms in the linear exposure costs $Cp \sum_j |r_j| = O(np^2)$, and the quadratic remainder costs $Cp^2(n + \sum_j |r_j|) = O(np^2)$. The leading optimizer $D_i / \sum_j b_j$ is bounded and admissible for all large $n$, so taking maxima preserves the remainder. The resulting label-dependent maximum is the standard normalized block-profile score. If $D_i = 0$, the derivative bound makes $t = 0$ optimal after combining the $\log t$ terms; use the continuous convention $0 \log 0 = 0$. □

**Proposition 4.8** (Uniform replacement by the true DC contrast)**.** *Fix $M, b, L$. Use the clamped theta estimates (39), the exact Bernoulli block fit, and the exact Bernoulli target profile on $R^h$. On the aggregate event and the local target condition (43), uniformly over every fitted partition in the $Mn/d$ basin,*

$$\widehat{L}_i^{ab} = [R^h w_{i,ab}]_i + O_{M,b,L}(1), \qquad (w_{i,ab})_j = \log \frac{q^{(i)}_{a,z_j}}{q^{(i)}_{b,z_j}} \quad (j \neq i). \tag{45}$$

*The true profiles $q^{(i)}$ are the self-excluded neighbor profiles defined in (114). The value of $w_{i,ab}$ at $i$ is immaterial since $R^h_{ii} = 0$.*

*Proof.* By lemma 4.7, the profile reduction costs $O(np^2) = O(1)$. Exact fitted-block normalization gives the fitted neighbor mass $\widehat{M}_c^{(i)} = \widehat{n}_c - \widehat{\theta}_i \mathbf{1}\{\widehat{z}_i = c\}$. The fitted sizes differ from the true sizes by $O_M(n/d)$, the fitted theta values are bounded, and (42) gives relative block error $O_M(1/d)$. Smoothness of normalization and logarithms on the fixed positive profile class therefore yields fitted log-profile error $O_M(1/d)$. Multiplication by the row absolute mass in (44) costs $O_M(1)$. Replacing the fitted labels by the true labels changes a bounded coefficient vector on at most $Mn/d$ coordinates; the second bound in (44) gives another $O_M(1)$. All inequalities hold simultaneously for every partition in the basin, so data-dependent fitting does not introduce an independence assumption. □

**Lemma 4.9** (A uniform basin for finite hard refits)**.** *There is a fixed $M$ such that, on the aggregate event and for all sufficiently large $n$, each hard nuisance-fit and profile-assignment update maps the $Mn/d$ error basin into itself under the same label matching. Outside a fixed target-exception set determined by the graph, every error at every updated iterate is contained in*

$$[R^h w_{i,ab}]_i \le C_M \quad \text{for some } b \ne z_i.$$

*The target-exception set can be taken to be the failure of the local conditions (43), with fixed sufficiently large thresholds. Thus a finite number of these specified hard refits, including adaptive finite stopping after at least one update, requires no union over iterations in the final risk bound.*

*Proof.* Fix sufficiently large constants $b, R$. The set $\{i : D_i > bd\}$ has at most $Cn/d$ elements by the degree energy in (33), since $b$ exceeds twice the largest population degree multiplier. Since $G_i \ge cd$, (35) gives $\|V_{i\cdot}\|_2 \le C/\sqrt{n}$. For $R > 2C$, the set $\{i : \sqrt{n}\,\|U_{i\cdot}\|_2 > R\}$ also has at most $Cn/d$ elements, by $\|U - V\|_F^2 \le C/d$. Every other target satisfies (43) for a fixed $L$, because $G_i \le C_b d$ there. This target-exception set depends only on the graph and its cardinality bound is independent of $M$.

Write $M_G = G^{-1/2}\Omega G^{-1/2}$ and $S_K = U\Lambda U^\top$. Best rank-$K$ approximation and (32) imply $\|S_K - M_G\|_{\mathrm{op}} \le C/\sqrt{d}$. Hence

$$\left\|G^{-1/2}(R_0 - \Omega)Z\right\|_F^2 \le \frac{C}{d}\left\|G^{1/2}Z\right\|_F^2 \le Cn.$$

Moreover $\left\|G^{-1/2}v\right\|_2^2 = \|S_{\mathrm{tail}}F_1\|_2^2 \le Cn/d$. The rank-one correction therefore preserves the bound

$$\left\|G^{-1/2}(R_1 - \Omega)Z\right\|_F^2 \le Cn. \tag{46}$$

The simple hollowing changes any bounded block-constant contrast at row $i$ by at most $C|(R_1)_{ii}|$. Since $\sum_i |(R_1)_{ii}| \le Cn$, at most $Cn/d$ rows can incur a hollowing contrast error larger than any prescribed fixed small multiple of $d$. This auxiliary counting set is needed only for the deterministic basin proof; it is not added as a probability-cost event in the sharp risk argument.

The true profile contrast has a uniform positive population margin $[\Omega w_{i,ab}]_i \ge c_{\mathrm{gap}} d$ for $a = z_i$, $b \ne a$, including its $O(p)$ self-exclusion correction. Extend each $w_{i,ab}$ to a block-constant vector at coordinate $i$; this does not affect $R^h w$ because $R^h_{ii} = 0$. Its $K$ block coefficients are uniformly bounded. For a regular target, proposition 4.8 bounds the fitted discrepancy by $C_M$, which is less than $c_{\mathrm{gap}} d/4$ for all sufficiently large $n$ (the threshold may depend on $M$). Outside the auxiliary hollowing set, a new erroneous label therefore forces

$$\|[(R_1 - \Omega)Z]_{i\cdot}\|_2 \ge cd.$$

For the regular targets, $G_i \le C_b d$. Summing the squared requirement divided by $G_i$ and applying (46) gives at most $C_0 n/d$ such targets. Including the two cardinality exceptions still gives $C_0 n/d$ with $C_0$ independent of $M$.

Choose $M$ at least both this $C_0$ and the initializer constant. The basin is invariant for all sufficiently large $n$. For every regular target, the canonical error-event inclusion follows directly from (45), independently of this counting argument. It is the same event for all partitions in the fixed basin. Induction therefore covers each finite hard refit and any data-dependent finite stopping rule without a new probability union or a stopping-time independence claim. □

# 5 Completion of the current DC risk bound

*Proof of the regularized DC risk bound below a fixed logarithmic scale.* Fix any constant $A_0 < \infty$ and first work on $d \le A_0 \log n$, still with $d \to \infty$. Let $\mathcal{E}$ collect the aggregate geometric

conditions of proposition 3.2, their recomputed versions after deletion of at most three vertices, two-sided total degree, and the true block-total event used in the fitting lemmas. None imposes a simultaneous lower bound on all raw degrees or a simultaneous incoherence bound. For every prescribed fixed $r$, its failure probability is $O(n^{-r})$ after increasing the fixed constants. For the block totals this also follows directly from the independent-edge Bernstein bound at a sufficiently large constant times $n$: its exponent is at least $cn/d$, and $np^2 \le C_1$ gives $n/d \ge d/C_1$; in the present range it is much larger than $\log n$.

The initializer and invariant-basin result in lemma 4.9 apply deterministically on $\mathcal{E}$. For a tested target $i$, impose only its upper-degree bound $D_i \le Ld$, its local normalized row bound, and the condition that its incident row hits no exceptional deleted eigenvector row. By lemma 3.3 and corollaries 3.4 and 3.5, the average probability cost of the two row conditions is $C_H e^{-Hd}$ for any prescribed fixed $H$, after choosing their fixed thresholds sufficiently large. For the LOO hit condition, the exceptional set consists of deleted rows with $\sqrt{n}\,\|(U_B)_{j\cdot}\|_2 > R$; the floor implies (24) on its complement. Its expected hit mass is bounded by the row-count argument in (22). The upper degree tail has the same property by the single-row Bernoulli Chernoff bound. Polynomial factors in $d$ are absorbed by choosing a larger exponential rate at the preceding step. No union over the $n$ tested targets is taken: their error probabilities are averaged.

On these local events, the fitted-score lemma proposition 4.8 replaces every estimated, possibly data-dependent contrast by a deterministic true block contrast at $O(1)$ cost. The local LOO comparison corollary 3.7 replaces its actual full-graph spectral action by an independent training action at another $O(1)$ cost. For the DC profile contrast, the new score is the true raw-row linear profile contrast plus

$$\sum_{j\neq i} A_{ij}e_j + O(1), \qquad e = (H_B - I)(w - \bar{w}\mathbf{1}).$$

The coefficients are measurable from the deleted graph and independent of the incident Bernoulli row. For each target use the analogous training event $\mathcal{E}_i$ formed only from that deleted graph and its further deletions. In applying the independent-row bound, drop the full-graph event after the deterministic error inclusion and condition only on the training graph on $\mathcal{E}_i$. The geometry cost is

$$\varepsilon_{\mathrm{geom}} = \mathbb{P}(\mathcal{E}^c) + \frac{1}{n}\sum_i \mathbb{P}(\mathcal{E}_i^c) \le C_r n^{-r}.$$

This does not condition the Bernoulli row on a full-graph event. The tilted mean and quadratic cost are bounded by (29); here $np^2$ is bounded. The exceptional coefficient hit-mass hypothesis of lemma 3.10 is now a conclusion of corollary 3.4, rather than an unproved premise. That sharp transfer gives, after averaging the local target exceptions and taking a fixed-$K$ union over competitors,

$$\mathbb{E}\ell(\widehat{z}, z) \le \frac{C}{n}\sum_i \sum_{b\neq z_i} d^{-1/2} e^{-J_{i,z_i\to b}} + C_H e^{-Hd} + \varepsilon_{\mathrm{geom}}.$$

Here $J_{i,a\to b}$ is the exact directional profile exponent in (120). By lemma 9.8, the same pairwise directional sharp scale is a lower bound, up to a constant, for the actual profile oracle. Since $K$ is fixed, the displayed sum is bounded by a constant times its average risk. Only the averaged target-row tail is used; no pointwise uniform tail for heterogeneous degree parameters is asserted.

Uniformly on the compact DC class, the reference average risk is at least $cd^{-1/2}e^{-C_J d}$. Choose $H > C_J + 1$ and then choose fixed $r > A_0(C_J + 1) + 2$. For $d \le A_0 \log n$,

$$\frac{e^{-Hd} + n^{-r}}{d^{-1/2}e^{-C_J d}} \longrightarrow 0.$$

This closes the unconditional expected-risk estimate throughout the entire range below any fixed multiple of $\log n$. In particular, no positive lower coefficient of $\log n$ is required. □

*Completion over the remaining degree scales.* Fix $A_0 > 0$ and use the preceding sparse-range argument when $d \le A_0 \log n$. For $d > A_0 \log n$, replace its polynomially accurate aggregate event by $\mathcal{E}_H$ from lemma 3.8. The proof of lemma 3.3 uses only the deterministic aggregate geometry on the event; the innovation concentration is applied for every training graph before restricting to the event. It therefore holds unchanged with $\mathcal{E}_H$ and the corresponding training-measurable events. Its coefficient-hit and target-row corollaries, the scalar LOO comparison, all fitted-score bounds, and the invariant basin likewise apply.

The resulting risk upper bound is the appropriate oracle term plus $C_{H'} e^{-H'd} + C_H e^{-Hd}$, where $H'$ can be prescribed by choosing the fixed local thresholds. On the fixed positive compact DC model class the oracle risk is at least $c d^{-1/2} e^{-C_J d}$. Choosing $H, H' > C_J + 1$ pays both remainders. This proves the stated all-scale profile-oracle bound. The known-target-degree comparison follows from the separate equal-exposure calculation in eq. (134). No logarithmic degree lower bound is imposed. □

# Part I
# Earlier decoders and additional results

## 6 Historical unregularized encoders and their separate results

This part preserves the separate analyses of the earlier unregularized algorithms. Its degree-range assumptions describe those historical encoders only; the current regularized theorem has no logarithmic lower bound. Their logarithmic-range hypotheses are not hypotheses of theorem 2.1. Their positive-degree fallback convention is replaced by the explicit floor in the primary algorithm.

When all degrees are positive, form

$$S = D^{-1/2} A D^{-1/2} = I - L.$$

Retain the degree eigenvector $D^{1/2}\mathbf{1}/\left\|D^{1/2}\mathbf{1}\right\|_2$ and the $K-1$ other eigenvectors with largest absolute eigenvalues in its orthogonal complement. Let $U^\top U = I_K$ and $SU = U\Lambda$, where $\Lambda = \operatorname{diag}(\mu_1, \dots, \mu_K)$. On the proved event this is exactly the informative $K$-dimensional space. The encoder supplies only $(D, U, \Lambda)$ to the decoder. Put

$$X = D^{-1/2}U, \qquad AX = D^{1/2}U\Lambda. \tag{47}$$

Thus row actions on the retained space require no later edge lookup. The normalized Laplacian eigenvalues contain the same information via $\mu = 1 - \lambda_L$. Exact arithmetic and algebraic retained spaces define the theorem; a fixed numerical rank threshold is not its definition.

The ordinary-SBM decoder uses the *signed* reconstruction

$$Q = \operatorname{orth}(X), \quad \Pi_X = QQ^\top, \quad R_E = A\Pi_X. \tag{48}$$

It fits free Bernoulli row profiles with the finite growing/replacement schedule and mandatory final E/M step specified in section 8. Only probabilities are clipped; $R_E$ is not clipped. Neither true labels nor a correct initial partition are inputs.

For DC-SBM we make the explicit degree-metric correction

$$H_D = D^{-1/2}UU^\top D^{1/2}, \qquad R_D = AH_D = D^{1/2}U\Lambda U^\top D^{1/2}. \tag{49}$$

With $h_i = \|U_{i\cdot}\|_2^2$ and $s_i = (R_D)_{ii}/(1-h_i)$, its hollow version is

$$R^\circ = R_D + \operatorname{diag}(s)(H_D - I). \tag{50}$$

It satisfies $\operatorname{diag}(R^\circ) = 0$, $R^\circ\mathbf{1} = D\mathbf{1}$ and $R^\circ X = AX$. It need not be symmetric after hollowing; a symmetric block parameter is fitted with the ordered-row working criterion divided by two.

For the initial partition, normalize every row of $U$ to unit length. Repeatedly select a remaining row having the largest number of remaining rows within distance $1/8$, and remove rows within distance $1/4$ of the selected row. After $K$ selections assign every row to the closest selected row. The separation argument in section 9 proves its $O(1/d)$ error; this is a deterministic procedure with $O(Kn^2)$ distance comparisons, not an assumed global k-means solution. Normalize degree estimates to mean one within each fitted group, fit the symmetric block matrix by the exact Bernoulli working criterion, and perform at least one exact Bernoulli profile-LRT assignment using $R^\circ$. One pass suffices. The same theorem permits a finite number of the stated hard refits, including adaptive stopping, by the invariant-basin argument proved below.

Both algorithms return a fixed partition, or their initial partition, if a prescribed operation is undefined. This includes zero degrees, insufficient retained rank, a hollow denominator zero, an empty initial group, or an inadmissible parameter fit. The required operations are all well-defined on the good event, so this convention costs only its actual failure probability.

**Theorem 6.1** (Earlier ordinary free-profile decoder)**.** *Fix $K \geq 2$, $\pi_- > 0$ and $\kappa > 0$. Suppose*

$$n_a \geq \pi_- n, \qquad p/2 \leq P_{ab} \leq p \leq 1/4, \qquad \sigma_{\min}(P) \geq \kappa p, \qquad d = np \geq 5 \log n.$$

*Use (48) and the finite growing/replacement likelihood decoder, with probability floor $\varepsilon = c_0 \max(1, n^{-1} \sum_i D_i)/n$ for a fixed sufficiently small $c_0 > 0$ (for example any $c_0 < 1/16$ on this class). There are constants $C, n_0$, depending only on the displayed fixed structural constants and $c_0$, such that for $n \geq n_0$,*

$$\mathbb{E}\ell(\widehat{z}_E, z) \leq C R_{\mathrm{SBM},*}(n). \tag{51}$$

*The algorithm uses neither $P$, $p$, nor the true partition. This includes fixed arbitrary $K$, unbalanced group sizes of comparable order, and general symmetric full-rank block probability matrices in the displayed range. It does not require all within-group probabilities, or all between-group probabilities, to be identical.*

**Theorem 6.2** (Earlier unregularized DC profile decoder)**.** *Assume (2)–(3). Fix $C_1 < \infty$ and suppose*

$$C_0 \log n \leq d = np, \qquad np^2 \leq C_1, \tag{52}$$

*where $C_0$ is sufficiently large depending only on the fixed structural bounds and $C_1$. The degree-metric decoder (49)–(50), with the deterministic initialization and estimated-parameter profile procedure just specified, satisfies*

$$\mathbb{E}\ell(\widehat{z}_D, z) \leq C R_{\mathrm{prof},*}(n) \tag{53}$$

*for all sufficiently large $n$, with $C$ independent of $n$. The statement is uniform over the displayed compact parameter class. No correct grouping, true degree parameters, true block matrix, sample splitting, or leave-one-out eigensystem is an input to the algorithm.*

**Corollary 6.3** (The known-target-degree oracle on equal exposures)**.** *Under theorem 6.2, suppose additionally that the actual finite-$n$ block exposures satisfy*

$$\sum_b (n_b/n) C_{ab} = S \quad \text{for all } a. \tag{54}$$

*Then*

$$R_{\mathrm{prof},*} \asymp R_{\theta,*}, \qquad \mathbb{E}\ell(\widehat{z}_D, z) \leq C R_{\theta,*}.$$

*Thus there are nontrivial fixed-$K$ SBM and DC-SBM classes on which the spectral-only fitted likelihood procedure attains a constant multiple of the original known-parameter local LRT risk. The ordinary free-profile decoder, with a sufficiently small floor coefficient for its structural parameter class, also satisfies (51) for general positive full-rank $C$ under (52) and (54), without the entry-ratio-two restriction of theorem 6.1.*

Exact equality in (54) can be relaxed to exposure differences whose contribution to all likelihood contrasts is uniformly $O(1)$. For example fixed $C\pi = S\mathbf{1}$ with $n_a = n\pi_a + O(1)$ suffices. Equality of limiting exposures with no rate is not by itself sufficient. The general known-$\theta_i$ claim without an additional restriction is false: theorem 11.1 gives an explicit diverging risk ratio for every full-graph algorithm.

*Remark* 6.4 (What is and is not constant). The factor may depend on signal conditioning, group proportions, the allowed degree-parameter range and $C_1$. It is independent of $n$ on the stated class. We do not assert a small numerical constant, a limit of the risk ratio, or uniformity when $\sigma_{\min}(C) \to 0$. The latter weak-signal extension needs separate quantitative estimates.

# 7 Historical unregularized spectral geometry and leave-one-out control

This section proves the spectral premises used by both fitted decoders. Leave-one-out (LOO) matrices occur only in the proof. The implemented encoder retains one full-graph eigensystem.

## 7.1 Raw concentration and deterministic spectral consequences

For any prescribed fixed $H > 0$, when $d \geq c \log n$ there is a fixed $L_H$ such that

$$\mathbb{P}\{\|A - \Omega\|_{\mathrm{op}} > L_H \sqrt{d}\} \leq e^{-Hd}. \tag{55}$$

Indeed, apply Corollary 3.12 and its centered-entry extension in Remark 3.13 of Bandeira–van Handel [2] to $A - \mathbb{E}A$. For bounded centered entries that result bounds the tail at $C\sqrt{\max_i \sum_j \mathrm{Var}(A_{ij})} + t$ by $n \exp(-t^2/C')$. Here the row variance sum is $O(d)$, and the entries have absolute value at most one. Choosing $t$ a sufficiently large fixed multiple of $\sqrt{d}$ absorbs both $H$ and $\log n$. The completed-mean diagonal costs $O(p)$. The centered-entry extension is needed: centered Bernoulli noise does not have a symmetric distribution.

Fix constants $a, b, L > 0$ and define the base event

$$\mathcal{E}_0 = \{\|A - \Omega\|_{\mathrm{op}} \leq L\sqrt{d}, \quad ad \leq D_i \leq bd \text{ for every } i\}. \tag{56}$$

Its minimum-degree probability is deliberately not included in (55); it is treated separately in section 10. On $\mathcal{E}_0$,

$$\left\|(D - \bar{D})\mathbf{1}\right\|_2 \leq L\sqrt{nd}, \qquad \sum_j \frac{(D_j - \bar{D}_j)^2}{D_j} \leq Cn, \quad \bar{D} = \mathrm{diag}(\Omega\mathbf{1}). \tag{57}$$

These follow by applying $A - \Omega$ to $\mathbf{1}$; no uniform vanishing relative degree error is required.

The observed-degree-normalized completed mean

$$M = D^{-1/2}\Omega D^{-1/2}$$

has rank $K$, column space $\mathrm{col}(D^{-1/2}\Theta Z)$, and its nonzero eigenvalues have absolute values at least a fixed positive constant. For the last statement, $Z^\top \Theta D^{-1} \Theta Z$ is diagonal with entries comparable to $n/d$, and the nonzero eigenvalues are those of

$$p(Z^\top \Theta D^{-1} \Theta Z)^{1/2} C (Z^\top \Theta D^{-1} \Theta Z)^{1/2}.$$

The lower singular-value assumption supplies the fixed gap. Since $\|S - M\|_{\mathrm{op}} \leq C/\sqrt{d}$, the retained eigenvalues are bounded away from zero, the discarded spectrum is $O(d^{-1/2})$, and

$$\left\|UU^\top - \Pi_M\right\|_{\mathrm{op}} \leq C/\sqrt{d}, \tag{58}$$

where $\Pi_M$ projects onto $\mathrm{col}(D^{-1/2}\Theta Z)$. These gap estimates follow directly by expanding a retained vector in the signal and complementary eigenbases, or by the usual subspace perturbation inequality. Positive and negative signal eigenvalues are both retained.

The same conclusions hold for every vertex-deleted graph with its degrees *recomputed*: deleting a vertex preserves the raw principal-submatrix norm bound, reduces each surviving degree by at most one, and changes a block mass by only a bounded amount. All constants can be chosen common for the full graph and these $n$ minors.

### 7.2 Exact generalized equations for one incident row

Put the tested vertex first and write

$$A = \begin{pmatrix} 0 & x^\top \\ x & B \end{pmatrix}, \qquad k = \|x\|_1, \quad \Delta = \operatorname{diag}(B\mathbf{1}), \quad D = \operatorname{diag}(k, \Delta + E_x), \quad E_x = \operatorname{diag}(x). \tag{59}$$

Here $x \in \{0,1\}^{n-1}$. Suppose the deleted degrees lie between fixed multiples of $d$, $0 < k \le bd$, and

$$S_B = \Delta^{-1/2} B \Delta^{-1/2} = U_B \Lambda_B U_B^\top + N, \quad U_B^\top U_B = I_K, \quad N U_B = 0,$$

with signal eigenvalues separated from zero and $\|N\|_{\mathrm{op}} \le C/\sqrt{d}$. Write

$$V = \Delta^{-1/2} U_B, \quad Q_\perp = I - U_B U_B^\top, \quad F = \Delta^{-1/2} E_x \Delta^{-1/2}, \quad h = \Delta^{-1/2} x.$$

Then $\|F\| \le C/d$ and $\|h\|^2 \le Ck/d$. The full normalized matrix differs from $\operatorname{diag}(0, S_B)$ by $O(d^{-1/2})$: its off-diagonal normalized row has squared norm at most $C/d$, and the lower degree-renormalization perturbation is $O(d^{-1})$. The full retained signal therefore has the same fixed gap.

Take full generalized eigenvectors $g_r = D^{-1/2} u_r$, so $A g_r = \mu_r D g_r$ and jointly $G^\top D G = I_K$. Decompose them exactly as

$$g_r = \begin{pmatrix} t_r \\ \Delta^{-1/2}(U_B y_r + z_r) \end{pmatrix}, \qquad U_B^\top z_r = 0.$$

Let $h_V = U_B^\top h$ and $h_Q = Q_\perp h$. The top row and complementary equations give

$$h^\top (U_B y_r + z_r) = \mu_r k t_r, \tag{60}$$

$$(\mu_r I - N + \mu_r Q_\perp F Q_\perp) z_r = t_r h_Q - \mu_r Q_\perp F U_B y_r. \tag{61}$$

The inverse $T_r$ of the complementary operator has bounded norm. Substituting into (60) gives

$$(\mu_r k - h_Q^\top T_r h_Q) t_r = h_V^\top y_r - \mu_r h_Q^\top T_r Q_\perp F U_B y_r. \tag{62}$$

Its denominator has magnitude at least $ck$: the second term is at most $Ck/d$. This remains true when $k$ is small.

Put $M_B = \|U_B\|_{2\to\infty}$. The coordinates $y_r$ have bounded norm because $\Delta^{1/2} G_{-i} = (I + F)^{-1/2} U_{-i}$ has norm at most one. Using

$$\|h_V\| \le C k M_B/\sqrt{d}, \qquad \|F U_B\|_F \le C M_B \sqrt{k}/d$$

in the exact equations yields

$$|t_r| \le C M_B/\sqrt{d}, \qquad \|z_r\|_2 \le C M_B \sqrt{k}/d. \tag{63}$$

The matrix $Y = (y_1, \dots, y_K)$ has bounded inverse: the full and padded deleted signal spaces have angle $O(d^{-1/2})$, and the extra multiplier $(I + F)^{-1/2}$ costs only $O(d^{-1})$.

**Lemma 7.1** (Normalized row bounds from model events)**.** *Under the structural model assumptions, for any prescribed $H > 0$ there are row-concentration events of total failure probability at most $2Kne^{-Hd}$ such that, on their intersection with $\mathcal{E}_0$, the full normalized retained vectors and every deleted normalized retained basis satisfy*

$$\|U\|_{2\to\infty} \le C_H/\sqrt{n}, \qquad \max_i \left\| U^{(-i)} \right\|_{2\to\infty} \le C_H/\sqrt{n}. \tag{64}$$

*They also satisfy the gap and angle bounds above. In particular, with $\Pi_0 = Z(Z^\top Z)^{-1} Z^\top$,*

$$\|\Pi_X - \Pi_0\|_{\mathrm{op}} \le C_H/\sqrt{d}, \quad \|Q\|_{2\to\infty} \le C_H/\sqrt{n}, \tag{65}$$

*and the same holds for all recomputed deleted projectors.*

*Proof.* Let $M_{\text{full}} = \|U\|_{2\to\infty}$ and $M_i = \left\|U^{(-i)}\right\|_{2\to\infty}$. The lower rows of the full eigenbasis are $(I+F)^{1/2}(U_B Y + Z_\perp)$, with $Z_\perp = (z_1, \dots, z_K)$. Using $Y^{-1}$ and (63) gives

$$M_i \le C M_{\text{full}} + C M_i/\sqrt{d}, \qquad M_i \le C' M_{\text{full}} \quad \text{for large } d.$$

This argument has not assumed row incoherence. For the tested full row, (62), $k \asymp d$ and $u_{i,r} = \sqrt{k}\, t_r$ imply the sharper estimate

$$\|U_{i\cdot}\|_2 \le \frac{C}{\sqrt{d}} \left\|x^\top \Delta^{-1/2} U_B\right\|_2 + C M_i/d. \tag{66}$$

Conditional on $B$, its eigenbasis and recomputed degrees are fixed and $x$ is an independent Bernoulli row. Its mean action has norm at most $C\sqrt{d/n}$, by $\|\mathbb{E}x\|_2 \le Cp\sqrt{n}$ and $\left\|\Delta^{-1/2} U_B\right\| \le C/\sqrt{d}$. We use the following weighted Bernoulli bound from Abbe–Fan–Wang–Zhong [1, Lemma 7, Eq. (C.2)]: for fixed $v$ and independent Bernoulli coordinates with success probabilities at most $q$,

$$\mathbb{P}\left\{\left|\sum_j v_j (X_j - \mathbb{E}X_j)\right| > \frac{(2+\alpha) nq \|v\|_\infty}{1 \vee \log(\sqrt{n}\,\|v\|_\infty / \|v\|_2)}\right\} \le 2e^{-\alpha nq}. \tag{67}$$

Apply it to each of the fixed $K$ columns of $\Delta^{-1/2} U_B$, taking $q = \min(Cp, 1)$, which is comparable to $p$ on the structural class, and selecting $\alpha$ to give rate $Hd$. The column norms are at most $C/\sqrt{d}$ and $CM_i/\sqrt{d}$ in $\ell_2$ and $\ell_\infty$, respectively. The function $b/(1 \vee \log(\sqrt{n} b/s))$ is nondecreasing in both admissible norm arguments. Thus outside conditional probability $2Ke^{-Hd}$,

$$\left\|x^\top \Delta^{-1/2} U_B\right\|_2 \le C\sqrt{d/n} + \frac{C_H \sqrt{d}\, M_i}{1 \vee \log(c\sqrt{n} M_i)}.$$

A union over $i$, followed by (66), yields

$$M_{\text{full}} \le C/\sqrt{n} + \frac{C'_H M_{\text{full}}}{1 \vee \log(c'\sqrt{n} M_{\text{full}})} + C' M_{\text{full}}/d.$$

For large $d$, absorb the last term. If $\sqrt{n} M_{\text{full}}$ were larger than a sufficiently large fixed constant, the logarithmic term could also be absorbed, giving a contradiction. This proves (64) and its deleted versions.

The conditioning is legitimate even though $\mathcal{E}_0$ depends on $x$: define the conditional row event for every deleted graph satisfying its measurable degree and gap premises, apply (67) there, and use $\mathcal{E}_0$ only for deterministic implications afterwards.

For (65), bounded-condition-number multiplication by $D^{-1/2}$ in (58) compares $\mathrm{col}(X)$ with $\mathrm{col}(D^{-1}\Theta Z)$ at angle $O(d^{-1/2})$. Multiplying block $a$ of $D^{-1}\Theta Z$ by $dS_a/\sqrt{n_a}$ gives entries $(\bar{D}_i/D_i)/\sqrt{n_a}$ on that block. By (57), this basis differs from $Z\,\mathrm{diag}(n_a^{-1/2})$ in Frobenius norm by $O(d^{-1/2})$. This proves the angle to the block space. Finally, $Q = X(X^\top X)^{-1/2}$ has the stated row bound since the eigenvalues of $X^\top X = U^\top D^{-1} U$ are comparable to $1/d$. The argument applies to DC-SBM as well as SBM. □

### 7.3 A deterministic full-to-LOO score lemma

**Lemma 7.2** (Both actual full-graph reconstructions have bounded LOO error)**.** *Assume the deleted degree, spectral gap and discarded-spectrum conditions following* (59)*, as well as* $\|U_B\|_{2\to\infty} \le C/\sqrt{n}$ *and* $k \le bd$*. For every* $w \in \mathbb{R}^n$ *with* $\|w\|_\infty \le W$*, let*

$$\Pi_B = V(V^\top V)^{-1} V^\top, \qquad H_B = VV^\top \Delta, \qquad V = \Delta^{-1/2} U_B.$$

*Uniformly over such incident rows and coefficient vectors,*

$$(R_E w)_i = x^\top \Pi_B w_{-i} + O(1), \tag{68}$$

$$(R_D w)_i = x^\top H_B w_{-i} + O(1). \tag{69}$$

*The constants depend only on the displayed fixed bounds and $W$. For the hollow degree-metric reconstruction on the model good event, (69) also holds with $R^\circ$ in place of $R_D$. The right-side coefficients use degrees recomputed from $B$. When $w$ is fixed, as in the oracle contrasts, or is measurable from $B$, those coefficients are independent of the tested row. Uniform scalar stability alone does not make a full-row-dependent $w$ independent.*

*Proof.* The case $k = 0$ has zero row action and can be given that convention. For $k > 0$, (63) now yields

$$\|t\| \le \frac{C}{\sqrt{nd}}, \qquad \left\|\Delta^{-1/2} Z_\perp\right\|_{\mathrm{op}} \le \frac{C\sqrt{k}}{d^{3/2}\sqrt{n}}. \tag{70}$$

For the Euclidean action right-multiply $G$ by $Y^{-1}(V^\top V)^{-1/2}$. This does not change its column space and gives a basis

$$\mathcal{F} = \begin{pmatrix} \alpha \\ Q_B + E \end{pmatrix}, \quad Q_B = V(V^\top V)^{-1/2}, \quad \|\alpha\| \le C/\sqrt{n}, \quad \|E\| \le C\sqrt{k}/(d\sqrt{n}).$$

Its Gram matrix is $I + J$, where $\|J\| \le C\{\sqrt{k}/(d\sqrt{n}) + 1/n\}$. The deleted Euclidean basis has row norm at most $C/\sqrt{n}$, so

$$\left\|x^\top Q_B\right\| \le Ck/\sqrt{n}, \qquad \left\|x^\top E\right\| \le Ck/(d\sqrt{n}),$$

$$\left\|Q_B^\top w_{-i}\right\| \le W\sqrt{n}, \qquad \left\|\mathcal{F}^\top w - Q_B^\top w_{-i}\right\| \le C_W(1/\sqrt{n} + \sqrt{k}/d).$$

Expand

$$(R_E w)_i = (x^\top Q_B + x^\top E)(I + J)^{-1}\mathcal{F}^\top w$$

about $x^\top Q_B Q_B^\top w_{-i}$. The errors are bounded by a constant times

$$k/d + k/n + k\sqrt{k}/(d\sqrt{n}) = O(1), \tag{71}$$

because $k \le bd$ and $d \le Cn$. This proves (68). A coarse $O(n^{-1/2})$ bound for the lower rows alone would not suffice; the stronger $E$ bound is essential.

For the degree-metric action, put $v_r = \Delta^{-1/2} z_r$, $f_r = V y_r + v_r$ and $w_V = V^\top \Delta w_{-i}$. The exact top equation gives $k\mu_r t_r = x^\top f_r$, whence

$$(R_D w)_i = \sum_r (x^\top V y_r + x^\top v_r)\{k t_r w_i + y_r^\top w_V + v_r^\top \Delta w_{-i} + f_r^\top E_x w_{-i}\}. \tag{72}$$

The normalization $G^\top D G = I$ gives

$$Y^\top Y + Z_\perp^\top Z_\perp + k t t^\top + (VY + \Delta^{-1/2} Z_\perp)^\top E_x (VY + \Delta^{-1/2} Z_\perp) = I.$$

All terms except $Y^\top Y$ have norm at most $Ck/(nd)$, by (70) and the row bound on $V$. Since $Y$ is square, $\left\|YY^\top - I\right\| \le Ck/(nd)$ as well. The leading term in (72) is $x^\top V Y Y^\top w_V$; its difference from $x^\top V V^\top \Delta w_{-i}$ is at most $C_W k^2/(nd)$. The other factors satisfy

$$\left\|x^\top V\right\| \le Ck/\sqrt{nd}, \qquad \|w_V\| \le C_W\sqrt{nd},$$

$$\left\|(x^\top v_r)_r\right\| \le Ck/(d^{3/2}\sqrt{n}), \qquad \left\|(v_r^\top \Delta w_{-i})_r\right\| \le C_W\sqrt{k/d},$$

$$\left\|(f_r^\top E_x w_{-i})_r\right\| \le C_W k/\sqrt{nd}, \qquad \|kt\| \le Ck/\sqrt{nd}.$$

Their products are bounded by constants times $k/d$, $k^2/(nd)$ and $k\sqrt{k}/(d\sqrt{n})$, all bounded. This proves (69) with negative as well as positive informative eigenvalues.

On the model good event, full row incoherence implies $|(H_D)_{ij}| \le C/n$, $|(R_D)_{ii}| \le Cd/n = Cp$ and $|s_i| \le Cp$. The hollow adjustment to a bounded row contrast is $s_i\{(H_D w)_i - w_i\} = O(p)$, proving the last assertion. □

The result compares the *actual* full-graph normalized algorithm to a row-independent score, with its changing degree normalizers included. It does not treat a full-graph eigenvector as independent of the row that created it. This is the dependence step needed before any conditional sharp-tail argument is applied.

# 8 The earlier ordinary free-profile decoder

We work under the ordinary-SBM assumptions of Theorem 6.1. In particular, $K \ge 2$ is fixed, $n_a \ge c_* n$, the block matrix is symmetric and uniformly full rank at scale $p$, and $p/2 \le P_{ab} \le p \le 1/4$. Write $d = np$, $C_a = \{i : z_i = a\}$, $M_0 = \Omega = ZPZ^\top$, and

$$\Pi_0 = Z(Z^\top Z)^{-1}Z^\top, \qquad p_{a*} = (P_{a,z_j})_{j=1}^n.$$

The diagonal of $M_0$ is retained in these comparison profiles; the observed graph has no self-loops. The resulting bounded diagonal correction is included below.

## 8.1 A finite decoder using only the retained information

Let $D = \mathrm{diag}(A\mathbf{1})$, and retain $K$ normalized adjacency eigenpairs $D^{-1/2}AD^{-1/2}U = U\Lambda$, including the Perron direction and selecting the remaining signal eigenvalues by absolute value. On graphs where the encoder is defined, put

$$X = D^{-1/2}U, \qquad Q = X(X^\top X)^{-1/2}, \qquad Y = DX\Lambda(X^\top X)^{-1/2}, \qquad R = YQ^\top.$$

These are exact algebraic operations. The generalized eigenrelation gives $Y = AQ$ and hence $R = A\Pi$, where $\Pi = QQ^\top$. The Perron direction implies $\Pi\mathbf{1} = \mathbf{1}$ and $R\mathbf{1} = A\mathbf{1}$. In particular, the entire decoder below is evaluable from $(D, U, \Lambda)$ without accessing an individual adjacency entry after compression. The exceptional-graph convention in the theorem makes the encoder and decoder total.

Fix $0 < c_0 < 1/16$, let $\bar{d} = n^{-1}\sum_i D_{ii}$, and set

$$\varepsilon = c_0 \max(1, \bar{d})/n, \qquad \mathcal{I} = [\varepsilon, 1-\varepsilon].$$

All clipping in this section is coordinatewise to $\mathcal{I}$. For $v \in \mathbb{R}^n$ and $\mu \in \mathcal{I}^n$, define

$$\ell(v;\mu) = \sum_j \{v_j \log \mu_j + (1-v_j)\log(1-\mu_j)\}, \qquad \ell_i(\mu) = \ell(R_{i*};\mu).$$

The entries of $R$ may be signed. The score is nevertheless finite and has the retained-factor expression

$$\ell_i(\mu) = Y_i Q^\top \mathrm{logit}(\mu) + \sum_j \log(1-\mu_j). \tag{73}$$

For $s$ fitted profiles $\mathcal{C} = (\mu_1, \dots, \mu_s)$ and simplex weights $\pi$, use

$$\mathcal{L}(\mathcal{C}, \pi) = \sum_i \log \sum_{a=1}^s \pi_a e^{\ell_i(\mu_a)}.$$

An E-step and its subsequent M-step are

$$\gamma_{ia} = \frac{\pi_a e^{\ell_i(\mu_a)}}{\sum_b \pi_b e^{\ell_i(\mu_b)}}, \qquad N_a = \sum_i \gamma_{ia},$$
$$\mu_{aj}^{\text{new}} = \text{clip}_{\mathcal{I}}\left(\frac{\sum_i \gamma_{ia} R_{ij}}{N_a}\right), \qquad \pi_a^{\text{new}} = N_a/n. \tag{74}$$

When $N_a = 0$, the old profile is retained and the new weight is zero. The inner fitting subroutine performs at most 100 exact E/M updates. It may stop earlier when its labels are unchanged and its objective increment is at most $10^{-12}(1 + |\mathcal{L}_{\text{new}}|)$. Ties in all maximizations are settled deterministically. The proof uses only finiteness and monotonicity of these inner updates.

The candidate attached to vertex $j$ is $\nu_j = \text{clip}_{\mathcal{I}}(R_{j*})$. Start from the clipped global mean and fit one component. For $s = 2, \dots, K$, append a candidate maximizing

$$\sum_i \left(\ell_i(\nu_j) - \max_{a<s} \ell_i(\mu_a)\right)_+,$$

excluding the vertex identifiers selected in previous growing stages, and fit all $s$ components from uniform weights. This gives $K$ profiles. Next perform at most $T = \lceil \log n \rceil$ replacement rounds. For each deletion $a$ from the round's incumbent, put

$$h_i^{(-a)} = \max_{b \neq a} \ell_i(\mu_b)$$

and choose a candidate maximizing $\sum_i (\ell_i(\nu_j) - h_i^{(-a)})_+$ over all vertices. Replace profile $a$ by this candidate, initialize all $K$ weights uniformly, and run the inner fit. After trying every deletion, retain the fit with largest $\mathcal{L}$, including the unchanged incumbent among the choices. A round without strict improvement terminates replacement.

Finally, evaluate the selected fit's responsibilities, perform one mandatory M-step in (74), and return

$$\widehat{z}_i \in \arg\max_a \{\ell_i(\mu_a^{\text{new}}) + \log \pi_a^{\text{new}}\}. \tag{75}$$

The reference to an E/M update here is ordered: the E-step is evaluated at the selected fit, its M-step follows, and the decoder uses the new profiles and weights.

## 8.2 A common box and deterministic reconstruction bounds

Throughout the deterministic arguments, intersect the base event $\mathcal{E}_0$ in (56) with the row-concentration events of Lemma 7.1. Its constants are fixed, independent of $n$. In addition to the degree bounds and normalized signal gap, its consequences used here are

$$\|A - M_0\|_{\text{op}} \le C\sqrt{d}, \qquad \|U\|_{2\to\infty} \le C/\sqrt{n}, \qquad \|\Pi - \Pi_0\|_{\text{op}} \le C/\sqrt{d}. \tag{76}$$

The same conclusions hold for the vertex-deleted graphs on their corresponding good events. Their probability costs are handled in the probability closure, after the sharp row calculation.

**Lemma 8.1** (Reconstruction and generated probability box)**.** *On the full-graph good event, for all sufficiently large $n$,*

$$\|R - M_0\|_F^2 \le Cd, \qquad \max_{i,j} |R_{ij}| \le Cp, \qquad \|R_i(I - \Pi_0)\|_2 \le C\sqrt{p}, \tag{77}$$
$$\|\Pi_0(A - R)\Pi_0\|_{\text{op}} \le C. \tag{78}$$

*There are fixed $a_-, a_+, \eta_- > 0$ such that every profile generated by the decoder, every clipped true-class mean, and every true comparison profile belongs to*

$$\mathcal{P}_n = \{\mu : \ a_- p \le \mu_j \le a_+ p, \quad \mu_j \le 1 - \eta_- \text{ for every } j\}. \tag{79}$$

*Every true probability is at distance at least $cp$ from both endpoints of the actual clipping interval $\mathcal{I}$.*

*Proof.* Write $E = A - M_0$. Since $R = A\Pi$,

$$R - M_0 = E\Pi + M_0(\Pi - I).$$

Each term has rank at most $K$, operator norm at most $C\sqrt{d}$, and hence Frobenius norm at most $C\sqrt{d}$, by (76) and $\|M_0\|_{\mathrm{op}} \le Cd$. For the other row bounds, write $G = X^\top X$. The degree bounds imply that the eigenvalues of $G$ are between constant multiples of $d^{-1}$. Thus $Q = XG^{-1/2}$ has row norms at most $C/\sqrt{n}$, and

$$AQ = DQT_0, \qquad T_0 = G^{1/2}\Lambda G^{-1/2}, \qquad \|T_0\|_{\mathrm{op}} \le C.$$

It follows that $\|(AQ)_i\|_2 \le Cd/\sqrt{n}$, which proves the entrywise bound for $R = (AQ)Q^\top$. Moreover,

$$\|R_i(I - \Pi_0)\|_2 \le \|(AQ)_i\|_2 \left\|Q^\top(I - \Pi_0)\right\|_{\mathrm{op}} \le C\sqrt{d/n}.$$

Both $\Pi$ and $\Pi_0$ are orthogonal projectors. Consequently,

$$\|\Pi_0(I - \Pi)\Pi_0\|_{\mathrm{op}} = \|(I - \Pi)\Pi_0\|_{\mathrm{op}}^2 \le C/d.$$

Using $M_0 = \Pi_0 M_0 \Pi_0$ gives

$$\Pi_0(A - R)\Pi_0 = M_0\Pi_0(I - \Pi)\Pi_0 + \Pi_0 E(I - \Pi)\Pi_0.$$

The two terms have norms at most $Cd/d$ and $C\sqrt{d}/\sqrt{d}$, respectively. This proves (78) without a commutation assumption on $A$ and $\Pi$.

The average degree satisfies

$$\left|\bar{d} - n^{-1}\mathbf{1}^\top M_0 \mathbf{1}\right| \le \|E\|_{\mathrm{op}} \le C\sqrt{d}.$$

Since $d/2 \le n^{-1}\mathbf{1}^\top M_0 \mathbf{1} \le d$, eventually $d/4 \le \bar{d} \le 2d$. In particular $\bar{d} > 1$ eventually, so

$$c_0 p/4 \le \varepsilon \le 2c_0 p < p/8. \tag{80}$$

Together with $P_{ab} \in [p/2, p]$ and $p \le 1/4$, this gives the claimed interior distance from the clipping endpoints.

Each active raw M-step coordinate is a convex combination of a column of $R$, even though that column may have negative entries. It therefore lies in $[-C_R p, C_R p]$. Clipping makes it at most $\max(C_R p, \varepsilon) \le C_\mu p$. The initial mean and all clipped candidates obey the same bound; an inactive component retains an earlier profile. These observations prove inductively that all generated profiles lie between $c_0 p/4$ and $C_\mu p$. If $p \le 1/(2C_\mu)$, their upper bound is $1/2$. Otherwise the lower bound in (80) gives $\varepsilon \ge c_0/(8C_\mu)$, providing a fixed upper gap through clipping. Enlarge the constants to include the true profiles and the clipped true-class means. □

The box in (79) describes profiles generated by this algorithm and its comparison profiles. Uniform statements below concern this common box; they do not concern arbitrary vectors in the larger nominal interval $\mathcal{I}^n$.

**Lemma 8.2** (Exact monotonicity for signed observations)**.** *Every exact update* (74) *increases or preserves* $\mathcal{L}$*.*

*Proof.* The log-sum-exp variational identity gives

$$\sum_{i,a} \gamma_{ia}\{\log \pi_a + \ell_i(\mu_a) - \log \gamma_{ia}\} \le \mathcal{L}(\mathcal{C}, \pi),$$

with equality at the E-step. For one profile coordinate and fixed responsibilities, put $a_j = \sum_i \gamma_{ia} R_{ij}$. Its objective is

$$a_j \log \mu + (N_a - a_j)\log(1 - \mu),$$

whose derivative has sign equal to $a_j/N_a - \mu$ when $N_a > 0$. Thus its global maximizer on $\mathcal{I}$ is the clipped weighted mean, whether or not $a_j$ lies between zero and $N_a$. The weight maximizer is $N_a/n$. Zero mass leaves the corresponding coordinate objective constant. Maximization of this variational lower bound proves the claim. □

### 8.3 The finite replacement certificate and common alignment

For a $K$-profile list and weights, define the empirical reference loss

$$F_R(\mathcal{C}, \pi) = \sum_i \ell(R_i; p_{z_i,*}) - \mathcal{L}(\mathcal{C}, \pi),$$

and its population version

$$F_0(\mathcal{C}, \pi) = -\sum_i \log \sum_a \pi_a e^{-\mathrm{KL}(p_{z_i,*} \| \mu_a)}.$$

Here a vector Bernoulli divergence means the coordinate sum

$$\mathrm{KL}(p_{c*} \| \mu) = \sum_j \mathrm{KL}(\mathrm{Bern}(P_{c,z_j}) \| \mathrm{Bern}(\mu_j)).$$

The first term defining $F_R$ is a comparison constant independent of the fitted list and weights. Hence every implemented likelihood comparison is precisely the reverse comparison of $F_R$. The population loss $F_0$ is nonnegative. Also put

$$G_R(\mathcal{C}) = \sum_i \{\ell(R_i; p_{z_i,*}) - \max_a \ell_i(\mu_a)\},$$
$$G_0(\mathcal{C}) = \sum_c n_c \min_a \mathrm{KL}(p_{c*} \| \mu_a).$$

These empirical reference losses are defined for signed $R$.

**Lemma 8.3** (Uniform comparison of signed and population objectives)**.** *On the full-graph good event, uniformly over center lists in $\mathcal{P}_n$ and simplex weights,*

$$|F_R - F_0| + |G_R - G_0| \le Cn\sqrt{d}. \tag{81}$$

*Always $G_R \le F_R$. With uniform weights, $F_R \le G_R + n \log K$. Every true class supplies a candidate $\nu_j$ with $\|\nu_j - p_{c*}\|_1 \le C\sqrt{d}$; all these candidates are available in every replacement trial.*

*Proof.* For a true class $c$ and fitted profile $\mu_a$, the difference $\ell(v; \mu_a) - \ell(v; p_{c*})$ is affine in $v$, with coefficient $\mathrm{logit}(\mu_a) - \mathrm{logit}(p_{c*})$ bounded uniformly on $\mathcal{P}_n$. The maximum and log-sum-exp operations are 1-Lipschitz for changes bounded in maximum norm. Therefore each comparison in (81) is at most $C \sum_i \|R_i - p_{z_i,*}\|_1$. By (77), this sum is at most $n \|R - M_0\|_F \le Cn\sqrt{d}$. The common $\log p$ terms have canceled in the score contrasts used in this argument.

The inequalities between hard and soft reference losses follow from

$$K^{-1} e^{\max_a s_a} \le K^{-1} \sum_a e^{s_a} \le e^{\max_a s_a}$$

for uniform weights, and $\sum_a \pi_a e^{s_a} \le e^{\max_a s_a}$ for arbitrary weights. Finally, each class has at least $c_* n$ rows, so one of its rows has squared error at most $Cd/n$ by the Frobenius bound. Its $\ell_1$ error is at most $C\sqrt{d}$. Clipping toward an interval containing the true profile contracts coordinatewise absolute error. □

**Lemma 8.4** (Finite replacement certificate)**.** *At the selected endpoint of the replacement stage,*

$$F_R \le \xi_n n d, \qquad \xi_n = C\{(1 - 1/K)^{\lceil \log n \rceil} + d^{-1/2}\} \longrightarrow 0. \tag{82}$$

*The same conclusion holds if replacement stops because a complete round does not strictly improve the incumbent.*

*Proof.* Assign each true profile a nearest fitted center in population divergence. A true class contributes at least $G_0/K$. If a fitted center is unused by this assignment, replacing that center by this true profile removes its contribution without increasing any other class's loss. If every center is used, the assignment between the $K$ true classes and $K$ fitted centers is a bijection; replacing the center of the highest-loss class has the same property. Consequently some ideal replacement has population hard loss at most $(1 - 1/K)G_0$.

The derivative of $\mathrm{KL}(p_{c*}\|\mu)$ in a parameter coordinate is $(\mu_j - P_{c,z_j})/[\mu_j(1-\mu_j)]$, uniformly bounded on the common box. Replacing the ideal profile by the available candidate from Lemma 8.3 costs at most $Cn\sqrt{d}$. Combining the objective comparisons gives a candidate list $\mathcal{C}'$ with

$$G_R(\mathcal{C}') \le (1 - 1/K)F_R(\mathcal{C}, \pi) + Cn\sqrt{d}.$$

For a fixed deletion $a$, the empirical hard loss after inserting $\nu_j$ is exactly

$$\sum_i \{\ell(R_i; p_{z_i,*}) - h_i^{(-a)} - (\ell_i(\nu_j) - h_i^{(-a)})_+\}.$$

Thus the implemented maximum positive gain minimizes this loss among candidates. All deletions are tried. Uniform trial weights add at most $n \log K$ to the hard loss, and the subsequent exact inner fit cannot increase $F_R$ by Lemma 8.2. Including the incumbent among the choices proves the recurrence

$$F_{R,t+1} \le (1 - 1/K)F_{R,t} + Cn\sqrt{d} + n \log K, \qquad F_{R,t+1} \le F_{R,t}. \tag{83}$$

Every generated profile lies in the common box, so each population divergence is at most $Cd$. Hence $F_{0,0} \le Cnd$ for arbitrary simplex weights, and the same bound holds for $F_{R,0}$ by (81). No preliminary accuracy guarantee for the growing stage is needed: its generated list supplies this finite starting bound. Iterating (83) for $\lceil \log n \rceil$ rounds gives (82), after enlarging its constant. If a complete round gives no strict improvement, its chosen loss is the incumbent's loss. The first inequality in (83) then immediately yields $F_R \le K(Cn\sqrt{d} + n \log K)$, which has the same conclusion. □

**Lemma 8.5** (Common matching and refined responsibilities)**.** *The selected fit admits a single permutation after which*

$$\max_a \|\mu_a - p_{a*}\|_1 = o(d), \qquad -\log \min_a \pi_a = o(d). \tag{84}$$

*Its next E-step satisfies*

$$\frac{1}{n} \sum_{i,a} |\gamma_{ia} - Z_{ia}| \le C/d. \tag{85}$$

*Proof.* The preceding lemmas give $F_0 \le \zeta_n nd$ with $\zeta_n \to 0$. Since $F_0 \ge G_0$, each true class has a fitted center whose divergence is $o(d)$. On the common box, Bernoulli entropy curvature and Cauchy–Schwarz imply

$$\mathrm{KL}(p_{a*}\|\mu) \ge \frac{\|p_{a*} - \mu\|_2^2}{Cp} \ge \frac{\|p_{a*} - \mu\|_1^2}{Cd}.$$

Thus each true profile has a fitted center at $o(d)$ in $\ell_1$. Uniform full rank and positive class sizes give

$$\min_{a \ne b} \|p_{a*} - p_{b*}\|_2^2 \ge cnp^2, \qquad \min_{a \ne b} \|p_{a*} - p_{b*}\|_1 \ge cd.$$

The nearby centers must therefore be distinct. There are exactly $K$ of them, proving one common matching and the first assertion of (84).

For a row of true class $a$, each wrong-center population divergence is at least $c_1 d$ eventually, by this matching and the common-box Lipschitz bound. Its population soft loss is at least

$-\log(\pi_a + e^{-c_1 d})$. It is the same for all $n_a$ rows of that class, and every population soft loss is nonnegative. Consequently

$$\pi_a + e^{-c_1 d} \geq e^{-\zeta_n nd/n_a}, \qquad \pi_a \geq \tfrac{1}{2} e^{-C\zeta_n d}$$

for all sufficiently large $n$. This proves the weight assertion.

Fix a sufficiently small constant $t_0 > 0$. If $\|R_i - p_{a*}\|_1 \leq t_0 d$ and $z_i = a$, the population separation, (84), and bounded contrast coefficients give a fitted correct-class margin at least $c_2 d$. The wrong responsibility mass on this row is at most $(K-1)e^{-c_2 d}$. Every remaining row has squared error at least $t_0^2 d^2/n$. The Frobenius bound limits their number to $Cn/d$. Summing their responsibility masses, which are at most one, proves (85). □

### 8.4 Fitted profiles change the relevant score by a bounded amount

Define the centered block fluctuation

$$\mathcal{B} = \max_{a,c} \left| \frac{\mathbf{1}_{C_a}^\top (A - \mathbb{E}A) \mathbf{1}_{C_c}}{\sqrt{n_a n_c}} \right|. \tag{86}$$

**Lemma 8.6** (Block moments and deletion)**.** *For every fixed $t \geq 0$, $\mathbb{E}e^{t\mathcal{B}} \leq C_t$. For a tested vertex $i$, let $\mathcal{B}_i$ be the analogous fluctuation calculated from $B = A_{-i,-i}$ and the deleted class sizes. Then $\mathbb{E}e^{t\mathcal{B}_i} \leq C_t$, and on $D_{ii} \leq Cd$,*

$$\mathcal{B} \leq C(1 + \mathcal{B}_i). \tag{87}$$

*Proof.* Each normalized block sum in (86) is a sum of independent centered upper-triangular Bernoulli entries with coefficients of magnitude at most $C/n$. Its sum of coefficient-squared edge probabilities is at most $Cp$. The elementary centered Bernoulli transform bound therefore gives, for each fixed real $t$,

$$\log \mathbb{E} \exp\left\{ t \frac{\mathbf{1}_{C_a}^\top (A - \mathbb{E}A) \mathbf{1}_{C_c}}{\sqrt{n_a n_c}} \right\} \leq Ct^2 e^{C|t|/n} p.$$

For example, this follows by applying $e^u - 1 - u \leq u^2 e^{|u|}/2$ to every independent edge and multiplying their transforms. Taking both signs and summing over the fixed number of block pairs bounds $\mathbb{E}e^{t\mathcal{B}}$.

Deleting one vertex preserves the positive class-size bounds, so the same argument applies to $\mathcal{B}_i$. The unnormalized difference between a full and deleted block sum consists of at most twice the tested row's centered edge sum. Its absolute value is at most $C(D_{ii} + d)$. Dividing by $\sqrt{n_a n_c} \geq cn$ gives $O(p)$. The factor converting a deleted normalized block sum to the full denominator is bounded by a constant. This proves (87). □

For each class define the raw true-class mean and its clipped version

$$\bar{r}_a = n_a^{-1} \sum_{l \in C_a} R_l, \qquad \bar{\mu}_a = \mathrm{clip}_{\mathcal{I}}(\bar{r}_a).$$

These are comparison quantities only. For $a \neq b$, also define

$$w_j^{ab} = \log \frac{P_{a,z_j}(1 - P_{b,z_j})}{P_{b,z_j}(1 - P_{a,z_j})}, \qquad r_j^{ab} = \log \frac{1 - P_{a,z_j}}{1 - P_{b,z_j}}. \tag{88}$$

**Lemma 8.7** (Uniform fitted-score comparison)**.** *On the full-graph good event, every M-step from responsibilities satisfying (85), after their common matching, has weights bounded below and obeys*

$$\left| \ell_i(\mu_a) - \ell_i(\mu_b) + \log \frac{\pi_a}{\pi_b} - \left\{ R_i w^{ab} + \sum_j r_j^{ab} \right\} \right| \leq C(1 + \mathcal{B}) \tag{89}$$

*uniformly over $i, a, b$ and all such responsibility matrices.*

*Proof.* Write $e = n^{-1}\sum_{l,a}|\gamma_{la} - Z_{la}| \le C/d$. Then $N_a = n_a + O(ne)$, so $N_a \ge cn$ eventually and the updated weights $N_a/n$ are bounded below. By (77), $\max_l \|R_l\|_1 \le Cd$. The difference of the raw responsibility-weighted mean and $\bar{r}_a$ can be written as a numerator perturbation and a denominator perturbation. Their combined $\ell_1$ norm is at most

$$\frac{Cd}{N_a}\sum_l |\gamma_{la} - Z_{la}| + \frac{Cd}{N_a}|N_a - n_a| \le Ced \le C.$$

Clipping is 1-Lipschitz, whence

$$\|\mu_a - \bar{\mu}_a\|_1 \le C. \tag{90}$$

Jensen's inequality and the Frobenius bound give

$$\|\bar{r}_a - p_{a*}\|_2^2 \le n_a^{-1}\sum_{l\in C_a}\|R_l - p_{a*}\|_2^2 \le Cp.$$

Clipping toward an interval containing $p_{a*}$ preserves this bound. The interior distance from Lemma 8.1 also gives the scalar inequality

$$|\operatorname{clip}_{\mathcal{I}}(x) - x| \le \frac{(x-u)^2}{cp}$$

for each true coordinate $u$. Thus $\|\bar{\mu}_a - \bar{r}_a\|_1 \le C$.

For a true block $C_c$, decompose the block sum of $\bar{r}_a - p_{a*}$ using $R - A$, $A - \mathbb{E}A$, and $\mathbb{E}A - M_0$. The first contributes at most

$$\sqrt{n_c/n_a}\,\|\Pi_0(R-A)\Pi_0\|_{\mathrm{op}} \le C,$$

by (78); the second is at most $C\mathcal{B}$. The diagonal correction contributes at most $P_{aa} \le p$. Including the clipping correction proves, for $\delta_a = \bar{\mu}_a - p_{a*}$,

$$\|\delta_a\|_2^2 \le Cp, \qquad \left|\sum_{j\in C_c}\delta_{aj}\right| \le C(1+\mathcal{B}). \tag{91}$$

For any profile in the common box, the score derivatives are

$$\frac{\partial \ell_i}{\partial \mu_j} = \frac{R_{ij} - \mu_j}{\mu_j(1-\mu_j)}, \qquad \frac{\partial^2 \ell_i}{\partial \mu_j^2} = -\frac{R_{ij}}{\mu_j^2} - \frac{1 - R_{ij}}{(1-\mu_j)^2}.$$

Their absolute values are at most $C$ and $C/p$, respectively. Therefore (90) changes the score by $O(1)$, and the second-order remainder in a Taylor expansion at $p_{a*}$ is at most $C\|\delta_a\|_2^2/p \le C$.

Split the first-order gradient at $p_{a*}$ into its projection onto $\mathrm{col}(Z)$ and its orthogonal complement. Its block-constant projection has bounded coordinates; paired with $\delta_a$, it is bounded by (91). The diagonal multiplier $[P_{a,z_j}(1 - P_{a,z_j})]^{-1}$ is block constant and commutes with $\Pi_0$. Hence the complementary gradient has norm at most

$$\frac{C}{p}\|R_i(I - \Pi_0)\|_2 \le C/\sqrt{p}.$$

Its inner product with $\delta_a$, whose norm is at most $C\sqrt{p}$, is $O(1)$. We have proved $|\ell_i(\mu_a) - \ell_i(p_{a*})| \le C(1+\mathcal{B})$. Apply the same calculation to $b$ and include the bounded updated log-weight ratio. The exact true-profile contrast is the expression in braces in (89), proving the result. □

### 8.5 Euclidean projected evidence at the oracle tilt

Fix a tested vertex $i$ and labels $a \neq b$. Let $B = A_{-i,-i}$, $x = A_{-i,i}$, and let $\Pi_B$ be the Euclidean projector onto the deleted generalized signal space from Lemma 7.2. Write $\Pi_{0,B}$ for the true block projector on the remaining vertices. For this paragraph, $w$ and $r$ denote the restrictions of (88) to these vertices. The exact oracle log likelihood ratio is

$$L_{i,ab} = b_{i,ab} + x^\top w, \qquad b_{i,ab} = \sum_{j \neq i} r_j^{ab}.$$

Let $I_{i,ab}$ be its exact Bernoulli Chernoff information in (13) and $t_*$ its optimizing parameter, using the transform under class $a$ with factor $e^{-t_* L_{i,ab}}$. Under the corresponding optimizing tilt, the row coordinates are independent with block-constant probabilities

$$q_j = \frac{P_{a,z_j}^{1-t_*} P_{b,z_j}^{t_*}}{P_{a,z_j}^{1-t_*} P_{b,z_j}^{t_*} + (1 - P_{a,z_j})^{1-t_*} (1 - P_{b,z_j})^{t_*}}. \tag{92}$$

These probabilities are comparable to $p$. The log-odds are bounded, and uniform full rank gives $\sum_{j \neq i} w_j^2 \asymp n$. For all tilt parameters in $[0,1]$, the log transform therefore has second derivative between $cd$ and $Cd$. Its two endpoint values are zero. It follows that $I_{i,ab} \asymp d$, the optimizing parameter stays inside a fixed compact subinterval of $(0,1)$, and its tilted variance is comparable to $d$. Thus the oracle conditions of Lemma 3.1 hold uniformly.

**Lemma 8.8** (Euclidean tilted cancellation)**.** *On the deleted-graph good event, let $\beta = (\Pi_B - I)w$. Then*

$$\max_j |\beta_j| \leq C, \qquad |q^\top \beta| \leq C, \qquad \sum_j q_j \beta_j^2 \leq C. \tag{93}$$

*Consequently, conditional on $B$, for every real $u$,*

$$\mathbb{P}_a\{b_{i,ab} + x^\top \Pi_B w \leq u \mid B\} \leq C \frac{e^{-I_{i,ab} + t_* u}}{\sqrt{d}}. \tag{94}$$

*Proof.* The deleted Euclidean basis has row norms at most $C/\sqrt{n}$, so bounded entries of $w$ imply bounded entries of $\Pi_B w$. Let $\delta = \|(I - \Pi_B)\Pi_{0,B}\|_{\mathrm{op}} \leq C/\sqrt{d}$, as follows from (65) for the deleted graph. Both $q$ and $w$ belong to $\mathrm{col}(\Pi_{0,B})$. Orthogonality gives

$$q^\top \beta = -\langle (I - \Pi_B)q, (I - \Pi_B)w \rangle .$$

Since $\|q\|_2 \leq Cp\sqrt{n}$ and $\|w\|_2 \leq C\sqrt{n}$, its absolute value is at most $Cnp\delta^2 \leq C$. Also $\|\beta\|_2^2 \leq \delta^2 \|w\|_2^2 \leq Cn/d$, so $\sum_j q_j \beta_j^2 \leq Cp \|\beta\|_2^2 \leq C$. This proves (93).

The conditional transform identity is

$$\mathbb{E}_a e^{-t_*(b_{i,ab} + x^\top \Pi_B w)} = e^{-I_{i,ab}} \mathbb{E}_q e^{-t_* x^\top \beta}.$$

Independence, $\log(1+y) \leq y$, and $e^v - 1 - v \leq v^2 e^{|v|}/2$ bound the final log transform by $C|q^\top \beta| + C \sum_j q_j \beta_j^2 \leq C$. Under tilting by this estimated score, the row is still a product of Bernoulli variables. Its probabilities and complements are comparable to their oracle-tilted counterparts because the coefficient errors are bounded. Its score variance is of order $d$: in the oracle-tilted quadratic form,

$$\sum_j q_j(1 - q_j)(w_j + \beta_j)^2 \geq \tfrac{1}{2} \sum_j q_j(1 - q_j) w_j^2 - \sum_j q_j \beta_j^2 \geq cd$$

eventually, and the corresponding upper bound is $Cd$. The bounded summands have total third absolute centered moment $O(d)$. The bounds (93) give all three premises in (10), with zero additive intercept error and fixed constants. Lemma 3.1 now gives (94). A threshold $u$ only contributes the displayed factor $e^{t_* u}$; it does not change this variance or the interval-concentration bound. □

**Proposition 8.9** (Sharp error bound for the finite fitted decoder)**.** *Align the selected fit by the common matching from Lemma 8.5. Let $\mathcal{E}_i$ be the intersection of the full-graph and vertex-deleted good events, including the tested degree bound, used in Lemmas 7.1 and 7.2. For a vertex with true class $a$ and any $b \neq a$, the final decoder (75) satisfies*

$$\mathbb{P}_z\{\widehat{z}_i \text{ is aligned to } b,\ \mathcal{E}_i\} \le C\frac{e^{-I_{i,ab}}}{\sqrt{d}}. \tag{95}$$

*In particular, summing over competitors and averaging over vertices bounds the good-event expected misclassification loss by*

$$\frac{C}{n}\sum_i \sum_{b\neq z_i} \frac{e^{-I_{i,z_i b}}}{\sqrt{d}}. \tag{96}$$

*Proof.* The finite replacement certificate and its mandatory final E/M update give the premise of Lemma 8.7. A wrong assignment from $a$ to $b$, including an unfavorable score tie, implies on the full-graph good event that

$$R_i w^{ab} + \sum_j r_j^{ab} \le C(1+\mathcal{B}).$$

The full-to-deleted Euclidean identity (68) applies to the fixed bounded vector $w^{ab}$. Together with (87), it gives the deterministic event containment

$$\{\widehat{z}_i \text{ is aligned to } b,\ \mathcal{E}_i\} \subseteq \{b_{i,ab} + x^\top \Pi_B w^{ab}_{-i} \le C(1+\mathcal{B}_i),\ B \text{ good}\}. \tag{97}$$

The nonedge intercept changes by $r_i^{ab} = O(p)$ when the tested diagonal is removed, which is absorbed into the constant.

Use the full-graph event only for this deterministic containment. The right-hand event does not condition the tested row on full-graph eigenvectors, fitted parameters, or a degree event. Conditional on $B$, the incident row has its original independent Bernoulli law. Applying (94) with $u = C(1+\mathcal{B}_i)$, integrating over good $B$, and using Lemma 8.6 gives

$$\mathbb{P}_z\{\widehat{z}_i \text{ is aligned to } b,\ \mathcal{E}_i\} \le C\frac{e^{-I_{i,ab}}}{\sqrt{d}}\mathbb{E}e^{C\mathcal{B}_i} \le C'\frac{e^{-I_{i,ab}}}{\sqrt{d}}.$$

This proves (95). The common permutation is the same for every vertex, so the loss minimized over permutations is at most the loss under this matching. Summing the pairwise bounds proves (96). □

The sharp Bernoulli oracle comparison in Lemma 3.1 identifies the terms in (96) with local oracle errors up to fixed constants, including the $d^{-1/2}$ factor [5]. The probability closure pays the complement of these good events at that same oracle scale and completes Theorem 6.1.

**Extension to general positive block profiles.** The deterministic and conditional arguments also apply to a fixed positive full-rank block class

$$c_- p \le P_{ab} \le c_+ p, \qquad \max_{a,b} P_{ab} \le 1-\epsilon_0,$$

with positive comparable class sizes and $d \to \infty$; $p$ need not tend to zero. In particular $p \le p_{\max} := (1-\epsilon_0)/c_-$. The average degree is eventually between $c_- d/2$ and $2c_+ d$. Choose

$$0 < c_0 < \min\{1/16,\ c_-/(4c_+),\ \epsilon_0/2\}.$$

Then $\varepsilon \le 2c_0 c_+ p < c_- p/2$, giving a distance at least $c_- p/2$ between every true probability and the lower clipping endpoint. For $n \ge 2$ the elementary graph bound $\bar{d} \le n-1$ also gives

$\varepsilon \le c_0 < \epsilon_0/2$. Hence the distance to the upper endpoint is at least $\epsilon_0/2 \ge \{\epsilon_0/(2p_{\max})\}p$. This proves the required interior condition for dense as well as sparse profiles.

The generated common-box proof, all likelihood derivatives, and all block-fluctuation moments now hold with fixed constants depending on $c_-, c_+, \epsilon_0$. The exact oracle-tilted probabilities remain between the two candidate probabilities. Both these means and the exact log-odds vectors are block constant, so the Euclidean tilted calculation uses no sparse approximation. All arguments above therefore apply on the proved good event to this general class. Proposition 10.1 pays that event's complement under an explicit information–degree margin; Corollary 6.3 supplies the sparse equal-exposure case where the margin is automatic.

# 9 The earlier unregularized DC profile decoder

This section proves the degree-corrected result in theorem 6.2. The reconstruction used here is the projection in the observed-degree inner product. It is an explicit decoder specification: the preceding Euclidean reconstruction is not silently identified with it. All likelihoods below are Bernoulli likelihoods. The condition $np^2 \le C_1$ controls their approximation at a bounded *log-likelihood* scale.

Write $Z_{ia} = \mathbf{1}\{z_i = a\}$, $\Theta = \operatorname{diag}(\theta_1, \dots, \theta_n)$, $B = pC$, and

$$\Omega = \Theta Z B Z^T \Theta, \qquad D_0 = \operatorname{diag}(\Omega \mathbf{1}), \qquad D = \operatorname{diag}(D_1, \dots, D_n), \quad D_i = \sum_j A_{ij}.$$

The matrix $\Omega$ includes its diagonal; the observed graph is hollow. Thus $\|\Omega - \mathbb{E}A\|_{\mathrm{op}} = O(p)$. In particular, every raw noise bound used below is unchanged up to its fixed constant by this completion. Throughout this section a generic constant may depend on the fixed parameter bounds in theorem 6.2, but never on $n$. Constants depending also on a fixed initialization-basin radius $M$ are denoted $C(M)$.

We use the common good event $\mathcal{G}$ supplied by lemma 7.1 and the probability calculation in section 10. Its consequences needed here are

$$cd \le D_i \le Cd, \qquad \|A - \Omega\|_{\mathrm{op}} \le C\sqrt{d}, \qquad \max_i \|U_{i\cdot}\|_2 \le Cn^{-1/2}, \tag{98}$$

$$\max_{a,b} \left| [Z^T(A - \Omega)Z]_{ab} \right| \le Cn, \tag{99}$$

together with the corresponding normalized spectral and degree consequences for the genuinely deleted graphs. The retained eigenvectors $U$ are orthonormal, and their eigenvalues $\mu_1, \dots, \mu_K$ are the $K$ largest in absolute value. The degree eigenvector is retained. The common lemmas establish these consequences from the model and pay their failure probabilities; they are not extra statistical assumptions. We now prove all remaining deterministic and local probabilistic steps.

## 9.1 The degree-metric reconstruction and its hollow correction

Put $S = D^{-1/2} A D^{-1/2}$, $\Pi = UU^T$, and

$$H = D^{-1/2} \Pi D^{1/2}, \qquad R = AH = D^{1/2} U \operatorname{diag}(\mu_1, \dots, \mu_K) U^T D^{1/2}. \tag{100}$$

The second formula evaluates $R$ from retained spectral information. Although $H$ need not be symmetric, $H^2 = H$ and $DH = H^T D$: it is an orthogonal projection for $\langle x, y \rangle_D = x^T D y$. Since the degree eigenvector is proportional to $D^{1/2}\mathbf{1}$, $H\mathbf{1} = \mathbf{1}$ and $R\mathbf{1} = D\mathbf{1}$.

For $h_i = R_{ii}$, define

$$s_i = \frac{h_i}{1 - \Pi_{ii}}, \qquad R^\circ = R + \operatorname{diag}(s_1, \dots, s_n)(H - I). \tag{101}$$

On $\mathcal{G}$, the denominators are positive for all sufficiently large $n$. Direct substitution gives

$$R^\circ_{ii} = 0, \qquad R^\circ \mathbf{1} = D\mathbf{1}, \qquad R^\circ Q = AQ \quad \text{if } \operatorname{col}(Q) \subseteq \operatorname{col}(D^{-1/2}U). \tag{102}$$

The last identity follows because $HQ = Q$. The hollow correction need not be symmetric; a symmetric block parameter is fitted by the ordered Bernoulli criterion, equivalently by symmetrizing its pair contributions. The row profile scores always use the original rows of $R^\circ$.

**Lemma 9.1** (Entry and absolute-row bounds)**.** *On $\mathcal{G}$,*

$$\max_{i,j} |R^\circ_{ij}| \le Cp, \qquad \max_i \sum_j |R^\circ_{ij}| \le Cd, \qquad \max_i |s_i| \le Cp. \tag{103}$$

*Every action in eqs.* (100) *and* (101) *is computable from $(D, U, \mu)$, with no access to individual edges after encoding.*

*Proof.* The normalized adjacency has operator norm at most one. Consequently

$$|R_{ij}| \le \sqrt{D_i D_j}\, \|U_{i\cdot}\|_2 \|U_{j\cdot}\|_2 \le Cd/n = Cp.$$

Moreover $|H_{ij}| \le \sqrt{D_j/D_i}\, \|U_{i\cdot}\|_2 \|U_{j\cdot}\|_2 \le C/n$. These estimates give $|s_i| \le Cp$ and the entry bound for $R^\circ$; summing its entries in absolute value gives the row bound. The displayed spectral formulas also prove the computational statement. □

### 9.2 A finite deterministic spectral initializer

The row-normalized population centers are particularly simple even when the $\theta_i$'s vary within blocks. To see this without replacing the observed degrees by their expectations, put

$$M_D = D^{-1/2}\Omega D^{-1/2}, \qquad G_D = Z^T \Theta D^{-1} \Theta Z.$$

The diagonal entries of $G_D$ are between constant multiples of $n/d$. The $K$ nonzero eigenvalues of $M_D$ have absolute values bounded away from zero, by the lower singular-value bound on $C$. An orthonormal basis of its range has the form

$$V = D^{-1/2}\Theta Z G_D^{-1/2} O \tag{104}$$

for an orthogonal $K$-by-$K$ matrix $O$. Each row has norm comparable to $n^{-1/2}$, and its normalization is exactly the $z_i$-th row of $O$. These $K$ unit centers have pairwise distance $\sqrt{2}$.

By eq. (98), $\|S - M_D\|_{\mathrm{op}} \le C/\sqrt{d}$. The signal/bulk gap and Davis–Kahan therefore give an aligned basis with $\|U - V\|_F^2 \le C/d$. The elementary normalization inequality

$$\left\| \frac{u}{\|u\|_2} - \frac{v}{\|v\|_2} \right\|_2 \le \frac{2\|u - v\|_2}{\|v\|_2}$$

holds when $u \neq 0$, and holds with a harmless change of constant under any fixed unit-vector convention for $u = 0$. For the normalized rows $x_i$ and true unit centers $c_a$, it follows that

$$\sum_{i=1}^n \|x_i - c_{z_i}\|_2^2 \le Cn/d. \tag{105}$$

The following initializer requires no numerical $K$-means certificate. Starting with all rows remaining, repeat $K$ times: choose a remaining row whose closed ball of radius $r = 1/8$ contains the largest number of remaining rows, break ties in a fixed order, record that row, and remove all remaining rows at distance at most $2r = 1/4$ from it. Finally assign every original row to its nearest recorded representative. If a step is undefined, use the algorithm's fixed fallback partition. The procedure uses $O(Kn^2)$ distance comparisons in fixed dimension.

**Lemma 9.2** (Ball-count initialization)**.** *Suppose every true block contains at least $\gamma n$ rows and the unit centers are separated by $\sqrt{2}$. Put $\epsilon = 1/32$. If $\sum_i \|x_i - c_{z_i}\|_2^2 \le E$ and $m = E/\epsilon^2 < \gamma n/3$, the procedure above selects exactly one representative for each true center and, after one common permutation of labels, makes at most $m$ errors. In particular, eq. (105) implies an initial error fraction at most $C/d$ on $\mathcal{G}$.*

*Proof.* Call a row good if it is within $\epsilon$ of its true center. There are at most $m$ bad rows. Inductively, all good rows of each previously selected block have been removed and all good rows of each unselected block remain. A radius-$r$ ball centered at a good row of any unselected block contains every good row of that block, since $2\epsilon < r$. Its count is at least $\gamma n - m > m$. Hence the maximizing ball contains a good row from an unselected block. Its selected representative is within $r + \epsilon = 5/32$ of that block's center. Every good row of this block is within $r + 2\epsilon = 3/16 < 2r$ of the representative, while every good row of another block is at least $\sqrt{2} - r - 2\epsilon > 2r$ away. This proves the induction. After $K$ selections, a good row is at most $3/16$ from its own representative and at least $\sqrt{2} - 3/16$ from every other one. Thus final nearest-representative assignment classifies all good rows correctly. □

### 9.3 Block-total accuracy and estimation of the nuisance parameters

We first record the residual estimate behind both the block fit and the tilted cancellation.

**Lemma 9.3** (Block residual in the degree metric)**.** *On $\mathcal{G}$, with $F = (I - \Pi)D^{1/2}Z$,*

$$\|F\|_{\mathrm{op}} \le \|F\|_F \le C\sqrt{n}, \qquad \sum_i \frac{(D_i - D_{0,i})^2}{D_i} \le Cn. \tag{106}$$

*The same conclusions hold for any genuinely deleted graph under its corresponding common good event.*

*Proof.* The degree-vector bound follows from

$$\|(D - D_0)\mathbf{1}\|_2 = \|(A - \Omega)\mathbf{1}\|_2 \le C\sqrt{nd}$$

and $\min_i D_i \ge cd$. Write

$$D^{1/2}Z = D^{-1/2}D_0Z + D^{-1/2}(D - D_0)Z.$$

Since $D_{0,i} = p\theta_i \sum_c n_c C_{z_i c}$, the first term belongs to $\mathrm{col}(D^{-1/2}\Theta Z) = \mathrm{col}(M_D)$. Its Frobenius norm is $O(\sqrt{nd})$, while the angle between this space and $\mathrm{col}(U)$ is $O(d^{-1/2})$. Its residual is therefore $O(\sqrt{n})$. The squared Frobenius norm of the second term is exactly the degree sum in eq. (106), because every row of $Z$ contains one unit entry. This proves the assertion. Deleting one vertex preserves all fixed group-mass bounds and the same calculation applies to the minor. □

**Lemma 9.4** (Canonical reconstructed block totals)**.** *Fix $M < \infty$. Uniformly over every possibly data-dependent partition $\widehat{Z}$ with at most $Mn/d$ mistakes after a common permutation,*

$$[\widehat{Z}^T R^{\circ} \widehat{Z}]_{ab} = n_a n_b p C_{ab} + O_M(n). \tag{107}$$

*The usual diagonal exclusion is included in the $O_M(n)$ term.*

*Proof.* Let $S_{\mathrm{tail}} = (I - \Pi)S(I - \Pi)$. Spectral invariance gives the exact quadratic identity

$$Z^T(R - A)Z = -F^T S_{\mathrm{tail}} F. \tag{108}$$

Because $M_D$ has rank $K$, Weyl's inequality gives $\|S_{\mathrm{tail}}\|_{\mathrm{op}} \le C/\sqrt{d}$. By lemma 9.3, the right side of eq. (108) has norm at most $Cn/\sqrt{d}$. The hollow correction changes a true block total by at most $Cd$: indeed its total absolute contribution is at most

$$\max_i |s_i| \sum_i \sum_j |(H - I)_{ij}| \le Cp\, Cn = Cd.$$

Together with eq. (99) and $Z^T \Omega Z = \mathrm{diag}(n_1, \dots, n_K)(pC)\,\mathrm{diag}(n_1, \dots, n_K)$, this proves the claim for true labels.

Changing at most $Mn/d$ labels changes at most $CMn^2/d$ ordered pair memberships. Each reconstructed entry has size at most $Cp$, so the additional error is at most $CMn$. This deterministic argument is uniform over the entire basin and requires no independence of $\widehat{Z}$. □

For fitted labels $\widehat{z}$, let $\widehat{n}_a$ be their sizes, $\overline{D}_a^{\widehat{z}}$ their mean degrees, and set

$$\widehat{\theta}_j = \frac{D_j}{\overline{D}^{\widehat{z}}_{\widehat{z}_j}}. \tag{109}$$

Thus $\sum_{\widehat{z}_j = a} \widehat{\theta}_j = \widehat{n}_a$ exactly. Fit the symmetric matrix $\widehat{B}$ by maximizing

$$\frac{1}{2} \sum_{i \ne j} \left\{ R^{\circ}_{ij} \log(\widehat{\theta}_i \widehat{\theta}_j B_{\widehat{z}_i \widehat{z}_j}) + (1 - R^{\circ}_{ij}) \log(1 - \widehat{\theta}_i \widehat{\theta}_j B_{\widehat{z}_i \widehat{z}_j}) \right\} \tag{110}$$

over positive symmetric $B$, under a fixed probability cap below one. This is a precisely defined signed Bernoulli working criterion. The next lemma proves that its relevant roots exist and lie in the interior on $\mathcal{G}$.

**Lemma 9.5** (Uniform fitted-parameter bounds)**.** *For every fixed $M$, uniformly over the basin in lemma 9.4, the degree estimates are bounded above and below by positive fixed constants and*

$$\frac{\|\widehat{\theta} - \theta\|_2}{\sqrt{n}} \le C(M) d^{-1/2}, \tag{111}$$

$$\max_{a,b} \left| \frac{\widehat{B}_{ab}}{pC_{ab}} - 1 \right| \le C(M)(d^{-1} + p) \le C(M) d^{-1}. \tag{112}$$

*The last inequality uses $dp = np^2 \le C_1$.*

*Proof.* Every fitted group has size comparable to $n$, and eq. (98) bounds its mean degree above and below by constant multiples of $d$. This proves boundedness of $\widehat{\theta}$. With true labels, write $D_i = dS_{z_i}\theta_i + \varepsilon_i$, where $S_a = \sum_c (n_c/n) C_{ac}$ and $\|\varepsilon\|_2 \le C\sqrt{nd}$. Division by the true-group mean degree, followed by subtraction of $\theta_i$, gives normalized squared error $O(d^{-1})$. Changing an $\eta$-fraction of labels changes each fitted-group mean degree by at most $Cd\eta$. For correctly labeled vertices this changes the estimated theta by $O(\eta)$; incorrectly labeled vertices are an $\eta$-fraction and have bounded error. Hence

$$\frac{\|\widehat{\theta} - \theta\|_2}{\sqrt{n}} \le C(d^{-1/2} + \eta + \sqrt{\eta}),$$

which yields eq. (111) for $\eta \le M/d$.

For the block fit, group the ordered pairs in eq. (110) belonging to the unordered block pair $\{a, b\}$. Denote this ordered set by $\mathcal{J}_{ab}$, put $\nu_{ab} = 2 - \mathbf{1}\{a = b\}$, $t_{ij} = \widehat{\theta}_i \widehat{\theta}_j$, and $E^{\circ}_{ab} = \sum_{\mathcal{J}_{ab}} R^{\circ}_{ij}$. Its score equation is

$$E^{\circ}_{ab} = b \sum_{(i,j) \in \mathcal{J}_{ab}} \frac{(1 - R^{\circ}_{ij}) t_{ij}}{1 - b t_{ij}}. \tag{113}$$

All $1 - R^\circ_{ij}$ are positive for large $n$. The right side is strictly increasing in $b$ below the cap. By lemma 9.4,

$$E^\circ_{ab} = \nu_{ab} n_a n_b p C_{ab} + O_M(n) \asymp n^2 p > 0.$$

For $b \asymp p$, bounded theta and $|R^\circ_{ij}| \le Cp$ give

$$\sum_{\mathcal{J}_{ab}} \frac{(1 - R^\circ_{ij}) t_{ij}}{1 - b t_{ij}} = \sum_{\mathcal{J}_{ab}} t_{ij}\{1 + O(p)\} = \nu_{ab} \widehat{n}_a \widehat{n}_b \{1 + O(p) + O(n^{-1})\}.$$

The last equality uses exact fitted-block mean normalization; for $a = b$, removing the diagonal subtracts only $\sum_{\widehat{z}_i = a} \widehat{\theta}_i^2 = O(n)$. Evaluating eq. (113) at sufficiently small and large fixed multiples of $p$ brackets its unique root, so

$$\widehat{B}_{ab} = \frac{E^\circ_{ab}}{\nu_{ab} \widehat{n}_a \widehat{n}_b} \{1 + O(p) + O(n^{-1})\}.$$

Since $|\widehat{n}_a - n_a| \le Mn/d$, this proves eq. (112). The root is $O(p)$, hence satisfies the fixed cap strictly for large $n$. □

### 9.4 Exact Bernoulli profiling at bounded score error

Fix a tested vertex $i$ with true block $a$. For candidate block $b$, define true neighbor masses and normalized profiles by

$$M_c^{(i)} = \sum_{\substack{j \ne i \\ z_j = c}} \theta_j, \qquad T_b^{(i)} = \sum_c M_c^{(i)} C_{bc}, \qquad q_{bc}^{(i)} = \frac{M_c^{(i)} C_{bc}}{T_b^{(i)}}. \tag{114}$$

The fitted analogues are

$$\widehat{M}_c^{(i)} = \sum_{\substack{j \ne i \\ \widehat{z}_j = c}} \widehat{\theta}_j = \widehat{n}_c - \widehat{\theta}_i \mathbf{1}\{\widehat{z}_i = c\}, \qquad \widehat{q}_{bc}^{(i)} = \frac{\widehat{M}_c^{(i)} \widehat{B}_{bc}}{\sum_e \widehat{M}_e^{(i)} \widehat{B}_{be}}. \tag{115}$$

The factor $p$ is not needed to evaluate the fitted quantity. All these probabilities are bounded away from zero. By lemma 9.5 and fitted group-size accuracy,

$$\max_{i,a,b,c} \left| \log \frac{\widehat{q}_{ac}^{(i)}}{\widehat{q}_{bc}^{(i)}} - \log \frac{q_{ac}^{(i)}}{q_{bc}^{(i)}} \right| \le C(M) d^{-1}. \tag{116}$$

Indeed normalization and logarithms are smooth on the fixed compact positive parameter set, and the relative mass error is $O_M(d^{-1} + n^{-1})$.

**Lemma 9.6** (Bernoulli profile reduction)**.** *Let a hollow row $r = (r_j)_{j \ne i}$ have $\sum_j r_j = D_i \ge 0$, $\sum_j |r_j| \le Ld$, and $D_i \le Ld$. Suppose either $r_j \in \{0, 1\}$, or $\max_j |r_j| \le Lp = o(1)$. For bounded positive neighbor parameters and block probabilities of order $p$, maximize the exact Bernoulli row criterion over the target parameter $t \ge 0$, with a fixed probability cap below one. After dropping terms independent of the candidate label, the profile contrast between $a$ and $b$ is*

$$L^{ab}_{r,\mathrm{prof}} = \sum_c N_c^r \log \frac{q_{ac}^{(i)}}{q_{bc}^{(i)}} + O(np^2), \qquad N_c^r = \sum_{\substack{j \ne i \\ z_j = c}} r_j. \tag{117}$$

*The same assertion holds with fitted labels, thetas, block probabilities, and profiles. Fixed positive candidate prior probabilities add their bounded log-ratio to both sides.*

*Proof.* For true parameters, omit the candidate-independent terms $\sum_j r_j \log \theta_j + D_i \log p$. The remaining score is

$$\ell_r(b,t) = D_i \log t + \sum_c N_c^r \log C_{bc} + \sum_{j \neq i} (1 - r_j) \log(1 - tp\theta_j C_{bz_j}).$$

For bounded $t$, expansion of the logarithm gives uniformly

$$\ell_r(b,t) = D_i \log t + \sum_c N_c^r \log C_{bc} - tpT_b^{(i)} + O(np^2). \tag{118}$$

The restored edge part of the linear exposure is bounded by $Cp \sum_j |r_j| \leq Cdp = Cnp^2$. The quadratic remainder is bounded by $Cp^2(n + \sum_j |r_j|) \leq Cnp^2$.

This expansion remains valid after maximizing. In both allowed row types $1 - r_j \geq 0$, so the exact score is concave in $t$. For a raw row, at least $n - O(d)$ nonedge terms remain; for a reconstructed row, every nonedge coefficient is at least $1/2$ for large $n$. Thus the nonedge part of its derivative has magnitude at least $cd$, whereas the positive term is $D_i/t$. The derivative is negative above a sufficiently large fixed $t_0$, uniformly in the candidate. The cap allows this bounded interval for large $n$, and the leading optimizer $t_b^0 = D_i/(pT_b^{(i)})$ is also bounded. Taking suprema in eq. (118) is therefore valid. The candidate-dependent leading maximum is $\sum_c N_c^r \log C_{bc} - D_i \log T_b^{(i)}$. Since $\sum_c N_c^r = D_i$, its difference is exactly the sum in eq. (117).

With fitted parameters replace $pC$ by $\widehat{B}$ and the true masses by their fitted analogues. Boundedness and $\widehat{B} \asymp p$ give the identical remainder calculation. When $D_i = 0$, use the continuous extension after combining the terms into $D_i \log t$, with $0 \log 0 = 0$. Nonnegative nonedge coefficients make $t = 0$ optimal, and the same contrast identity holds. In particular a raw zero row gives zero contrast. □

Exact preservation of $D_i$ is essential in this argument: it fixes the coefficient of $\log t$ and the profiled total-degree term. The proof needs no $O(d^{-1})$ estimate of an individual target theta.

**Lemma 9.7** (Uniform replacement of all fitted stages)**.** *Fix $M$. For every partition in the $M/d$ basin, fitted by eqs. (109) and (110), its actual row-profile contrast satisfies, simultaneously for all $i, a, b$,*

$$\widehat{L}_i^{ab} = [R^\circ w_{i,ab}]_i + O_M(1), \qquad (w_{i,ab})_j = \log \frac{q_{a,z_j}^{(i)}}{q_{b,z_j}^{(i)}}. \tag{119}$$

*The formula is uniform over data-dependent labels and parameters obtained by these fits, under the same common label permutation.*

*Proof.* The profile remainder is $O(np^2) = O(1)$. Replacing $\widehat{Z}$ by $Z$ in a bounded block coefficient vector changes row $i$'s score by at most

$$\max_j |R_{ij}^\circ| \, C \#\{j : \widehat{z}_j \neq z_j\} \leq Cp\,CMn/d = C(M).$$

Replacing the fitted coefficients by their true counterparts costs at most $\sum_j |R_{ij}^\circ| \, C(M) d^{-1} \leq C(M)$ by eq. (116). These are deterministic bounds holding for every partition in the basin, which proves the claimed uniformity. □

### 9.5 The profile reference and its sharp directional tails

For the fixed target $i$, true block $a$, and competitor $b$, write $w_j = (w_{i,ab})_j$, $v = \theta_i$, and

$$p_j = vp\theta_j C_{az_j}, \qquad T_{i,ab} = \sum_{j \neq i} w_j A_{ij}, \qquad K_{i,ab}(s) = \sum_{j \neq i} \log\{1 + p_j(e^{-sw_j} - 1)\}.$$

Set

$$J_{i,a\to b} = -\min_{s>0} K_{i,ab}(s). \tag{120}$$

This is the exact Bernoulli transform of the leading profiled contrast. It is not asserted to be the simple-hypothesis Chernoff information for the known-target-theta oracle.

**Lemma 9.8** (Uniform profile saddle and reference scale)**.** *The minimizing $s = s_{i,ab}$ stays in a fixed compact subinterval of $(0,1)$. Under the exact tilted product law,*

$$r_j = \frac{p_j e^{-sw_j}}{1 - p_j + p_j e^{-sw_j}}, \qquad \sum_j r_j w_j = 0, \qquad \sum_j r_j(1-r_j)w_j^2 \asymp d. \tag{121}$$

*Uniformly for bounded real thresholds $x$,*

$$\mathbb{P}_a(T_{i,ab} < x) \asymp d^{-1/2}\exp\{-J_{i,a\to b}\}. \tag{122}$$

*The exact full-edge Bernoulli profile-reference error at vertex $i$ is comparable, by fixed constants, to*

$$\sum_{b\neq a} d^{-1/2}\exp\{-J_{i,a\to b}\}. \tag{123}$$

*Proof.* Uniformly on bounded $s$ intervals, the transform and its first two derivatives satisfy

$$d^{-1}K_{i,ab}(s) = v\,\frac{T_a^{(i)}}{n}\left\{\sum_c (q_{ac}^{(i)})^{1-s}(q_{bc}^{(i)})^s - 1\right\} + O(p).$$

The derivative of the leading expression is strictly negative at zero and strictly positive at one. Positivity, the lower singular-value bound on $C$, and the group-mass bounds make these signs and the positive second derivative uniform. Indeed two normalized rows could coincide only if two rows of $C$ were proportional; compactness of the stated parameter class excludes even an arbitrarily small normalized separation. Strict convexity now gives the interior saddle and tilted variance of order $d$.

The bounded-weight, variance, and third-moment hypotheses of lemma 3.1 hold, so its bounded-shift tail conclusion proves eq. (122). In particular, no nonlattice assumption is necessary. Alternatively, under the displayed tilt, Berry–Esseen bounds every unit-interval mass by $C/\sqrt{d}$; summing against $e^{sT}$ gives the upper bound. A sufficiently long fixed interval immediately below $x$ has Gaussian mass larger than twice the Berry–Esseen error, giving the lower bound.

The probability of $D_i > Ld$ can be made $\exp(-Ld\,c_L)$ with arbitrarily large fixed exponential rate by increasing the fixed upper cutoff. On its complement, lemma 9.6 compares the exact reference contrast to $T_{i,ab}$ up to a fixed additive constant. The upper-degree exception is negligible relative to the displayed tail scales. Each strict pairwise defeat of the true label implies an error, while an error is contained in the union of weak pairwise defeats. Fixed $K$, the two bounded-shift tail bounds, and fixed ties therefore prove eq. (123). □

### 9.6 The complete degree-weighted tilted cancellation

We now work on the genuinely deleted graph $A^{(-i)}$. To simplify notation within this subsection, $D_-$, $U_-$, $\Pi_-$, and

$$H_- = D_-^{-1/2}\Pi_- D_-^{1/2}$$

refer only to this independent training graph. Its completed mean is $\Omega_-$, and $D_{0,-} = \operatorname{diag}(\Omega_- \mathbf{1})$. The tested row remains independent Bernoulli conditional on the deleted graph. Let $w = (w_j)_{j\neq i}$ be the true block coefficient vector above, and define $e = (H_- - I)w$.

**Lemma 9.9** (Bounded tilted coefficient error)**.** *On the deleted-graph good event, the exact profile tilt in eq.* (121) *satisfies*

$$\max_j |e_j| \le C, \qquad \sum_j r_j e_j^2 \le C, \qquad \left|\sum_j r_j e_j\right| \le C. \tag{124}$$

*All constants are uniform over the vertex and ordered pair of distinct candidate blocks.*

*Proof.* The residual lemma on the deleted graph gives

$$\|(I - \Pi_-)D_-^{1/2} Z_-\|_{\mathrm{op}} \le C\sqrt{n}.$$

Consequently, for every bounded block vector $w$,

$$\|(I - H_-)w\|_{D_-} \le C\sqrt{n}, \qquad \|(I - H_-)w\|_2^2 \le Cn/d. \tag{125}$$

Incoherence bounds the coordinates of $H_- w$:

$$|(H_- w)_j| \le D_{-,j}^{-1/2} \|U_{-,j\cdot}\|_2 \|U_-^T D_-^{1/2} w\|_2 \le (C/\sqrt{d})(C/\sqrt{n})C\sqrt{nd} \le C.$$

This proves bounded coefficients. Since $r_j \le Cp$, eq. (125) gives $\sum_j r_j e_j^2 \le Cp\,Cn/d \le C$.

For the mean, exact Bernoulli tilting gives

$$r_j = vp\theta_j C_{az_j} e^{-sw_j} + \varepsilon_j, \qquad \max_j |\varepsilon_j| \le Cp^2. \tag{126}$$

For a remaining vertex in block $c$, write

$$T_c^- = \sum_{\ell \ne i} \theta_\ell C_{cz_\ell}, \qquad h_j = \frac{vC_{ac}e^{-sw_c}}{T_c^-}.$$

Then $h$ is block constant with $\|h\|_\infty \le C/n$, and the leading vector in eq. (126) is exactly $D_{0,-}h$. Because $H_-$ is $D_-$-orthogonal,

$$\begin{aligned} r^T(H_- - I)w = & -\langle (I - H_-)h, (I - H_-)w\rangle_{D_-} \\ & + \big((D_{0,-} - D_-)h\big)^T (H_- - I)w + \varepsilon^T (H_- - I)w. \end{aligned} \tag{127}$$

This identity retains the random-degree correction explicitly. The first factor in its first term is at most $C/\sqrt{n}$: apply the block residual estimate to the block coefficients of $h$, which have magnitude $O(n^{-1})$. The second factor is at most $C\sqrt{n}$, so that term is $O(1)$.

For the next term, the degree aggregate in eq. (106) gives

$$\|(D_{0,-} - D_-)h\|_{D_-^{-1}}^2 = \sum_j \frac{(D_{0,-,j} - D_{-,j})^2 h_j^2}{D_{-,j}} \le C/n.$$

Cauchy–Schwarz in the degree metric and eq. (125) again give $O(1)$. Finally, bounded $e_j$ and eq. (126) bound the last term by $Cnp^2 = O(1)$. This proves the tilted mean bound and the lemma. □

The middle term of eq. (127) is why replacing random degrees by population degrees without a remainder is invalid. The product of two residual bounds is what reduces the leading tilted bias to a constant.

## 9.7 Completion of the risk bound on the common good event

By lemma 9.2, the initial partition lies in a fixed $M/d$ basin on $\mathcal{G}$. Apply one nuisance fit and one profile assignment. The uniform comparison eq. (119) and the actual full-to-deleted-graph scalar lemma lemma 7.2 give

$$\widehat{L}_i^{ab} = \sum_{j \neq i} A_{ij}(H_- w_{i,ab,-i})_j + O_M(1) = T_{i,ab} + \sum_{j \neq i} A_{ij} e_j + O_M(1). \tag{128}$$

The leave-one-out graph appears only in this proof identity; the implemented classifier uses the single full-graph encoding. The $O_M(1)$ error is a deterministic uniform bound, so it need not be asserted to be independent of the tested row.

Condition on the genuinely deleted graph. On its measurable good event, lemma 9.9 supplies all three coefficient conditions of the sharp transfer lemma lemma 3.1. Therefore, for a fixed bounded threshold large enough to absorb eq. (128),

$$\mathbb{P}_a\left\{ \sum_{j \neq i} A_{ij}(H_- w_{i,ab,-i})_j \leq C(M) \,\middle|\, A^{(-i)} \right\} \leq C(M) d^{-1/2} e^{-J_{i,a\to b}}. \tag{129}$$

The common full event implies the required minor event by the common geometry lemmas. Hence restricting an actual error to $\mathcal{G}$ is contained in the event on the left of eq. (129) together with the measurable minor event. This does not condition the tested row on a dependent full-graph event.

Integrating over the deleted graph and summing over $b \neq a$ gives

$$\mathbb{P}\{\widehat{z}_i \neq z_i, \mathcal{G}\} \leq C \sum_{b \neq z_i} d^{-1/2} e^{-J_{i,z_i \to b}}.$$

Here labels are aligned by the common initial permutation. The permutation-invariant misclassification loss is no larger than the loss under that alignment. By lemma 9.8, the last sum is at most a constant times the exact full-edge profile-reference error at $i$. Averaging proves

$$\mathbb{E}\{\ell(\widehat{z}, z)\mathbf{1}\{\mathcal{G}\}\} \leq C R_{\mathrm{prof},*}. \tag{130}$$

The calculation in section 10 proves $\mathbb{P}(\mathcal{G}^c) \leq C R_{\mathrm{prof},*}$, including the minimum-degree exception at the correct exponential scale. Because the fallback loss is at most one, adding that contribution to eq. (130) completes theorem 6.2.

## 9.8 Optional repeated hard fitting and finite adaptive stopping

The same theorem holds if, after initialization, the algorithm repeatedly recomputes eqs. (109) and (110) and makes a hard profile assignment using the same $R^\circ$. It may return any data-dependent finitely stopped iterate after at least one completed profile assignment. A finite iteration cap is one explicit termination rule.

**Lemma 9.10** (A uniform hard-refit basin)**.** *There is a fixed $M$ such that, on $\mathcal{G}$ and for all sufficiently large $n$, every hard fit-and-profile update maps the $M/d$ error basin into itself, under the same label matching. Every error at every updated iterate is contained in*

$$[R^\circ w_{i,ab}]_i \leq C(M) \quad \textit{for some } b \neq z_i. \tag{131}$$

*Proof.* First,

$$\|R^\circ - \Omega\|_F^2 \leq C_{\mathrm{spec}} d. \tag{132}$$

Indeed $S_K = U \operatorname{diag}(\mu) U^T$ is a best rank-$K$ operator-norm approximation to $S$. Since $M_D$ has rank $K$, $\|S_K - M_D\|_{\mathrm{op}} \leq 2\|S - M_D\|_{\mathrm{op}} \leq C/\sqrt{d}$. Their difference has rank at most $2K$, so its

Frobenius norm is also $O(d^{-1/2})$. Multiplication by $D^{1/2}$ on both sides gives $\|R-\Omega\|_F \le C\sqrt{d}$. The hollow term has Frobenius norm at most

$$\max_i |s_i|\,\|H-I\|_F \le Cp\,C\sqrt{n} = Cd/\sqrt{n},$$

whose square is at most $Cd$, proving eq. (132).

The completed population contrast has a uniform positive margin: with $\lambda_i = \sum_{j\neq i} p\theta_i\theta_j C_{z_i z_j}$,

$$[\Omega w_{i,ab}]_i = \lambda_i\,\mathrm{KL}(q_a^{(i)}\|q_b^{(i)}) + O(p) \ge c_{\mathrm{gap}}d, \qquad a = z_i,\ b\neq a. \tag{133}$$

The $O(p)$ term is the completed diagonal. The uniform KL gap follows from the same compactness and full-rank argument as in lemma 9.8. Also $\|w_{i,ab}\|_2 \le C\sqrt{n}$.

For an input partition in a fixed $M/d$ basin, lemma 9.7 bounds its fitted-contrast discrepancy by $C(M)$. For large $n$, this is at most $c_{\mathrm{gap}}d/2$. A new erroneous assignment therefore requires

$$\|(R^\circ-\Omega)_{i\cdot}\|_2 \ge \frac{c_{\mathrm{gap}}d}{2C\sqrt{n}}.$$

Summing this squared requirement and applying eq. (132) shows that the new iterate has at most $C_{\mathrm{new}}n/d$ errors. The constant $C_{\mathrm{new}}$ depends only on the fixed spectral and model constants, not on $M$. Choose $M$ at least both $C_{\mathrm{new}}$ and the initialization constant. All sufficiently large $n$ then give basin invariance. Group indices remain matched to the same true indices throughout: the uniform score comparison and its population margin use that fixed matching at every step.

Finally every erroneous assignment directly satisfies the fixed-threshold canonical event in the statement by lemma 9.7. The threshold is the same for all partitions in the basin. □

By induction, lemma 9.10 applies to every iterate on the single graph event $\mathcal{G}$. The final error of an adaptively chosen finite iterate is thus contained in the same canonical event used to prove eq. (129). Applying that probability bound once proves the same sharp risk bound; there is no union over iterations and no stopping-time independence assertion. This corollary concerns these specified hard updates. It does not assert the same result for an unrelated soft update or a reconstruction changed during iteration.

### 9.9 Equal exposures and the original known-target-theta oracle

We finish by proving corollary 6.3. Suppose

$$\sum_c \frac{n_c}{n}C_{ac} = S \quad \text{for every } a. \tag{134}$$

The block means of theta equal one, so $T_a^{(i)} = nS - \theta_i C_{a,z_i}$. In particular $T_a^{(i)} - T_b^{(i)} = O(1)$ and $\log(T_a^{(i)}/T_b^{(i)}) = O(n^{-1})$. Let $v = \theta_i$ be supplied to the full-edge known-target-theta oracle. Its exact Bernoulli contrast is

$$L^{ab}_{A,\mathrm{known}} = \sum_c N_c \log\frac{C_{ac}}{C_{bc}} + \sum_{j\neq i}(1-A_{ij})\log\frac{1-vp\theta_j C_{az_j}}{1-vp\theta_j C_{bz_j}}.$$

On $D_i \le Ld$, the same direct Bernoulli expansion used above gives

$$L^{ab}_{A,\mathrm{known}} = \sum_c N_c \log\frac{C_{ac}}{C_{bc}} - vp(T_a^{(i)} - T_b^{(i)}) + O(np^2).$$

By lemma 9.6, the exact profile contrast is

$$L^{ab}_{A,\mathrm{prof}} = \sum_c N_c \log\frac{C_{ac}}{C_{bc}} - D_i \log\frac{T_a^{(i)}}{T_b^{(i)}} + O(np^2).$$

Thus, using eq. (134),

$$|L^{ab}_{A,\mathrm{prof}} - L^{ab}_{A,\mathrm{known}}| \le C(np^2 + p) = O(1). \tag{135}$$

Fixed prior log-ratios, if present, are identical on both sides. The bounded-shift upper and lower tails in eq. (122) show that the two directional error probabilities are comparable by constants. The upper-degree exception has arbitrarily large fixed exponential rate after enlarging $L$, and is negligible for both references. Finite unions over competitors and averaging over vertices yield

$$R_{\mathrm{prof},*} \asymp R_{\theta,*}.$$

Combining this with theorem 6.2 proves corollary 6.3. The condition permits heterogeneous theta, unequal block sizes, and general positive full-rank symmetric $C$ obeying its exposure constraints. Its role is precisely to remove the label information carried by the leading total-degree exposure, which profiling otherwise changes.

### 9.10 The classical balanced DC-SBM oracle and its sharp risk

The fundamental local testing problem used by Gao, Ma, Zhang and Zhou [3] fixes the target degree parameter under both candidate labels. Its exact simple-hypothesis likelihood test therefore corresponds to $R_{\theta,*}$ in section 1.2. The next proposition explicitly connects that reference, the profile reference and true-group neighbor voting in the balanced planted-partition model. This is an oracle comparison with its own assumptions: it uses neither the spectral good event $\mathcal{G}$ nor a logarithmic lower bound on $np$.

**Proposition 9.11** (Sharp alignment of the classical DC-SBM references)**.** *Fix an arbitrary $K \ge 2$, positive bounds $\theta_-, \theta_+$, and $0 < r_- < r_+ < 1$. Let $n = Km$, let every group have $m$ vertices, and suppose*

$$\theta_- \le \theta_i \le \theta_+, \qquad \sum_{i:z_i=a} \theta_i = m \quad \text{for all } a.$$

*Specialize the block matrix $B = pC$ to*

$$B_{aa} = p, \qquad B_{ab} = q \ (a \ne b), \qquad r_- \le q/p \le r_+. \tag{136}$$

*Assume only, for the present oracle comparison,*

$$d = np \longrightarrow \infty, \qquad np^2 \le C_1 < \infty. \tag{137}$$

*Define $R_{\mathrm{vote},*}$ as the average error of the local raw-row rule which knows the neighbor labels and selects the group with the largest incident block count. Use any fixed tie ordering. Then, uniformly over this parameter class,*

$$\begin{aligned} R_{\theta,*} &\asymp R_{\mathrm{prof},*} \asymp R_{\mathrm{vote},*} \\ &\asymp \frac{1}{n}\sum_{i=1}^{n} \frac{1}{\sqrt{np}} \exp\left\{-\theta_i \frac{n}{K}(\sqrt{p} - \sqrt{q})^2\right\}. \end{aligned} \tag{138}$$

*The profile reference is exactly the Bernoulli profile in (6), with candidate probabilities capped at $1/2$. The same comparison holds if the voting counts are normalized by the self-excluded neighbor-group sizes. Constants depend only on the displayed fixed bounds, not on $n$. In particular, (137) permits $np = o(\log n)$.*

*Proof.* Fix a target $i$, its true group $a$, and a competitor $b \ne a$. For this proof write

$$N_c = \sum_{\substack{j \ne i \\ z_j = c}} A_{ij}, \qquad M_c = \sum_{\substack{j \ne i \\ z_j = c}} \theta_j = m - \theta_i \mathbf{1}\{c = a\}, \qquad L = \log(p/q).$$

Thus $L$ is bounded above and below by positive constants. Put

$$V_{i,ab} = L(N_a - N_b), \qquad J_i = \theta_i m(\sqrt{p} - \sqrt{q})^2.$$

The assumptions imply $J_i \asymp d$, uniformly in $i$. They also imply $p \to 0$, so every true candidate edge probability is eventually less than $1/2$.

**The exact Bernoulli count transform.** Under the actual true-row law, independence gives

$$\begin{aligned}\Psi_{i,ab}(s) &:= \log \mathbb{E}_a e^{-sV_{i,ab}} \\ &= \sum_{\substack{j\neq i \\ z_j=a}} \log\{1 + \theta_i\theta_j p(e^{-sL} - 1)\} + \sum_{j:z_j=b} \log\{1 + \theta_i\theta_j q(e^{sL} - 1)\}. \end{aligned} \tag{139}$$

Taylor expansion, uniformly for $s \in [0, 1]$, gives

$$\Psi_{i,ab}(s) = m\theta_i\{p(e^{-sL} - 1) + q(e^{sL} - 1)\} + O(np^2 + p). \tag{140}$$

The same error bound holds for the first two derivatives. Bounded theta and bounded $L$ make the sum of quadratic Bernoulli remainders $O(np^2)$. The additional $O(p)$ term accounts for the omitted $\theta_i$ in $M_a = m - \theta_i$.

The leading transform in (140) is minimized at $s = 1/2$, with minimum $-J_i$. Its endpoint derivatives have opposite signs, each of magnitude at least $cd$, and its second derivative is between positive constant multiples of $d$. Since $np^2 + p = o(d)$, the exact minimizing saddle stays in a fixed compact subinterval of $(0, 1)$. Convexity makes it the global minimum over $s \geq 0$. In particular,

$$I^{\mathrm{cnt}}_{i,ab} := -\min_{s\geq 0} \Psi_{i,ab}(s) = J_i + O(np^2 + p) = J_i + O(1). \tag{141}$$

One can also obtain this bound directly by taking minima in the uniform transform approximation.

Under the exact optimizing tilt, the Bernoulli probabilities on the two relevant groups are comparable to $p$. Their sum is $O(d)$, and the variance of $V_{i,ab}$ is comparable to $d$. Consequently lemma 3.1 applies to (139) without any spectral event. For every fixed $x_0 < \infty$, uniformly over $|x| \leq x_0$,

$$c_{x_0} d^{-1/2} e^{-J_i} \leq \mathbb{P}_a(V_{i,ab} < x) \leq \mathbb{P}_a(V_{i,ab} \leq x) \leq C_{x_0} d^{-1/2} e^{-J_i}. \tag{142}$$

The strict lower bound from that lemma uses an open interval of fixed length under the tilt. Thus lattice atoms and an arbitrary fixed tie ordering cause no loss of the lower bound.

**Comparison with the known-target likelihood.** Let

$$\mathcal{T}_c = \sum_{j\neq i} \theta_j B_{c,z_j} = pT^{(i)}_c.$$

By the exact group normalization,

$$\begin{aligned}\mathcal{T}_a &= m[p + (K-1)q] - \theta_i p, \\ \mathcal{T}_b &= m[p + (K-1)q] - \theta_i q.\end{aligned} \tag{143}$$

In particular $\mathcal{T}_a, \mathcal{T}_b \asymp d$, $\mathcal{T}_a - \mathcal{T}_b = O(p)$, and $\log(\mathcal{T}_a/\mathcal{T}_b) = O(n^{-1})$.

Fix a sufficiently large constant $L_0$ to be chosen below, and consider the single-target event

$$\mathcal{E}_i = \{D_i \leq L_0 d\}.$$

The exact known-target Bernoulli contrast is

$$L^{ab}_{i,\mathrm{known}} = V_{i,ab} + \sum_{j\neq i}(1 - A_{ij}) \log \frac{1 - \theta_i\theta_j B_{a,z_j}}{1 - \theta_i\theta_j B_{b,z_j}}.$$

Expansion of these nonedge logarithms on $\mathcal{E}_i$ gives

$$\begin{aligned} L^{ab}_{i,\mathrm{known}} &= V_{i,ab} - \theta_i(\mathcal{T}_a - \mathcal{T}_b) + O(D_i p + np^2) \\ &= V_{i,ab} + O(np^2 + p). \end{aligned} \tag{144}$$

The $O(D_i p)$ term restores the incident-edge terms in the linear exposure; the quadratic remainder is $O(np^2)$.

**Comparison with the capped Bernoulli profile.** For a candidate $c$, put $b_{cj} = \theta_j B_{c,z_j}$. The admissible interval in (6) is

$$0 \le t \le t_{\max,c} := \frac{1}{2\max_{j\neq i} b_{cj}}.$$

Because $b_{cj} \asymp p$, its upper endpoint tends to infinity. For $t > 0$ the derivative of the exact raw-row likelihood is

$$\frac{D_i}{t} - \sum_{j\neq i}(1 - A_{ij})\frac{b_{cj}}{1 - tb_{cj}}.$$

On $\mathcal{E}_i$, there are at least $n - 1 - L_0 d \ge n/2$ nonedges for all sufficiently large $n$. Their total intensity is at least $cd$, uniformly over candidates. Hence the derivative is negative for every admissible $t$ above a fixed constant depending only on $L_0$ and the structural bounds. Every exact maximizing $t$ is therefore bounded. The cap at $1/2$ is inactive on this bounded interval for all large $n$.

Uniformly over the resulting bounded $t$ interval, the likelihood is

$$D_i \log t + \sum_{j\neq i} A_{ij} \log b_{cj} - t\mathcal{T}_c + O(np^2). \tag{145}$$

Indeed, restoring the edge terms in the linear exposure costs $O(D_i p)$, and the sum of quadratic nonedge remainders costs $O(np^2)$. Both are $O(np^2)$ on $\mathcal{E}_i$. The leading optimizer $D_i/\mathcal{T}_c$ is bounded as well, so the same error bound holds after maximization. Terms $D_i \log D_i - D_i$ and $\sum_{j\neq i} A_{ij} \log \theta_j$ cancel in a contrast. Thus the exact capped profile satisfies

$$\begin{aligned} L^{ab}_{i,\mathrm{prof}} &= V_{i,ab} - D_i \log(\mathcal{T}_a/\mathcal{T}_b) + O(np^2) \\ &= V_{i,ab} + O(np^2 + p) \quad \text{on } \mathcal{E}_i. \end{aligned} \tag{146}$$

For $D_i = 0$, the raw row is identically zero; $t = 0$ maximizes every candidate likelihood with value zero. The displayed formulas then hold with the continuous convention $0 \log 0 = 0$.

**Single-row exceptions, directional errors, and averaging.** Every mean degree is at most $\lambda_+ d$ for a fixed $\lambda_+$. For any prescribed fixed $H$, a Bernoulli Chernoff bound permits a fixed choice of $L_0$ such that

$$\sup_i \mathbb{P}(\mathcal{E}_i^c) \le e^{-Hd}. \tag{147}$$

Also $J_i \le C_J d$ uniformly. Choose $H > C_J + 1$. It follows from $d \to \infty$ that

$$\sup_i \frac{\mathbb{P}(\mathcal{E}_i^c)}{d^{-1/2}e^{-J_i}} \le \sqrt{d}\, e^{-(H-C_J)d} \longrightarrow 0.$$

This calculation concerns one target at a time; it introduces no factor $n$ and requires no logarithmic lower bound on $d$.

By (144) and (146), a fixed constant $B_0$ bounds the difference between $V_{i,ab}$ and either exact contrast on $\mathcal{E}_i$. Thus for either $L^{ab}_i = L^{ab}_{i,\mathrm{known}}$ or $L^{ab}_i = L^{ab}_{i,\mathrm{prof}}$,

$$\begin{aligned} \mathbb{P}_a(V_{i,ab} < -B_0) - \mathbb{P}_a(\mathcal{E}_i^c) &\le \mathbb{P}_a(L^{ab}_i < 0) \\ &\le \mathbb{P}_a(L^{ab}_i \le 0) \\ &\le \mathbb{P}_a(V_{i,ab} \le B_0) + \mathbb{P}_a(\mathcal{E}_i^c). \end{aligned}$$

Applying (142) and then (147) proves that both the strict and nonstrict directional errors of either oracle are comparable to $d^{-1/2}e^{-J_i}$. These are unconditional event inclusions; no independence from $\mathcal{E}_i$ is asserted or needed.

For the raw block-count vote, the pairwise contrast is $V_{i,ab}$ itself. If counts are normalized by the self-excluded group sizes, multiplying all candidate scores by $mL$ changes the pairwise contrast by at most $CLD_i/n = O(p)$ on $\mathcal{E}_i$, since those sizes are $m$ or $m-1$. The same tail comparison therefore applies to either voting convention.

For each of these coherent families of candidate scores, a strict pairwise defeat of the true group implies an error, and an error implies a nonstrict defeat by some competitor. Taking the maximum of the strict pair probabilities for a lower bound and their nonstrict sum for an upper bound proves that each per-target error is comparable to $d^{-1/2}e^{-J_i}$, since $K$ is fixed. Averaging these bounds over $i$ proves (138). □

*Remark* 9.12 (Oracle alignment and the algorithmic range). The degree-dependent exponent in (138) is the balanced DC-SBM exponent appearing in the minimax analysis of Gao, Ma, Zhang and Zhou [3]. Their exponent-level minimax expression does not, by itself, give the sharp prefactor or a constant-ratio theorem for every parameter point. The proposition supplies the exact local-oracle comparison on the fixed-separation, bounded-theta class stated here.

For (136), the finite-$n$ exposures in (54) are exactly equal. Hence, on the overlap with the algorithmic range of theorem 6.2, corollary 6.3 and proposition 9.11 give

$$\mathbb{E}\ell(\widehat{z}_D, z) \le CR_{\theta,*}, \qquad R_{\theta,*} \asymp R_{\mathrm{prof},*}.$$

The oracle proposition alone does not remove the logarithmic restriction from the earlier unregularized theorem 6.2. The required sparse spectral and likelihood-transfer estimates are now supplied in section 2 for the explicit regularized decoder. Consequently theorem 2.1 give the displayed known-target oracle bound at every $np \to \infty$ with bounded $np^2$ on this classical class.

# 10 Probability closure for the historical decoders

The preceding sections prove fitted-score comparisons on events whose deterministic premises are consequences of raw noise, degree and row concentration bounds. This section completes the unconditional theorems. It is not enough that those events have probability tending to one.

## 10.1 Events with an arbitrarily large fixed exponential rate

For any fixed $H > 0$, choose the raw norm threshold $L_H\sqrt{d}$ in (55). Since $\lambda_i \le \lambda_+ d$ for a fixed $\lambda_+$, the elementary Bernoulli Chernoff inequality also gives, for a sufficiently large fixed $b_H$,

$$\mathbb{P}\{\max_i D_i > b_H d\} \le ne^{-Hd}.$$

For example the exponent in the Poisson upper bound is $b_H \log(b_H/\lambda_+) - b_H + \lambda_+$, which grows without bound as $b_H$ grows. Here this is a bound on a Bernoulli transform, not a change of sampling model. The conditional events in lemma 7.1 contribute at most $2Kne^{-Hd}$. The constants in the deterministic lemmas may increase with $H$, but remain independent of $n$.

For DC-SBM we additionally impose

$$\max_{a,b} |[Z^\top(A-\Omega)Z]_{ab}| \le L_B n. \tag{148}$$

Each centered block total is a sum of independent bounded upper-triangular edge variables, with coefficients at most two and variance at most $Cnd$. The completed diagonal contributes only $O(d)$. Scalar Bernstein therefore implies, for sufficiently large fixed $L_B$ and all large $n$,

$$\mathbb{P}\{(148) \text{ fails}\} \le 2K^2 \exp\left\{-\frac{cL_B^2 n}{d + L_B}\right\}.$$

Under $d^2 \le C_1 n$, the exponent is at least $c' L_B^2 d/C_1$ once $d \ge L_B$. Thus $L_B$ can be chosen to make this probability at most $2K^2 e^{-Hd}$ for any prescribed fixed $H$. Ordinary free-profile fitting does not need this event: it integrates the block-fluctuation exponential moment instead, as proved in section 8.

Let $\mathcal{G}$ be the intersection of these required events and the minimum-degree event $D_i \ge a_0 d$. All spectral, initialization and fitting premises in the earlier sections follow on $\mathcal{G}$. In particular all tested deleted graphs satisfy their own measurable training premises there. Apart from the minimum-degree event, the failure probability is at most

$$C_K n e^{-Hd}. \tag{149}$$

### 10.2 Ordinary SBM on the explicit parameter class

For $P_{ab} \in [p/2, p]$ and $p \le 1/4$, the log-odds coefficient of any pairwise oracle contrast satisfies

$$|w_c^{ab}| \le W := \log(7/3).$$

Every tilted Bernoulli probability between the two candidates is at most $p$, so the second derivative of the log affinity in (13) is at most $dW^2$. A twice differentiable convex function vanishing at zero and one with second derivative at most $M$ is at least $-Mt(1-t)/2$. Consequently

$$I_{i,ab} \le d\,\frac{\log^2(7/3)}{8} < 0.09d. \tag{150}$$

For all sufficiently large $n$, every mean degree is at least $0.49d$. Choose $a_0 = 1/16$. The Bernoulli lower-tail transform yields

$$\begin{aligned}\mathbb{P}(D_i < d/16) &\le \exp\left\{-d\left[0.49 - \frac{1}{16} - \frac{1}{16}\log(16\cdot 0.49)\right]\right\} \\ &\le e^{-0.298d}.\end{aligned}$$

$$\mathbb{P}(\min_i D_i < d/16) \le n e^{-0.298d}. \tag{151}$$

Take $H = 1$, increasing the other fixed thresholds if necessary. Equations (149) and (151) give $\mathbb{P}(\mathcal{G}^c) \le C n e^{-0.298d}$. The information bound (150) applies to every vertex and every competitor. Thus the sharp pairwise lower bound (14), followed by averaging over all vertices, gives

$$R_{\mathrm{SBM},*} \ge \frac{1}{n}\sum_{i=1}^{n} \frac{c}{\sqrt{d}} e^{-0.09d} = \frac{c}{\sqrt{d}} e^{-0.09d}.$$

No additional balancing assumption is needed: the uniform coefficient bound, group-proportion bound and full-rank separation already hold at every vertex. Consequently

$$\frac{\mathbb{P}(\mathcal{G}^c)}{R_{\mathrm{SBM},*}} \le C n \sqrt{d}\, e^{-0.208d}.$$

The function $\sqrt{d}\, e^{-0.208d}$ is decreasing once $d > 1/(2\cdot 0.208)$. Therefore, for $d \ge 5\log n$ and all sufficiently large $n$,

$$n\sqrt{d}\, e^{-0.208d} \le \sqrt{5\log n}\, n^{1-5\cdot 0.208} = \sqrt{5\log n}\, n^{-0.04} = o(1).$$

The earlier coefficient 20 was a sufficient numerical choice obtained by retaining only one vertex in the oracle lower bound and then bounding $\sqrt{d}$ by $\sqrt{n}$. Both losses are unnecessary. The improved coefficient 5 changes neither the encoder nor the likelihood decoder; it does not remove the global minimum-degree event from this proof. The good-event comparison in section 8 and (15) complete theorem 6.1.

### 10.3 The automatic margin for a general DC profile oracle

Let $J_{i,a\to b}$ be the exact Bernoulli directional profile information defined in section 9. For the self-excluded neighbor masses $M_c = \sum_{j\neq i: z_j = c} \theta_j$, set

$$T_a = \sum_c M_c C_{ac}, \qquad q_{ac} = \frac{M_c C_{ac}}{T_a}, \qquad h^*_{ab} = \min_{0\le s\le 1} \sum_c q_{ac}^{1-s} q_{bc}^{s}.$$

Uniform positive group proportions and entries of $C$ give $q_{ac} \ge q_- > 0$ for all large $n$. Hence

$$h^*_{ab} \ge \sum_c \min(q_{ac}, q_{bc}) \ge K q_- > 0. \tag{152}$$

The exact log-transform expansion, uniformly in $s \in [0,1]$, is

$$\sum_{j\neq i} \log\{1 + \theta_i\theta_j p C_{az_j}(e^{-s w_{z_j}} - 1)\} = \theta_i p T_a \left[\sum_c q_{ac}^{1-s} q_{bc}^{s} - 1\right] + O(np^2).$$

Minimizing therefore gives

$$J_{i,a\to b} = \lambda_i(1 - h^*_{ab}) + O(np^2), \qquad \lambda_i = \theta_i p T_a. \tag{153}$$

This calculation uses the exact Bernoulli transform. Under $np^2 \le C_1$, (152) and $\lambda_i \ge \lambda_- d$ imply, for a fixed $\eta > 0$ and all large $n$,

$$J_{\min} := \min_i \min_{b\neq z_i} J_{i,z_i\to b} \le \lambda_{\min} - \eta d. \tag{154}$$

To see it, choose a vertex with minimum mean degree and any competitor. Equal candidate exposures are not required for this profile margin.

Choose $a_0 > 0$ sufficiently small that

$$a_0\{1 + \log(\lambda_+/a_0)\} \le \eta/2, \qquad a_0 < \lambda_-.$$

For each vertex, the Bernoulli lower-tail bound is

$$\mathbb{P}(D_i < a_0 d) \le \exp\{-\lambda_i + a_0 d + a_0 d \log(\lambda_i/(a_0 d))\}.$$

The exponent is monotone in $\lambda_i$ above $a_0 d$. Together with (154), it follows that

$$\mathbb{P}(\min_i D_i < a_0 d) \le n \exp\{-J_{\min} - \eta d/2\}. \tag{155}$$

The full profile reference has the lower bound

$$R_{\mathrm{prof},*} \ge \frac{c}{n\sqrt{d}} e^{-J_{\min}}.$$

Thus the ratio of (155) to that reference is at most

$$C n^2 \sqrt{d}\, e^{-\eta d/2} = o(1)$$

when $d \ge C_0 \log n$ with, for example, $C_0 > 6/\eta$ and sufficiently large $n$. We used only $d \le n$ in bounding the polynomial factor. The constants can be enlarged without changing their independence of $n$.

Also $J_{\min} \le C_J d$ for a fixed $C_J$ by its bounded-weight transform. Choosing $H > C_J + 1$ in (149), and enlarging $C_0$ if needed, pays those remaining failures at the same reference scale. Hence $\mathbb{P}(\mathcal{G}^c) = O(R_{\mathrm{prof},*})$. The good-event theorem and invariant-basin result in section 9 now prove theorem 6.2, including its finite hard-refit version.

Under the equal-exposure condition, the profile and known-target-degree oracle risks are comparable by the bounded-contrast proof in section 9. This proves the DC assertion of corollary 6.3. For the native Euclidean ordinary-SBM decoder, the same sparse transform expansion gives $I_{i,ab} = \lambda_i(1 - h^*_{ab}) + O(np^2 + p)$ under equal exposures. Thus its information–degree margin is also automatic. The general-positive-profile version of section 8, with its block fluctuation integrated instead of imposing (148), therefore proves the remaining assertion of corollary 6.3.

### 10.4 An explicit extension for ordinary SBM with unequal exposures

The entry-ratio-two restriction in theorem 6.1 is a simple sufficient way to ensure the failure-event comparison. The deterministic and conditional likelihood proof covers a larger parameter range.

**Proposition 10.1** (A deterministic information–degree margin suffices)**.** *Suppose ordinary SBM has fixed $K$, positive comparable group sizes, $c_- p \le P_{ab} \le c_+ p$, $\max P_{ab} \le 1 - \epsilon_0$, and $\sigma_{\min}(P) \ge \kappa p$, for fixed positive constants. If for some fixed $\eta > 0$ and all sufficiently large $n$,*

$$\lambda_{\min} - I_{\min} \ge \eta d, \qquad d = np \ge C_0 \log n,$$

*then the native Euclidean signed likelihood decoder, with sufficiently small structural floor coefficient, satisfies $\mathbb{E}\ell(\widehat{z}_E, z) \le C R_{\mathrm{SBM},*}$ when $C_0$ is sufficiently large. No condition $np^2 = O(1)$ is needed for this ordinary-SBM proposition.*

*Proof.* All true Bernoulli likelihood coefficients and tilted means are exactly block constant, so the Euclidean projection, fitted-score and sharp-tail arguments of section 8 apply without a sparse Bernoulli approximation. Generated profiles remain comparable to $p$ and uniformly away from one when the floor is chosen for the structural class. The displayed margin, the same small degree cutoff used above and an arbitrarily strong choice of the other exponential bounds imply $\mathbb{P}(\mathcal{G}^c) = O(R_{\mathrm{SBM},*})$. Here the polynomial comparison uses $d \le Cn$, which follows from $c_- p \le P_{ab} \le 1 - \epsilon_0$; it does not require $p \le 1$. □

This proposition is a condition on the deterministic model parameters, not an assumed success event or a hypothesis that the desired risk ratio already holds. It is nevertheless an additional restriction. For model families violating it, a refined analysis of exceptional low-degree graphs or a different normalized encoder is needed; the crude loss bound one on all such graphs does not prove a constant ratio.

## 11 Why the known-target oracle cannot be the general DC-SBM benchmark

The profile reference and an oracle supplied with the target's true degree parameter are different experiments. The distinction cannot be removed by a more accurate spectral reconstruction: the following obstruction holds even when the classifier observes the entire adjacency matrix. The example has bounded positive degree parameters, exact block normalization, balanced groups, a positive definite block matrix, and logarithmic expected degree. Every probability calculation below is for the Bernoulli graph itself.

**The local known-target reference.** For a fixed DC-SBM parameter point $\tau = (z^\tau, \theta^\tau)$, write $\mathbb{P}_\tau$ for the graph law. At a target vertex $v$, give a local oracle the values of $p, C, \theta_v^\tau$ and all neighbor parameters $(z_j^\tau, \theta_j^\tau)$, $j \ne v$. Its candidate row laws are

$$q_{vj}^{(a,\tau)} = p\theta_v^\tau \theta_j^\tau C_{a z_j^\tau}, \qquad L_v^{(a,\tau)}(A_{v,-v}) = \prod_{j \ne v} (q_{vj}^{(a,\tau)})^{A_{vj}} (1 - q_{vj}^{(a,\tau)})^{1 - A_{vj}}. \tag{156}$$

Let $\delta_v^{\mathrm{known},\tau}$ maximize these two likelihoods with equal candidate priors and a fixed tie rule, and define

$$R_{\mathrm{known}}^\tau(n) = \frac{1}{n} \sum_{v=1}^n \mathbb{P}_\tau\{\delta_v^{\mathrm{known},\tau} \ne z_v^\tau\}. \tag{157}$$

The target parameter is held at its supplied true value under both candidate labels. The local comparison does not impose a global exact-group-size or block-normalization constraint after revealing the other labels; an exact group-size constraint would itself determine a missing label. Thus (157) is the nondegenerate local likelihood benchmark, not a globally constrained oracle experiment.

**Theorem 11.1** (An exact Bernoulli obstruction for the known-target oracle)**.** *Let $n = 2m$ tend to infinity through even integers, and set*

$$C = \begin{pmatrix} 4 & 2 \\ 2 & 11/10 \end{pmatrix}, \qquad p = \frac{M \log n}{n}, \qquad M = 100. \tag{158}$$

*There are two explicit parameter sequences, denoted $+$ and $-$, for the independent-edge Bernoulli DC-SBM with probabilities $p\theta_u\theta_v C_{z_u z_v}$, such that both sequences satisfy*

$$n_1 = n_2 = m, \qquad \sum_{v:z_v=a} \theta_v = m \quad (a = 1, 2), \qquad \frac{1}{2} \le \theta_v \le \frac{3}{2}. \tag{159}$$

*For every possibly randomized clustering algorithm based on the full graph and information common to the two parameter points, put*

$$R^{\tau}_{\mathrm{alg}}(n) = \mathbb{E}_{\tau}\left[\frac{1}{n} \min_{\sigma \in \mathfrak{S}_2} \sum_{v=1}^{n} \mathbf{1}\{\sigma(\widehat{z}_v) \neq z^{\tau}_v\}\right], \qquad \tau \in \{+, -\}. \tag{160}$$

*Define*

$$c = \frac{(2 - \sqrt{2})^2}{8}, \qquad \gamma = \frac{(\sqrt{11/10} - 1)^2}{2}. \tag{161}$$

*Then*

$$\max_{\tau \in \{+,-\}} \frac{R^{\tau}_{\mathrm{alg}}(n)}{R^{\tau}_{\mathrm{known}}(n)} \ge \frac{1 + o(1)}{4}\, n^{M(c-2\gamma)-1}. \tag{162}$$

*The exponent is strictly positive; at $M = 100$,*

$$M(c - 2\gamma) - 1 = 3.05109151537556\ldots > 0. \tag{163}$$

*In particular, no algorithm with unknown vertex degree parameters can have a bounded risk ratio to (157) along every admissible parameter sequence in this class. This includes every algorithm based only on retained spectral information and observed degrees.*

*Proof.* **Construction and exact normalization.** Distinguish vertices $i = 1$ and $k = 2$. Partition the remaining vertices into two fixed anchor sets $V_1, V_2$, each of size $m - 1$. Every anchor in $V_a$ has label $a$ under both parameter points. Give anchors in $V_1$ the common degree parameter

$$t_m = 1 + \frac{1}{2(m-1)},$$

and anchors in $V_2$ degree parameter one. At the two parameter points set

$$\begin{aligned} +: &\quad (z_i, \theta_i) = (1, 1/2), \quad (z_k, \theta_k) = (2, 1), \\ -: &\quad (z_i, \theta_i) = (2, 1), \qquad (z_k, \theta_k) = (1, 1/2). \end{aligned} \tag{164}$$

Each group has exactly $m$ vertices. Its total degree parameter is exactly $m$, since

$$(m-1)t_m + \frac{1}{2} = m, \qquad (m-1) + 1 = m.$$

The stated degree bounds hold for $m \ge 2$. The matrix $C$ has positive entries, positive trace, and determinant $2/5 > 0$, so it is positive definite with a fixed positive smallest eigenvalue. All edge probabilities belong to $(0, 1)$ for sufficiently large $n$.

**Only $n - 2$ Bernoulli coordinates distinguish the graph laws.** All anchor–anchor probabilities agree between the two parameter points. The edge $i$–$k$ has probability $p$ under

either point. For an edge from either distinguished vertex to an anchor of $V_1$, the two candidate intensities agree exactly:

$$\frac{1}{2} \cdot 4 \cdot t_m = 1 \cdot 2 \cdot t_m = 2t_m.$$

Thus the only distinct coordinates are the $2(m-1) = n-2$ edges from $\{i,k\}$ to $V_2$. Their two probabilities are $p$ and $(11/10)p$, exchanged between $+$ and $-$.

For probability mass functions $P, Q$, let $\mathcal{A}(P,Q) = \sum_x \sqrt{P(x)Q(x)}$ be their affinity. Independence gives the exact graph affinity

$$\mathcal{A}_n = \mathcal{A}(\mathbb{P}_+, \mathbb{P}_-) = \left[\sqrt{11/10}\, p + \sqrt{(1-p)(1-(11/10)p)}\right]^{n-2}. \tag{165}$$

The factor in brackets has the elementary expansion

$$\sqrt{11/10}\, p + \sqrt{(1-p)(1-(11/10)p)} = 1 - \gamma p + O(p^2).$$

Since $np^2 = M^2(\log n)^2/n = o(1)$,

$$\mathcal{A}_n^2 = \exp\{-2M\gamma \log n + o(1)\} = (1+o(1))n^{-2M\gamma}. \tag{166}$$

The elementary Cauchy–Schwarz bound $\mathrm{TV}(P,Q) \le \sqrt{1-\mathcal{A}(P,Q)^2}$ implies that every test $\phi$ of the two graph laws satisfies

$$\frac{\mathbb{P}_+(\phi = -) + \mathbb{P}_-(\phi = +)}{2} \ge \frac{1 - \mathrm{TV}(\mathbb{P}_+, \mathbb{P}_-)}{2} \tag{167}$$

$$\ge \frac{\mathcal{A}_n^2}{4} = \frac{1+o(1)}{4} n^{-2M\gamma}, \tag{168}$$

where the second inequality uses $1 - \sqrt{1-y} \ge y/2$. Independent randomization by the test or the clustering algorithm does not alter this bound: adjoining a common random seed leaves the affinity unchanged.

**From testing to graph-average permutation loss.** Orient the two output labels of an arbitrary estimated partition by the permutation minimizing disagreement with the fixed anchor labels, using a fixed tie rule. This is one measurable rule under both parameter points, since the anchors and their labels are the same. Define a test $\phi$ to declare $+$ if the oriented label of vertex $i$ is 1, and $-$ otherwise.

If the estimated partition has zero permutation-invariant error, the anchor orientation recovers its correct permutation uniquely: both anchor sets are nonempty. Therefore an erroneous test outcome implies at least one error in the permutation-invariant count. Pathwise, under either point,

$$\frac{1}{n} \min_{\sigma \in \mathfrak{S}_2} \sum_v \mathbf{1}\{\sigma(\widehat{z}_v) \ne z_v^\tau\} \ge \frac{1}{n} \mathbf{1}\{\phi \ne \tau\}. \tag{169}$$

Taking expectations and using (168) gives

$$\frac{R_{\mathrm{alg}}^+(n) + R_{\mathrm{alg}}^-(n)}{2} \ge \frac{1+o(1)}{4} n^{-2M\gamma-1}. \tag{170}$$

The anchors are needed only to turn a partition into a test in this proof. Indeed, their labels and degree parameters, the block matrix, the exact group sizes, and the block-normalization constraints could all be revealed to the clustering algorithm: they are common to the two parameter points, and (168) would still apply.

**An upper bound on every known-target local error.** For any target $v$, at least $m-2$ anchors in $V_1$ are different from $v$. For each such anchor $j$, its two local candidate edge probabilities in (156) are

$$u = 4p\theta_v\theta_j, \qquad v' = 2p\theta_v\theta_j.$$

They satisfy

$$(\sqrt{u}-\sqrt{v'})^2 = p\theta_v\theta_j(2-\sqrt{2})^2 \geq \frac{p}{2}(2-\sqrt{2})^2.$$

For any Bernoulli probabilities $u, v'$, the arithmetic–geometric mean inequality for their nonedge probabilities yields

$$\sqrt{uv'}+\sqrt{(1-u)(1-v')} \leq 1-\frac{(\sqrt{u}-\sqrt{v'})^2}{2} \leq \exp\left\{-\frac{(\sqrt{u}-\sqrt{v'})^2}{2}\right\}. \tag{171}$$

Under either local true label, Markov's inequality applied to the square root of the competing likelihood ratio bounds the LRT's directional error by the product affinity of all its row coordinates. Discarding coordinates outside these $m-2$ anchors can only increase that upper bound. Hence, uniformly over targets and both parameter points,

$$\mathbb{P}_\tau\{\delta_v^{\mathrm{known},\tau} \neq z_v^\tau\} \leq \exp\left\{-\frac{(m-2)p}{4}(2-\sqrt{2})^2\right\} \tag{172}$$

$$= (1+o(1))n^{-Mc}. \tag{173}$$

In particular,

$$R^+_{\mathrm{known}}(n) + R^-_{\mathrm{known}}(n) \leq 2(1+o(1))n^{-Mc}. \tag{174}$$

Finally, the maximum of two nonnegative risk ratios is at least their denominator-weighted average. Combining (170) and (174) therefore proves (162). The constants in (161) are

$$c = 0.04289321881345246\ldots, \qquad \gamma = 0.00119115182984846\ldots,$$

which give the exponent in (163). Its positivity does not depend on numerical evaluation: $c > 1/25$ and $2\gamma < 1/400$, so $100(c-2\gamma)-1 > 11/4 > 0$. □

*Remark* 11.2 (Scope of the obstruction). The same proof works for every fixed $M > (c-2\gamma)^{-1}$, so its logarithmic-degree multiplier can exceed any prescribed fixed lower threshold. The example also satisfies $np^2 \to 0$. Thus neither a larger logarithmic expected degree nor the sparse Bernoulli approximation removes the nuisance-information gap.

The result concerns a uniform comparison to the known-target reference, not the full-edge profile benchmark used by the positive DC-SBM theorem. It also does not assert that every fixed DC-SBM has an unbounded ratio: it proves that at least one of the two explicit parameter sequences has an unbounded limsup ratio for any algorithm. Additional restrictions can make the stronger benchmark attainable. For example, the balanced block strengths in this counterexample are $C(1/2,1/2)^T = (3, 31/20)^T$, which are unequal; the example lies outside the equal-exposure corollary.

The counterexample is also outside the assortative parameter class of Gao, Ma, Zhang and Zhou [3]: its smallest diagonal entry is 1.1, whereas its off-diagonal entry is 2. Their class requires every within-group strength to exceed every between-group strength. Therefore this obstruction is not a negative result for their classical balanced planted-partition benchmark. That benchmark is explicitly covered by proposition 9.11 on the stated bounded-parameter class. Their classwise minimax theorem and a pointwise comparison to a known-target oracle on every arbitrary block matrix are different claims.

Finally, a lower bound for an experiment retaining only block counts would not by itself establish a lower bound for arbitrary data-dependent real spectral coordinates. The present theorem needs no such compression claim: it is a lower bound for the full adjacency experiment with unknown target degree parameters.

# 12 Additional arguments, implementation scope, and retained records

## 12.1 Why a whole-graph zero-degree fallback fails

**Proposition 12.1** (A whole-graph zero-degree fallback is invalid in the sparse regime)**.** *Consider a balanced two-group SBM with within probability $p$, between probability $p/2$, and $d = np = \log\log n$. The probability of at least one isolated vertex tends to one. For every deterministic fallback partition fixed in advance, there is a balanced true partition for which an algorithm returning that fallback whenever an isolated vertex exists has expected permutation-invariant misclassification bounded below by $1/4 + o(1)$. Its local oracle risk tends to zero.*

*Proof.* For every vertex the isolation probability is $(1-p)^{n/2-1}(1-p/2)^{n/2} = \exp\{-3d/4 + o(1)\}$. If $N_0$ counts isolated vertices, then $\mathbb{E}N_0 = n\exp\{-3d/4 + o(1)\} \to \infty$. For distinct vertices, their two isolation events share only the edge between them, giving

$$\frac{\mathrm{Var}(N_0)}{(\mathbb{E}N_0)^2} \le \frac{1}{\mathbb{E}N_0} + \frac{p}{1-p} \longrightarrow 0.$$

Choose a balanced true partition whose overlap with the fixed fallback has error at least $1/4$ after either label permutation. Such a partition exists by splitting each fallback class approximately evenly. The loss lower bound follows. Standard incident-row Chernoff comparison gives oracle risk at most $Ce^{-cd} \to 0$ on this separated model. □

Thus the logarithmic restriction cannot be removed merely by deleting text from the unregularized theorem. The valid repair is to restore a positive floor and local risk accounting. With the regularized encoder, a zero-degree vertex has a zero reconstructed row and causes no whole-graph fallback.

## 12.2 The existing degree-plus-spectrum action remains valid

Write $\mathbf{d} = A\mathbf{1}$ for the raw degree vector. For any strictly positive diagonal matrix $W$, retain

$$WAWU = U\Lambda, \qquad U^\top U = I, \qquad X = [\mathbf{1}, WU], \qquad Y = [\mathbf{d}, W^{-1}U\Lambda].$$

Then the exact identities

$$AX = Y, \qquad YX^\dagger = A\Pi_X, \qquad A\Pi_X\mathbf{1} = \mathbf{d} \tag{175}$$

hold for every graph, including graphs with zero degrees. Here $\Pi_X$ is the Euclidean orthogonal projector onto the column space of $X$; no full-column-rank assumption is necessary. The earlier choice $W_{ii} = \min(1, 2\bar{D}/D_i)$ for $D_i > 0$, with $W_{ii} = 1$ for $D_i = 0$ and $\bar{D} = n^{-1}\sum_i D_i$, supplies a positive reweighting when $\bar{D} > 0$. The all-zero graph can be assigned a fixed output. The block-count error is exactly $-A(I - \Pi_X)Z$.

These facts need no logarithmic expected-degree assumption. They are computational identities, however: their validity does not give the sharp tail of the remaining random block-count error. In DC-SBM a normalization must also preserve the effective likelihood coefficients, as opposed to preserving only a population mean.

## 12.3 Algorithm distinctions and scope

The current ordinary article uses the regularized block Bernoulli LRT, with fixed degree parameters equal to one and no target profiling. The current DC result in this companion uses clamped, group-normalized nuisance estimates and profiles the target separately for each candidate. The ordinary article retains the eigensystem of $WAW$ with $W_i = \sqrt{\min(1, b_0\bar{D}/D_i)}$ and a sufficiently

large fixed $b_0$. This leaves every vertex below the cap unweighted. The current DC result retains the small-floor normalized eigensystem, with $G_i = \max(D_i, a_0\bar{D})$ and sufficiently small fixed $a_0$. Both retain raw degrees, apply exact degree correction and simple hollowing, and control finite hard refits through a uniform invariant basin. The large-threshold ordinary proof uses the population vector $\tau G^{-1/2}Z$; it does not supply the DC argument involving $G^{-1/2}\Theta Z$. The small-floor DC geometry and its probability bounds are therefore established separately in Section 3.

The older ordinary free-profile growing/replacement/soft-EM schedule in section 8 is a different finite algorithm. Its separate proof also covers probabilities outside bounded $np^2$. The old common-box proof uses a simultaneous entry bound and must not be attached without change to the present sparse encoder. In particular, exact degree preservation together with $|R_{ij}| \le Cp$ would force all raw degrees to be at most $Cd$, which is not a valid sparse uniform premise. The logarithmic restrictions written in the historical part describe those earlier unregularized proofs, not the current theorem.

The earlier DC reconstruction $AH_D$ is different from the Euclidean $A\Pi_X$, even though both can be evaluated from the same retained input. Earlier experimental curves do not automatically evaluate either the new empirical floor or its clamped nuisance fit. The present claims are mathematical risk bounds, not conclusions inferred from simulations. Numerical eigensolver tolerances and optimization tolerances require their own control of induced score errors; exact arithmetic defines these algorithms. The largest absolute eigenvalues are retained so that a full-rank indefinite signal is included.

The constants may depend on the fixed signal gap, group proportions, bounded degree-parameter range, and $C_1$. A bounded risk ratio is not a claim that the ratio tends to one. General DC-SBM with unbounded $np^2$, a vanishing signal gap, or degree parameters approaching zero is outside the present theorem. The general known-target-degree benchmark is obstructed by theorem 11.1; equal exposures retain the positive result.

### 12.4 Sparse proof history and review records

The 9 October degree-plus-spectrum draft contained the sparse truncated row-energy mechanism at $d \to \infty$, $d = o(\log n)$. Its score errors were $O(\sqrt{d})$ or $o(d)$ and therefore established a leading exponent, not the present constant risk ratio. The current argument keeps its same-threshold mechanism, upgrades the local full-to-deleted score comparison and fitted contrast errors to $O(1)$, and preserves the sharp $d^{-1/2}$ prefactor. The earlier source remains in the repository at commit `de60b64`.

The source package includes the `audits/` directory and its index. It retains the sparse-history audit, the independent local-floor LOO review, the same-threshold tail and conditioning review, the high-scale probability review, the ordinary algorithm-scope review, the final combined review, and earlier DC and regularization audit snapshots. These are internal working checks, not external peer review. Statements of then-open gaps in older snapshots are superseded by the current proofs. The full pre-split revision is preserved separately in the repository; it is an archive rather than the reading order of these notes.

An inventory in `README.md` records where each old source module now belongs. The original implementation/audit narrative is retained as `archive/scope_audit.tex`; the obsolete end-of-revision status narrative is retained as `archive/sparse_interfaces.tex`. These source records are not assertions of a remaining logarithmic lower bound.

The ordinary normalized version dated 20 October is preserved as a complete source-only snapshot in `archive/ordinary_normalized_20261020/`. Its algorithm and proof preceded the adjacency-based first article. The snapshot is an archival alternative, not an additional hypothesis on the current ordinary theorem or an unproved extension of the DC theorem.

## References


[1] Emmanuel Abbe, Jianqing Fan, Kaizheng Wang, and Yiqiao Zhong. Entrywise eigenvector analysis of random matrices with low expected rank, 2017. arXiv:1709.09565; Lemma 7, equation (C.2), version accessed 20 October 2026.

[2] Afonso S. Bandeira and Ramon van Handel. Sharp nonasymptotic bounds on the norm of random matrices with independent entries. *The Annals of Probability*, 44(4):2479–2506, 2016. Corollary 3.12 and Remark 3.13; arXiv:1408.6185v4.

[3] Chao Gao, Zongming Ma, Anderson Y. Zhang, and Harrison H. Zhou. Community detection in degree-corrected block models. *The Annals of Statistics*, 46(5):2153–2185, 2018. Local testing formulation checked in the author manuscript arXiv:1607.06993.

[4] Can M. Le, Elizaveta Levina, and Roman Vershynin. Concentration and regularization of random graphs. *Random Structures & Algorithms*, 51(3):538–561, 2017. Theorem 2.1 and Remark 2.2; arXiv:1506.00669.

[5] Zhixin Zhou and Ping Li. Rate optimal Chernoff bound and application to community detection in the stochastic block models. *Electronic Journal of Statistics*, 14(1):1302–1347, 2020.